

# Disorders of Potassium Balance

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## CHAPTER OUTLINE

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| POTASSIUM DISORDERS, 471                          | CAUSES OF HYPOKALEMIA, 484      |
| NORMAL POTASSIUM BALANCE, 471                     | TREATMENT OF HYPOKALEMIA, 495   |
| CONSEQUENCES OF HYPERKALEMIA AND HYPOKALEMIA, 481 | HYPERKALEMIA, 498               |
|                                                   | MANAGEMENT OF HYPERKALEMIA, 505 |

## POTASSIUM DISORDERS

The diagnosis and management of potassium disorders are central skills in clinical nephrology, relevant to not only consultative nephrology but also dialysis and renal transplantation. An understanding of the underlying physiology is critical to the diagnostic and management approach to hyperkalemic and hypokalemic patients. This chapter reviews those aspects of the physiology of potassium homeostasis judged to be relevant to the understanding of potassium disorders; a more detailed review of renal potassium transport is provided in [Chapter 6](#).

The pathophysiology of potassium disorders continues to evolve. The expanding list of drugs with the potential to affect plasma potassium concentration ( $K^+$ ) has both complicated clinical management and provided new insight. In addition, the evolving molecular understanding of both common and rare disorders affecting plasma  $K^+$  continues to uncover novel pathways of regulation.<sup>1-11</sup> These advances can be incorporated into an increasingly mechanistic, molecular understanding of potassium disorders.

## NORMAL POTASSIUM BALANCE

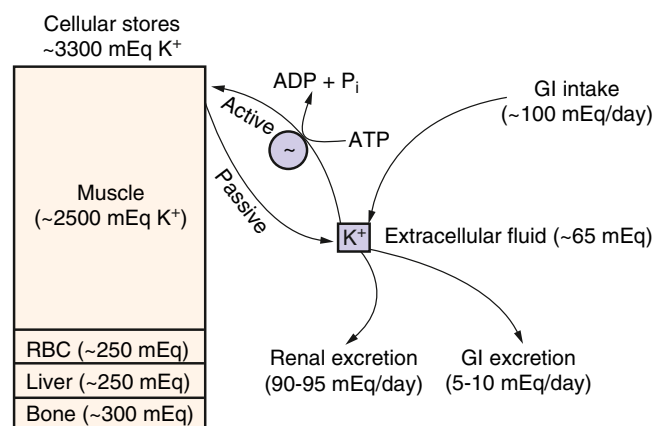
The dietary intake of potassium ranges from  $<35$  to  $>110$  mmol/day in U.S. men and women. Despite this widespread variation in intake, homeostatic mechanisms serve to precisely maintain plasma  $K^+$  between 3.5 and 5.0 mmol/L. In a healthy individual at steady state, the entire daily intake of potassium is excreted, approximately 90% in the urine and 10% in the stool. More than 98% of total-body potassium is intracellular, chiefly in muscle ([Fig. 16.1](#)). Buffering of extracellular  $K^+$  by this large intracellular pool plays a crucial role in the regulation of plasma  $K^+$ .<sup>12</sup> Thus within 60 minutes of an intravenous load of 0.5 mmol/kg of  $K^+$ -Cl<sup>-</sup>, only 41% appears in the urine yet serum  $K^+$  rises by no more than 0.6 mmol/L<sup>13</sup>; adding the equivalent of 35 mmol exclusively to the extracellular space of a 70-kg man would

be expected to raise serum  $K^+$  by  $\sim 2.5$  mmol/L. Changes in cellular distribution also serve to defend plasma  $K^+$  during  $K^+$  depletion. For example, military recruits have been shown to maintain a normal serum  $K^+$  after 11 days of basic training, despite a profound  $K^+$  deficit generated by renal and extrarenal loss.<sup>14</sup> The rapid exchange of intracellular  $K^+$  with extracellular  $K^+$  plays a crucial role in maintaining plasma  $K^+$  within such a narrow range; this is accomplished by overlapping and synergistic<sup>15</sup> regulation of a number of renal and extrarenal transport pathways.

## CELLULAR POTASSIUM TRANSPORT MECHANISMS

The intracellular accumulation of  $K^+$  against its electrochemical gradient is an energy-consuming process, mediated by the ubiquitous  $Na^+/K^+$ -ATPase enzyme. The  $Na^+/K^+$ -ATPase functions as an electrogenic pump since the stoichiometry of transport is three intracellular  $Na^+$  ions to two extracellular  $K^+$  ions. The enzyme complex is made up of a tissue-specific combination of multiple  $\alpha$ -,  $\beta$ -, and  $\gamma$ -subunits, which are further subject to tissue-specific patterns of regulation.<sup>16</sup> Cardiac glycosides (i.e., digoxin and ouabain) bind to the  $\alpha$  subunits of  $Na^+/K^+$ -ATPase at an exposed extracellular hairpin loop that also contains the major binding sites for extracellular  $K^+$ .<sup>17</sup> The binding of digoxin and  $K^+$  to the  $Na^+/K^+$ -ATPase complex is thus mutually antagonistic, explaining in part the potentiation of digoxin toxicity by hypokalemia.<sup>18</sup> Ouabain-like molecules have been postulated to play a role in hypertension and cardiovascular disease. Thus modulation of the  $K^+$ -dependent binding of circulating ouabain-like compounds to  $Na^+/K^+$ -ATPase may underlie at least some of cardiovascular complications of hypokalemia<sup>19</sup> (see “Consequences of Hypokalemia” later).

Skeletal muscle contains as much as 75% of body potassium (see [Fig. 16.1](#)) and exerts considerable influence on extracellular  $K^+$ . Exercise is thus a well-described cause of transient hyperkalemia; interstitial  $K^+$  in human muscle can reach levels as high as 10 mM after fatiguing exercise.<sup>20</sup> Not surprisingly, therefore, changes in skeletal muscle



**Fig. 16.1 Body  $K^+$  distribution and cellular  $K^+$  flux.** (From Wingo CS, Weiner ID. Disorders of potassium balance. In: Brenner BM, ed. *The Kidney*. vol 1. WB Saunders; 2000:998–1035.)

$Na^+/K^+$ -ATPase activity and abundance are major determinants of the capacity for extrarenal  $K^+$  homeostasis. Hypokalemia induces a marked decrease in muscle  $K^+$  content and  $Na^+/K^+$ -ATPase activity, an “altruistic”<sup>12</sup> mechanism to regulate plasma  $K^+$ . This is primarily due to dramatic decreases in the protein abundance of the  $\alpha$ -2 subunit of  $Na^+/K^+$ -ATPase.<sup>21</sup> In contrast, hyperkalemia due to potassium loading is associated with adaptive increases in muscle  $K^+$  content and  $Na^+/K^+$ -ATPase activity.<sup>22</sup> These interactions are reflected in the relationship between physical activity and the ability to regulate extracellular  $K^+$  during exercise.<sup>23</sup> For example, exercise training is associated with increases in muscle  $Na^+/K^+$ -ATPase concentration and activity, with reduced interstitial  $K^+$  in trained muscles<sup>24</sup> and an enhanced recovery of plasma  $K^+$  after defined amounts of exercise.<sup>23</sup>

Potassium can also accumulate in cells by coupling to the gradient for  $Na^+$  entry, entering via the electroneutral  $Na^+K^+2Cl^-$  cotransporters NKCC1 and NKCC2. The NKCC2 protein is found only at the apical membrane of thick ascending limb (TAL) and macula densa cells (Fig. 16.2 and Fig. 16.3), where it functions in transepithelial salt transport and tubular regulation of renin release.<sup>25</sup> In contrast, NKCC1 is widely expressed in multiple tissues<sup>25</sup> including muscle.<sup>26</sup> The cotransport of  $K^+Cl^-$  by the four  $K^+Cl^-$  cotransporters (KCC1-4) can also function in the transfer of  $K^+$  across membranes; although the KCCs typically function as efflux pathways, they can mediate influx when extracellular  $K^+$  increases.<sup>25</sup>

The efflux of  $K^+$  out of cells is largely accomplished by  $K^+$  channels, which comprise the largest family of ion channels in the human genome. There are three major subclasses of mammalian  $K^+$  channels; the six-transmembrane domain (TMD) family, which encompasses both the voltage-sensitive and  $Ca^{2+}$ -activated  $K^+$  channels, the two-pore, four TMD family, and the two TMD family of inward-rectifying  $K^+$  (Kir) channels.<sup>27</sup> There is tremendous genomic variety in human  $K^+$  channels, with at least 26 separate genes encoding principal subunits of the voltage-gated Kv channels and 17 genes encoding the principal Kir subunits. Further complexity is generated by the presence of multiple accessory subunits and alternative patterns of mRNA splicing. Not surprisingly, an increasing number and variety of  $K^+$  channels have been implicated in the control of  $K^+$

homeostasis and the membrane potential of excitable cells, such as muscle and heart, with important, evolving roles in the pathophysiology of potassium disorders.<sup>2,28-30</sup>

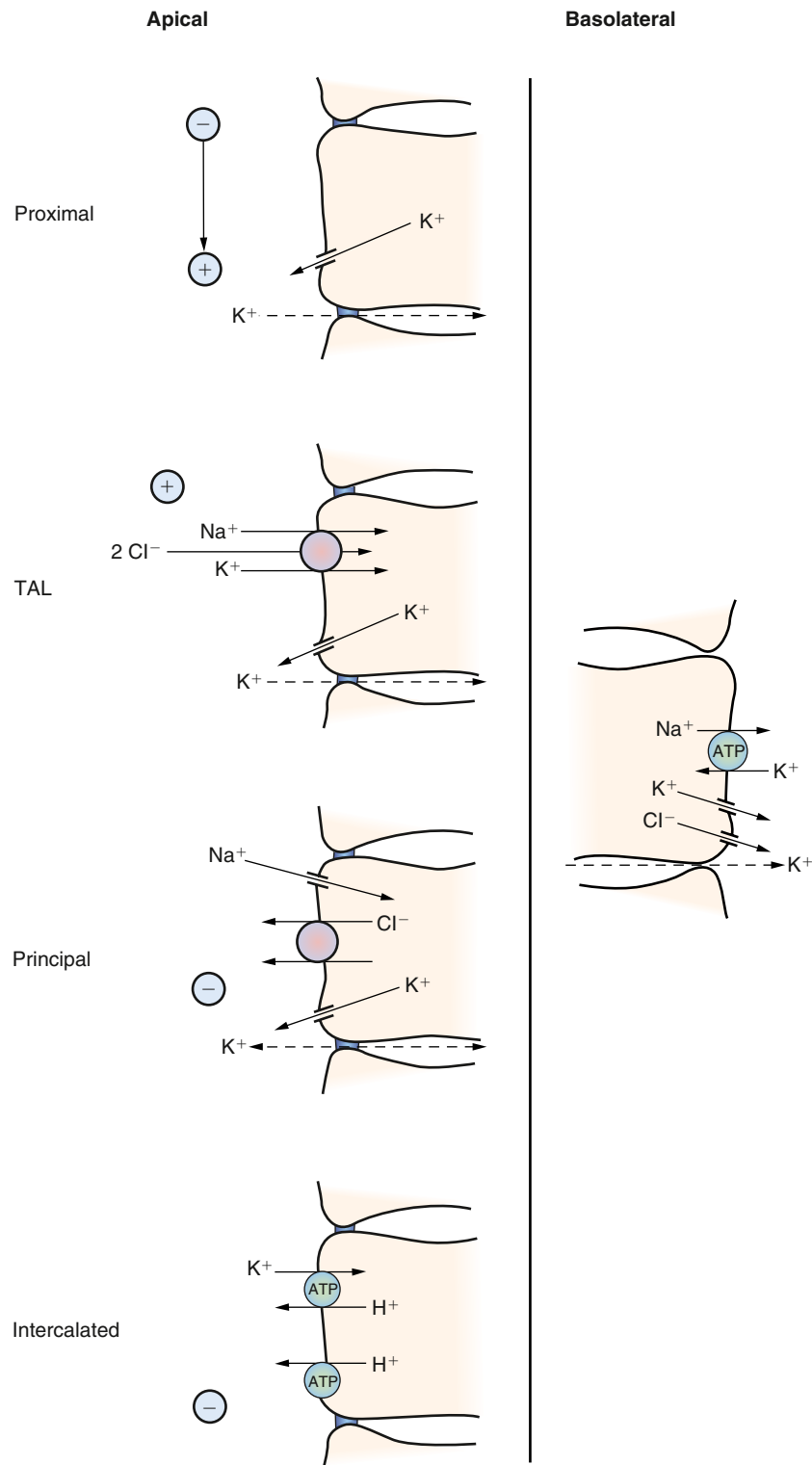
## FACTORS AFFECTING INTERNAL DISTRIBUTION OF POTASSIUM

A number of hormones and physiologic conditions have acute effects on the distribution of  $K^+$  between the intracellular and extracellular space (Table 16.1). Some of these factors are of particular clinical relevance and are therefore reviewed in detail.

### INSULIN

The hypokalemic effect of insulin has been known since the early 20th century.<sup>31</sup> The impact of insulin on plasma  $K^+$  and plasma glucose is separable at multiple levels, suggesting independent mechanisms. For example, despite impaired glucose uptake, peripheral  $K^+$  uptake is not impaired in humans with type 2 diabetes.<sup>32</sup> Notably, the hypokalemic effect of insulin is not kidney dependent.<sup>33</sup> Insulin and  $K^+$  appear to form a feedback loop of sorts, in that increases in plasma  $K^+$  have a marked stimulatory effect on insulin levels.<sup>34</sup> Insulin-stimulated  $K^+$  uptake, measured in rats using a “ $K^+$  clamp” technique, is rapidly reduced by 2 days of  $K^+$  depletion, before a modest drop in plasma  $K^+$ ,<sup>35</sup> and in the absence of a change in plasma  $K^+$  in rats subject to a lesser  $K^+$  restriction for 14 days.<sup>15</sup> Insulin-mediated  $K^+$  uptake is thus modulated by the factors that serve to preserve plasma  $K^+$  in the setting of  $K^+$  deprivation (see also “Control of Potassium Secretion: The Effect of Potassium Intake” later). Inhibition of basal insulin secretion in normal subjects by somatostatin infusion increases serum  $K^+$  by up to 0.5 mmol/L, in the absence of a change in urinary excretion, emphasizing the crucial role of circulating insulin in the regulation of plasma  $K^+$ .<sup>36</sup> Clinically, inhibition of insulin secretion by the somatostatin agonist octreotide can cause significant hyperkalemia in both anephric patients<sup>37</sup> and patients with normal renal function.<sup>38</sup>

Insulin stimulates the uptake of  $K^+$  by several tissues, most prominently liver, skeletal muscle, cardiac muscle, and fat. It does so by activating several  $K^+$  transport pathways, with particularly well-documented effects on the  $Na^+/K^+$ -ATPase.<sup>39</sup> Insulin activates  $Na^+H^+$  exchange and/or  $Na^+K^+2Cl^-$  cotransport in several tissues; although the ensuing increase in intracellular  $Na^+$  was postulated to have a secondary activating effect on  $Na^+/K^+$ -ATPase,<sup>27</sup> it is clear that this is not the primary mechanism in most cell types.<sup>40</sup> Insulin induces translocation of the  $Na^+/K^+$ -ATPase  $\alpha$ -2 subunit to the plasma membrane of skeletal muscle cells, with a lesser effect on the  $\alpha$ -1 subunit.<sup>41</sup> This translocation is dependent on the activity of phosphoinositide-3 kinase (PI-3) kinase,<sup>41</sup> which itself also binds to a proline-rich motif in the N-terminus of the  $\alpha$  subunit.<sup>42</sup> The activation of PI3-kinase by insulin thus induces phosphatase enzymes to dephosphorylate a specific serine residue adjacent to the PI3-kinase binding domain. Trafficking of  $Na^+/K^+$ -ATPase to the cell surface also appears to require the phosphorylation of an adjacent tyrosine residue, perhaps catalyzed by the tyrosine kinase activity of the insulin receptor itself.<sup>43</sup> Finally, the “serum and glucocorticoid-induced kinase-1” (SGK1) plays a critical role in insulin-stimulated  $K^+$  uptake,

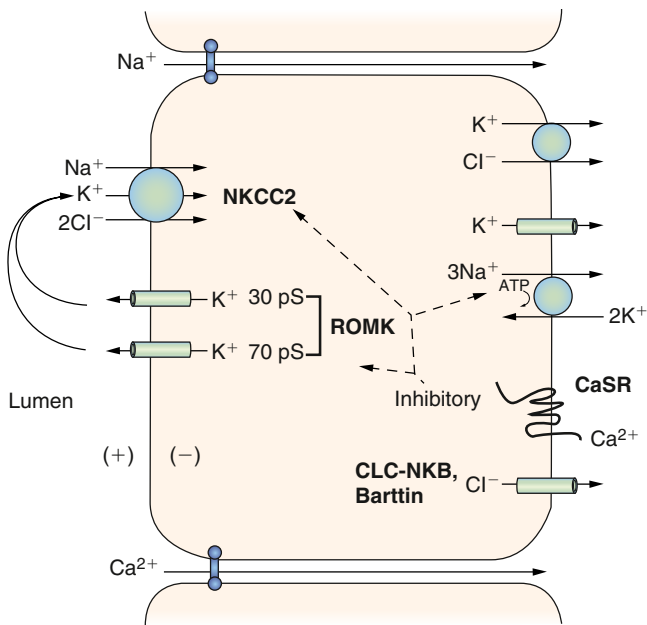


**Fig. 16.2 Schematic cell models of potassium transport along the nephron.** Cell types are as specified. Note the differences in luminal potential difference along the nephron. TAL, Thick ascending limb. (From Giebisch G. Renal potassium transport: mechanisms and regulation. *Am J Physiol.* 1998;274:F817–833.)

presumably via the known stimulatory effects of this kinase on Na<sup>+</sup>/K<sup>+</sup>-ATPase activity and/or Na<sup>+</sup>-K<sup>+</sup>-2Cl<sup>-</sup> cotransport.<sup>44</sup> The hypokalemic effect of insulin plus glucose is blunted in SGK1 knockout mice, with a marked reduction in hepatic insulin-stimulated K<sup>+</sup> uptake.<sup>44</sup>

#### SYMPATHETIC NERVOUS SYSTEM

The sympathetic nervous system plays a prominent role in regulating the balance between extracellular and intracellular K<sup>+</sup>. Again, as is the case for insulin, the effect of catecholamines on plasma K<sup>+</sup> has been known



**Fig. 16.3 Bartter syndrome and the thick ascending limb.** Bartter syndrome can result from loss-of-function mutations in the Na<sup>+</sup>-K<sup>+</sup>-2Cl<sup>-</sup> cotransporter NKCC2, the K<sup>+</sup> channel subunit ROMK, or the Cl<sup>-</sup> channel subunits CLC-NKB and Barttin (Bartter syndrome types I to IV, respectively). Gain-of-function mutations in the calcium-sensing receptor CaSR can also cause a Bartter syndrome phenotype (type V); the CaSR has an inhibitory effect on salt transport by the thick ascending limb, targeting several transport pathways. ROMK encodes the low conductance 30 pS K<sup>+</sup> channel in the apical membrane and also appears to function as a critical subunit of the higher conductance 70 pS channel. The loss of K<sup>+</sup> channel activity in Bartter syndrome type II leads to reduced apical K<sup>+</sup> recycling and reduced Na<sup>+</sup>-K<sup>+</sup>-2Cl<sup>-</sup> cotransport. Decreased apical K<sup>+</sup> channels also lead to a decrease in the lumen-positive potential difference, which drives paracellular Na<sup>+</sup>, Ca<sup>2+</sup>, and Mg<sup>2+</sup> transport.

for some time<sup>45</sup>; however, a complicating issue is the differential effect of stimulating  $\alpha$ - and  $\beta$ -adrenergic receptors. Uptake of K<sup>+</sup> by liver and muscle, with resultant hypokalemia, is stimulated via  $\beta_2$  receptors.<sup>27</sup> The hypokalemic effect of catecholamines appears to be largely independent of changes in circulating insulin<sup>27</sup> and has been reported in nephrectomized animals.<sup>46</sup> The cellular mechanisms whereby catecholamines induce K<sup>+</sup> uptake in muscle include activation of the Na<sup>+</sup>/K<sup>+</sup>-ATPase,<sup>47</sup> likely via increases in cyclic-AMP.<sup>48</sup> However,  $\beta$ -adrenergic receptors in skeletal muscle also activate the inwardly directed Na<sup>+</sup>-K<sup>+</sup>-2Cl<sup>-</sup> cotransporter NKCC1, which may account for as much as one-third of the uptake response to catecholamines.<sup>26,49</sup>

In contrast to  $\beta$ -adrenergic stimulation,  $\alpha$ -adrenergic agonists impair the ability to buffer increases in K<sup>+</sup> induced via intravenous loading or by exercise<sup>50</sup>; the cellular mechanisms whereby this occurs are not known. It is thought that  $\beta$ -adrenergic stimulation increases K<sup>+</sup> uptake during exercise to avoid hyperkalemia, whereas  $\alpha$ -adrenergic mechanisms help blunt the ensuing postexercise nadir.<sup>50</sup> The clinical consequences of the sympathetic control of extrarenal K<sup>+</sup> homeostasis are reviewed elsewhere in this chapter.

**Table 16.1 Factors Affecting K<sup>+</sup> Distribution Between Intracellular and Extracellular Compartments**

| Acute                      |                            |
|----------------------------|----------------------------|
| Factor                     | Effect on Potassium        |
| Insulin                    | Enhanced cell uptake       |
| $\beta$ -Catecholamines    | Enhanced cell uptake       |
| $\alpha$ -Catecholamines   | Impaired cell uptake       |
| Acidosis                   | Impaired cell uptake       |
| Alkalosis                  | Enhanced cell uptake       |
| External potassium balance | Loose correlation          |
| Cell damage                | Impaired cell uptake       |
| Hyperosmolality            | Enhanced cell efflux       |
| Chronic                    |                            |
| Factor                     | Effect on ATP Pump Density |
| Thyroid                    | Enhanced                   |
| Adrenal steroids           | Enhanced                   |
| Exercise (training)        | Enhanced                   |
| Growth                     | Enhanced                   |
| Diabetes                   | Impaired                   |
| Potassium deficiency       | Impaired                   |
| Chronic renal failure      | Impaired                   |

From Giebisch G. Renal potassium transport: mechanisms and regulation. *Am J Physiol.* 1998;274:F817–833.

ACID-BASE STATUS

The association between changes in pH and plasma K<sup>+</sup> was observed some time ago.<sup>51</sup> It has long been held that acute disturbances in acid-base equilibrium result in changes in serum K<sup>+</sup>, such that alkalemia shifts K<sup>+</sup> into cells, whereas acidemia is associated with K<sup>+</sup> release.<sup>52,53</sup> It is thought that this apparent K<sup>+</sup>-H<sup>+</sup> exchange and/or K<sup>+</sup>-HCO<sub>3</sub><sup>-</sup> cotransport serves to help maintain extracellular pH.<sup>54</sup> Several different transport mechanisms together result in net exchange of K<sup>+</sup> with H<sup>+</sup> including functional coupling between Na<sup>+</sup>-H<sup>+</sup> exchangers with the Na<sup>+</sup>/K<sup>+</sup>-ATPase, coupling between Na<sup>+</sup>-2HCO<sub>3</sub><sup>-</sup> cotransport with the Na<sup>+</sup>/K<sup>+</sup>-ATPase, and coupling between Cl<sup>-</sup>-HCO<sub>3</sub><sup>-</sup> exchange and K<sup>+</sup>-Cl<sup>-</sup> cotransporters.<sup>54</sup>

Rather limited data exist for the durable concept that a change of 0.1 unit in serum pH will result in 0.6 mmol/L change in serum K<sup>+</sup> in the opposite direction.<sup>55</sup> However, despite the complexities of changes in K<sup>+</sup> homeostasis associated with various acid-base disorders, a few general observations can be made. The induction of metabolic acidosis by the infusion of mineral acids (NH<sub>4</sub><sup>+</sup>-Cl<sup>-</sup> or H<sup>+</sup>-Cl<sup>-</sup>) consistently increases serum K<sup>+</sup>,<sup>53</sup> whereas organic acidosis generally fails to increase serum K<sup>+</sup>.<sup>53,56-58</sup> Notably, a newer report failed to detect an increase in serum K<sup>+</sup> in normal human subjects with acute acidosis secondary to duodenal NH<sub>4</sub><sup>+</sup>-Cl<sup>-</sup> infusion, in which a modest acidosis was accompanied by an increase in circulating insulin.<sup>59</sup> However, as noted by Adrogué and Madias,<sup>60</sup> the concomitant infusion of 350 mL of D5W in these fasting subjects may have served to increase circulating insulin, thus blunting the potential hyperkalemic response to NH<sub>4</sub><sup>+</sup>-Cl<sup>-</sup>. Clinically, use of the oral phosphate binder sevelamer-hydrochloride in patients with

end-stage renal disease (ESRD) is associated with acidosis, due to effective gastrointestinal absorption of  $\text{H}^+\text{-Cl}^-$ ; in hemodialysis patients, this acidosis has been associated with an increase in serum  $\text{K}^+$ , which can be ameliorated by an increase in dialysis bicarbonate concentration.<sup>61</sup> Of note, hyperkalemia is not an expected complication of sevelamer-carbonate, which has generally supplanted sevelamer-hydrochloride as a phosphate binder.

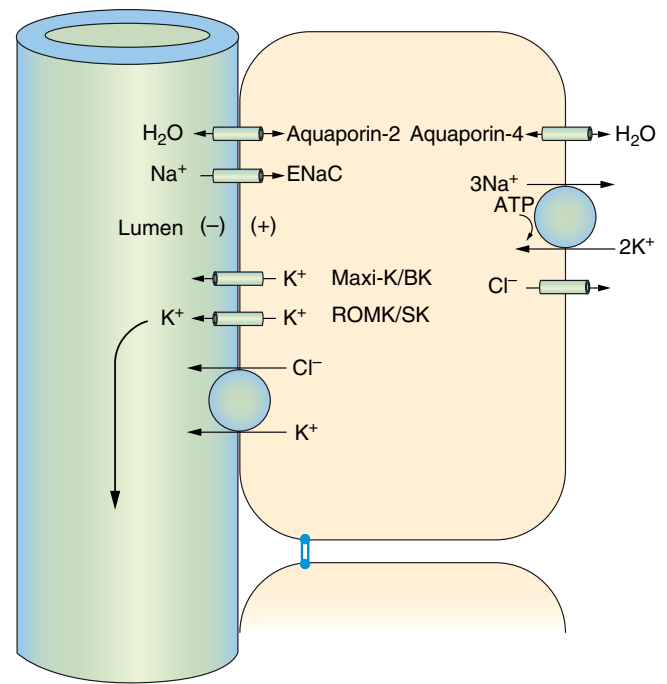
Metabolic alkalosis induced by sodium-bicarbonate infusion usually results in a modest reduction in serum  $\text{K}^+$ .<sup>52,53,55</sup> Respiratory alkalosis reduces plasma  $\text{K}^+$ , by a magnitude comparable with that of metabolic alkalosis.<sup>52,53,55,62</sup> Finally, acute respiratory acidosis increases plasma  $\text{K}^+$ ; the absolute increase is smaller than that induced by metabolic acidosis secondary to inorganic acids.<sup>52,53,55</sup> Again, however, some studies have failed to show a change in serum  $\text{K}^+$  following acute respiratory acidosis.<sup>53,63</sup> The smaller increments in serum  $\text{K}^+$  with respiratory acidosis are explained in part by the elevated serum  $\text{HCO}_3^-$  levels in respiratory acidosis, which blunt pH effects on the  $\text{Na}^+\text{-}2\text{HCO}_3^-$  cotransporter, resulting in lesser apparent  $\text{K}^+\text{-H}^+$  exchange than in metabolic acidosis.<sup>54</sup>

## RENAL POTASSIUM EXCRETION

### POTASSIUM TRANSPORT IN THE DISTAL NEPHRON

The proximal tubule and loop of Henle mediate the bulk of potassium reabsorption, such that a considerable fraction of filtered potassium is reabsorbed before entry into the superficial distal tubules.<sup>64</sup> Renal potassium excretion is primarily determined by regulated secretion in the distal nephron, specifically within the connecting segment (CNT) and cortical collecting duct (CCD). The principal cells of the CNT and CCD play a dominant role in  $\text{K}^+$  secretion; the relevant transport pathways are shown in Figs. 16.2 and 16.4. Apical  $\text{Na}^+$  entry via the amiloride-sensitive epithelial sodium channel (ENaC)<sup>65</sup> results in the generation of a lumen-negative potential difference in the CNT and CCD, which drives passive  $\text{K}^+$  exit through apical  $\text{K}^+$  channels. A critical, clinically relevant consequence of this relationship is that  $\text{K}^+$  secretion is dependent on delivery of adequate luminal  $\text{Na}^+$  to the CNT and CCD<sup>66,67</sup>;  $\text{K}^+$  secretion by the CCD essentially ceases as luminal  $\text{Na}^+$  drops below 8 mmol/L.<sup>68</sup> Selective increases in thiazide-sensitive  $\text{Na}^+\text{-Cl}^-$  cotransport in the distal convoluted tubule (DCT), as seen in familial hyperkalemia with hypertension (FHHt, see “Hyperkalemia: Hereditary Tubular Defects and Potassium Excretion” later), reduce  $\text{Na}^+$  delivery to principal cells in the downstream CNT and CCD, leading to hyperkalemia.<sup>69</sup> Dietary  $\text{Na}^+$  intake also influences  $\text{K}^+$  excretion, such that excretion is enhanced by excess  $\text{Na}^+$  intake and reduced by  $\text{Na}^+$  restriction (Fig. 16.5).<sup>66,67</sup> Basolateral exchange of  $\text{Na}^+$  and  $\text{K}^+$  is mediated by the  $\text{Na}^+\text{-K}^+\text{-ATPase}$ , providing the driving force for both  $\text{Na}^+$  entry and  $\text{K}^+$  exit at the apical membrane (see Figs. 16.2 and 16.4).

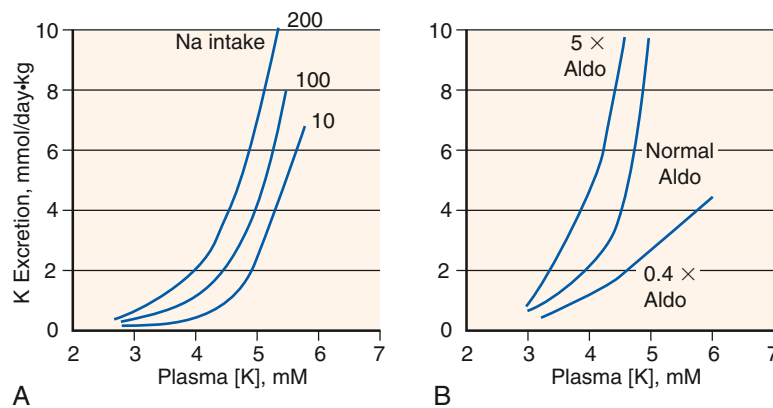
Under basal conditions of high  $\text{Na}^+\text{-Cl}^-$  and low  $\text{K}^+$  intake, the bulk of aldosterone-stimulated  $\text{Na}^+$  and  $\text{K}^+$  transport occurs in the CNT, before the entry of tubular fluid into the CCD.<sup>70</sup> The density of both  $\text{Na}^+$  and  $\text{K}^+$  channels is thus considerably greater in the CNT than in the CCD<sup>71,72</sup>; the capacity of the CNT for  $\text{Na}^+$  reabsorption may be as much as 10 times greater than that of the CCD.<sup>73</sup> The recruitment of



**Fig. 16.4**  $\text{K}^+$  secretory pathways in principal cells of the connecting segment (CNT) and cortical collecting duct (CCD). The absorption of  $\text{Na}^+$  via the amiloride-sensitive epithelial sodium channel (ENaC) generates a lumen-negative potential difference, which drives  $\text{K}^+$  excretion through the apical secretory  $\text{K}^+$  channel ROMK. Flow-dependent  $\text{K}^+$  secretion is mediated by an apical voltage-gated, calcium-sensitive BK channel. Chloride-dependent, electroneutral  $\text{K}^+$  secretion is likely mediated by a  $\text{K}^+\text{-Cl}^-$  cotransporter. Water transport in principal cells occurs via aquaporin-2 (Aqp-2) and aquaporins-3/4 (Aqp-3/4).

ENaC subunits in response to dietary  $\text{Na}^+$  restriction begins in the CNT, with progressive recruitment of subunits to the apical membrane of the CCD at lower levels of dietary  $\text{Na}^+$ .<sup>73</sup> The activity of secretory  $\text{K}^+$  channels in the CNT is also influenced by changes in dietary  $\text{K}^+$  with axial extension distally<sup>75</sup>; again, this is consistent with progressive, axial recruitment of transport capacity for the absorption of  $\text{Na}^+$  and secretion of  $\text{K}^+$  along the distal nephron.

Electrophysiologic characterization has documented the presence of several subpopulations of apical  $\text{K}^+$  channels in the CCD and CNT, most prominently a small-conductance (SK) 30 pS channel<sup>71,76</sup> and a large-conductance,  $\text{Ca}^{2+}$ -activated 150 pS (BK) channel<sup>71,77</sup> (see Fig. 16.4). The SK channel is thought to mediate  $\text{K}^+$  secretion under baseline conditions, hence its designation as the “secretory”  $\text{K}^+$  channel. SK channel activity is mediated by the ROMK (Renal Outer Medullary  $\text{K}^+$  channel) protein, encoded by the *Kenj1* gene; targeted deletion of this gene in mice results in complete loss of SK activity within the CCD.<sup>77</sup> Increased distal flow has a significant stimulatory effect on  $\text{K}^+$  secretion, due in part to both enhanced delivery and absorption of  $\text{Na}^+$  and to increased removal of secreted  $\text{K}^+$ .<sup>66,67</sup> The apical  $\text{Ca}^{2+}$ -activated BK channel plays a critical role in flow-dependent  $\text{K}^+$  secretion by the CNT and CCD.<sup>76</sup> BK channels have a heteromeric structure, with  $\alpha$ -subunits that form the ion channel pore and modulatory  $\beta$ -subunits.<sup>76</sup> The  $\beta 1$  subunits of BK channels are restricted to principal cells within the CNT,<sup>76,78</sup> whereas  $\beta 4$  subunits are detectable



**Fig. 16.5 (A) Relationship between steady-state plasma K<sup>+</sup> and urinary K<sup>+</sup> excretion in the dog, as a function of dietary Na<sup>+</sup> intake (mmol/day).** Animals were adrenalectomized and replaced with aldosterone, and dietary K<sup>+</sup> and Na<sup>+</sup> content were varied as specified. **(B) Relationship between steady-state plasma K<sup>+</sup> and urinary K<sup>+</sup> excretion as a function of circulating aldosterone.** Animals were adrenalectomized and variably replaced with aldosterone, dietary K<sup>+</sup> content was varied. (A from Young DB, Jackson TE, Tipayamontri U, Scott RC. Effects of sodium intake on steady-state potassium excretion. *Am J Physiol.* 1984;246:F772–778. B from Young DB. Quantitative analysis of aldosterone's role in potassium regulation. *Am J Physiol.* 1988;255:F811–822.)

at the apical membranes of TAL, DCT, and intercalated cells.<sup>78</sup> Flow-dependent K<sup>+</sup> secretion is reduced in mice with targeted deletion of the  $\alpha 1$  and  $\beta 1$  subunits,<sup>76,79,80</sup> consistent with a dominant role for BK channels. The Ca<sup>2+</sup>-permeable TRPV4 channel increases intracellular Ca<sup>2+</sup> concentrations in response to tubular flow, resulting in the activation of BK channels.<sup>81</sup> Additional work has demonstrated a critical role for the PIEZO1 mechanosensitive basolateral calcium-permeable channel in activating flow-dependent K<sup>+</sup> secretion.<sup>11</sup>

In addition to apical K<sup>+</sup> channels, considerable evidence implicates apical K<sup>+</sup>-Cl<sup>-</sup> cotransport in distal K<sup>+</sup> secretion.<sup>66,82,83</sup> Pharmacologic studies of perfused tubules are consistent with K<sup>+</sup>-Cl<sup>-</sup> cotransport mediated by the KCC proteins,<sup>82</sup> potentially mediated by KCC3a.<sup>9</sup> A particularly provocative study underlines the importance of ENaC-independent K<sup>+</sup> excretion, be it mediated by apical K<sup>+</sup>-Cl<sup>-</sup> cotransport and/or by other mechanisms.<sup>84</sup> Rats were infused with amiloride via osmotic minipumps, generating urinary concentrations considered sufficient to inhibit >98% of ENaC activity. Whereas amiloride almost abolished K<sup>+</sup> excretion in rats on a normal K<sup>+</sup> intake, acute and long-term high-K<sup>+</sup> diets led to an increasing fraction of K<sup>+</sup> excretion that was independent of ENaC activity (~50% after 7–9 days on a high-K<sup>+</sup> diet).<sup>84</sup>

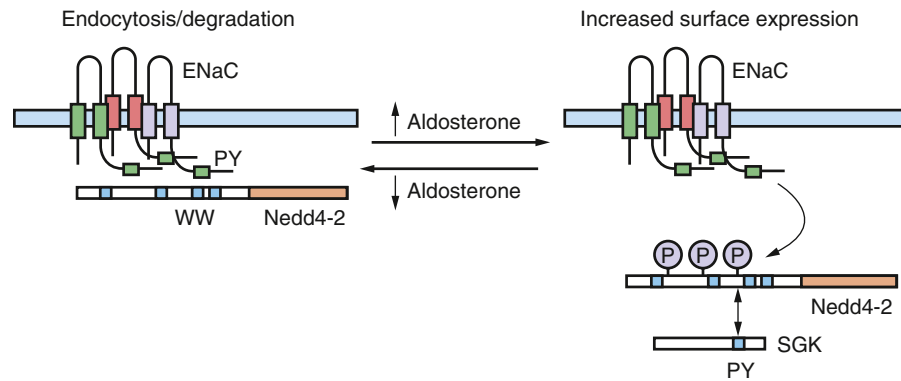
In addition to secretion, the distal nephron is capable of considerable reabsorption of K<sup>+</sup>, particularly during restriction of dietary K<sup>+</sup>.<sup>49,64,79,86</sup> This reabsorption is accomplished primarily by intercalated cells in the outer medullary collecting duct (OMCD), via the activity of apical H<sup>+</sup>/K<sup>+</sup>-ATPase pumps (see Fig. 16.2).

#### CONTROL OF POTASSIUM SECRETION: ALDOSTERONE

Aldosterone is well established as an important regulatory factor in K<sup>+</sup> excretion, and increases in plasma K<sup>+</sup> are an important stimulus for aldosterone secretion (see also “Regulation of Renal Renin and Adrenal Aldosterone” later). However, an important principle is that aldosterone plays a permissive, synergistic, but not essential role in K<sup>+</sup> homeostasis.<sup>87–89</sup> This is reflected clinically in the frequent absence of hyperkalemia or hypokalemia in disorders

associated with a deficiency or an overabundance of circulating aldosterone, respectively (see “Hyperaldosteronism” and “Hypoaldosteronism”). Regardless, it is clear that aldosterone and downstream effectors of this hormone have clinically relevant effects on plasma K<sup>+</sup> levels and that the ability to excrete K<sup>+</sup> is modulated by systemic aldosterone concentrations (see Fig. 16.5).

Aldosterone has no effect on the density of apical SK channels in the CCD or CNT; rather, the hormone induces a marked increase in the density of apical Na<sup>+</sup> channels,<sup>90</sup> thus increasing the driving force for apical K<sup>+</sup> excretion. The apical amiloride-sensitive epithelial Na<sup>+</sup> channel (ENaC) is composed of three subunits,  $\alpha$ -,  $\beta$ -, and  $\gamma$ -, that assemble together to synergistically traffic to the cell membrane and mediate Na<sup>+</sup> transport. Aldosterone activates ENaC channel complexes by multiple mechanisms. First, it induces transcription of the  $\alpha$ -ENaC subunit,<sup>91,92</sup> increasing the availability for coassembly with the more abundant  $\beta$  and  $\gamma$  subunits.<sup>93</sup> Second, aldosterone and dietary Na<sup>+</sup>-Cl<sup>-</sup> restriction stimulate a significant redistribution of ENaC subunits in the CNT and early CCD, from a largely cytoplasmic location during dietary Na<sup>+</sup>-Cl<sup>-</sup> excess to a purely apical distribution after aldosterone or Na<sup>+</sup>-Cl<sup>-</sup> restriction.<sup>73,94,95</sup> Third, aldosterone induces the expression of a serine-threonine kinase called SGK-1 (serum and glucocorticoid-induced kinase-1)<sup>96</sup>; coexpression of SGK-1 with ENaC subunits results in increased expression at the plasma membrane.<sup>94</sup> SGK-1 modulates membrane expression of ENaC by interfering with regulated endocytosis of its channel subunits. Specifically, the kinase interferes with interactions between ENaC subunits and the ubiquitin-ligase Nedd4-2.<sup>93</sup> The so-called “PPXY” domains in the C-termini of all three ENaC subunits bind to WW domains of Nedd4-2<sup>97</sup>; these PPXY domains are deleted, truncated, or mutated in patients with Liddle syndrome<sup>98</sup> (see “Liddle Syndrome” later), leading to a gain-of-function in channel activity.<sup>99</sup> Nedd4-2 ubiquitinates ENaC subunits, thus inducing removal of channel subunits from the cell membrane followed by degradation in lysosomes and the proteasome.<sup>93</sup> A PPXY domain in SGK-1 also binds to Nedd4-2, which is a phosphorylation substrate for the



**Fig. 16.6 Coordinated regulation of ENaC by the aldosterone-induced SGK kinase and the ubiquitin-ligase Nedd4-2.** Nedd4-2 binds via its WW domains to ENaC subunits via their “PPXY” domains (denoted PY here), ubiquitinating the channel subunits and targeting them for removal from the cell membrane and destruction in the proteasome. Aldosterone induces the SGK kinase, which phosphorylates and inactivates Nedd4-2, thus increasing surface expression of ENaC channels. Mutations that cause Liddle syndrome affect the interaction between ENaC and Nedd4-2. (From Snyder PM, Olson DR, Thomas BC. Serum and glucocorticoid-regulated kinase modulate Nedd4-2-mediated inhibition of the epithelial Na<sup>+</sup> channel. *J Biol Chem.* 2002;277:5–8.)

kinase; phosphorylation of Nedd4-2 by SGK-1 abrogates the inhibitory effect of this ubiquitin ligase on ENaC subunits<sup>3</sup> (Fig. 16.6). This signaling axis appears to be activated in a K<sup>+</sup>-dependent fashion by the upstream mTORC2 kinase.<sup>10</sup>

Another mechanism whereby aldosterone activates ENaC involves proteolytic cleavage of the channel proteins by serine proteases. A “channel-activating protease” that increases channel activity of ENaC was initially identified in *Xenopus laevis* A6 cells.<sup>100</sup> The mammalian ortholog, denoted CAP1 (channel-activating protease-1) or prostaticin, is an aldosterone-induced protein in principal cells.<sup>101</sup> Urinary excretion of CAP1 is increased in hyperaldosteronism, with a reduction after adrenalectomy.<sup>101</sup> CAP1 is membrane associated, via a glycosylphosphatidylinositol (GPI) linkage<sup>100</sup>; mammalian principal cells also express two transmembrane proteases, denoted CAP2 and CAP3, with homology to CAP1.<sup>102</sup> These and other proteases (furin, plasmin, etc.) activate ENaC by excising extracellular inhibitory domains from the  $\alpha$  and  $\gamma$  subunits, increasing the open probability of channels at the plasma membrane.<sup>102,103</sup> This proteolytic cleavage of ENaC appears to activate the channel by removing the “self-inhibitory” effect of external Na<sup>+</sup><sup>104</sup>; in the case of furin-mediated proteolysis of  $\alpha$ -ENaC, this appears to involve the removal of an inhibitory domain from within the extracellular loop.<sup>105</sup> Extracellular Na<sup>+</sup> appears to interact with a specific acidic cleft in the cleaved extracellular loop of  $\alpha$ -ENaC, causing inhibition of the channel.<sup>106</sup> Since SGK-1 increases channel expression at the cell surface,<sup>94</sup> one would expect synergistic activation by coexpressed CAP1-3 and SGK; this is indeed the case.<sup>102</sup> Therefore aldosterone activates ENaC by at least three separate synergistic mechanisms; induction of  $\alpha$ -ENaC, induction of SGK-1 and repression of Nedd4-2, and induction of channel-activating proteases. Clinically, the inhibition of channel-activating proteases by the protease inhibitor nafamostat causes hyperkalemia, due to inhibition of ENaC activity.<sup>107,108</sup>

#### CONTROL OF POTASSIUM SECRETION: THE EFFECT OF POTASSIUM INTAKE

Changes in K<sup>+</sup> intake strongly modulate K<sup>+</sup> channel activity in the CNT and CCD (secretory capacity), in addition to

H<sup>+</sup>/K<sup>+</sup>-ATPase activity in the OMCD (reabsorptive capacity). Increased dietary K<sup>+</sup> rapidly increases the activity of SK channels in the CCD and CNT,<sup>74,109</sup> along with a modest increase in Na<sup>+</sup> channel (ENaC) activity<sup>90</sup>; this is associated with an increase in apical expression of the ROMK channel protein.<sup>110</sup> The increase in ENaC and SK channel density in the CCD occurs within hours of assuming a high-K<sup>+</sup> diet, with a minimal associated increase in circulating aldosterone.<sup>109</sup> BK channels in the CNT and CCD are also activated by dietary K<sup>+</sup> loading. Trafficking of BK subunits is thus affected by dietary K<sup>+</sup>, with largely intracellular distribution of  $\alpha$ -subunits in K<sup>+</sup>-restricted rats and prominent apical expression in K<sup>+</sup>-loaded rats.<sup>111</sup> Again, aldosterone does not directly contribute to the regulation of BK channel activity or expression in response to a high-K<sup>+</sup> diet.<sup>112</sup> However, aldosterone does participate in marked upregulation of the TRPV4 channel in response to a high-K<sup>+</sup> diet; Ca<sup>2+</sup> entry via TRPV4 increases intracellular Ca<sup>2+</sup> concentrations in response to tubular flow, thus activating BK channels.<sup>81</sup> TRPV4 knockout mice demonstrate an impaired adaptation to a high-K<sup>+</sup> diet.<sup>81</sup>

A complex, synergistic mix of signaling pathways regulates K<sup>+</sup> channel activity in response to changes in dietary K<sup>+</sup>. In particular, the WNK (With No Lysine) kinases play a critical role in modulating distal K<sup>+</sup> secretion. WNK1 and WNK4 were initially identified as the causative genes for FHHt (see also “Hyperkalemia: Hereditary Tubular Defects and Potassium Excretion” later). ROMK expression at the membrane of *Xenopus* oocytes is reduced by coexpression of WNK4; FHHt-associated mutations increase this effect, suggesting a direction inhibition of SK channels in FHHt.<sup>113</sup> Transcription of the *WNK1* gene generates several different isoforms; the predominant intrarenal WNK1 isoform is generated by a distal nephron transcriptional site that bypasses the N-terminal exons that encode the kinase domain, yielding a kinase-deficient “short” isoform<sup>114</sup> (“WNK1-S”). Full-length WNK1 (WNK1-L) inhibits ROMK by inducing endocytosis of the channel protein; the shorter, kinase-deficient WNK1-S isoform inhibits this effect of WNK1-L,<sup>115,116</sup> and ROMK activity is impaired in WNK1-S knockout mice.<sup>117</sup> The ratio of WNK1-S to WNK1-L transcripts is reduced by K<sup>+</sup> restriction (greater endocytosis

of ROMK)<sup>116,118</sup> and increased by K<sup>+</sup> loading (reduced endocytosis of ROMK),<sup>115,118</sup> suggesting that this ratio between WNK1-S and WNK1-L functions as a molecular “switch” to regulate distal K<sup>+</sup> secretion. The BK channel is also regulated by the WNK kinases.<sup>119-121</sup>

The membrane trafficking of ROMK is also modulated by tyrosine phosphorylation of the channel protein, such that tyrosine phosphorylation stimulates endocytosis and tyrosine dephosphorylation induces exocytosis.<sup>122</sup> Intrarenal activity of the cytoplasmic tyrosine kinases, c-Src and c-Yes, is inversely related to dietary K<sup>+</sup> intake, with a decrease under high-K<sup>+</sup> conditions and a marked increase after several days of K<sup>+</sup> restriction.<sup>27,123</sup> Several studies have implicated the intrarenal generation of superoxide anions in the activation of cytoplasmic tyrosine kinases by K<sup>+</sup> depletion.<sup>124-126</sup> Potential candidates for the upstream hormonal signals include angiotensin-II and growth factors such as IGF-1.<sup>124</sup> In particular, angiotensin-II inhibits ROMK activity in K<sup>+</sup>-restricted rats, but not rats on a normal K<sup>+</sup> diet.<sup>127</sup> This inhibition by angiotensin-II involves downstream activation of superoxide production and c-Src activity, such that the induction of angiotensin-II by low-K<sup>+</sup> diet appears to play a major role in reducing distal tubular K<sup>+</sup> secretion.<sup>128</sup>

#### INTEGRATED REGULATION OF DISTAL SODIUM ABSORPTION AND POTASSIUM SECRETION

Under certain physiologic conditions associated with marked induction of aldosterone, such as dietary sodium restriction, Na<sup>+</sup> balance can be maintained without significant effects on K<sup>+</sup> excretion. Yet by activating ENaC and generating a more lumen-negative potential difference, increases in aldosterone should lead to an obligatory kaliuresis; how is this physiologic consequence avoided? The mechanisms that underlie this “aldosterone paradox,” the independent regulation of Na<sup>+</sup> and K<sup>+</sup> handling by the aldosterone-sensitive distal nephron has emerged. The major factors that allow for integrated but independent control of Na<sup>+</sup> and K<sup>+</sup> transport appear to include electroneutral thiazide-sensitive Na<sup>+</sup>-Cl<sup>-</sup> transport within the CCD,<sup>129-131</sup> ENaC-independent K<sup>+</sup> excretion within the distal nephron,<sup>84</sup> and the differential regulation of various signaling pathways by aldosterone, angiotensin-II, and dietary K<sup>+</sup>.<sup>132,133</sup> (see also Chapter 6).

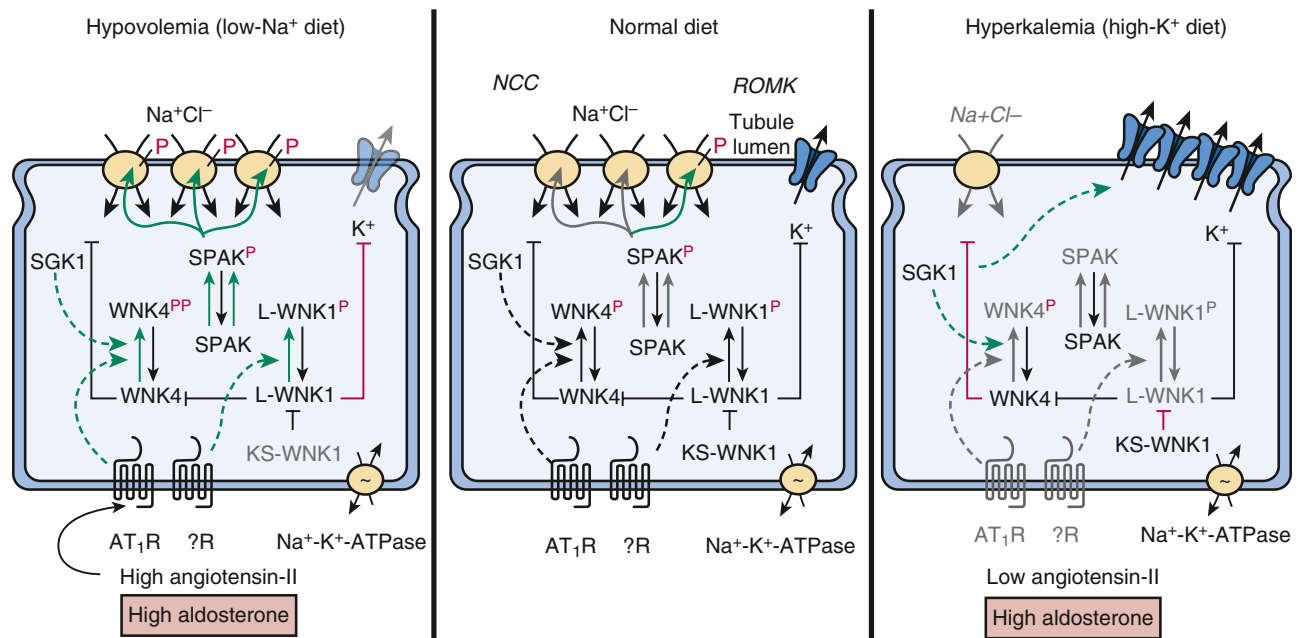
Electroneutral Na<sup>+</sup>-Cl<sup>-</sup> transport in the CCD and ENaC-independent K<sup>+</sup> secretion may play important roles in disconnecting Na<sup>+</sup> and K<sup>+</sup> transport within the distal nephron. Electroneutral, thiazide-sensitive, and amiloride-resistant Na<sup>+</sup>-Cl<sup>-</sup> transport within the CCD<sup>129-131</sup> is mediated by the combined activity of the Na<sup>+</sup>-dependent Cl<sup>-</sup>-HCO<sub>3</sub><sup>-</sup>-SLC4A8 Cl<sup>-</sup>-HCO<sub>3</sub><sup>-</sup> exchanger and the SLC26A4 Cl<sup>-</sup>-HCO<sub>3</sub><sup>-</sup> exchanger<sup>131</sup> (see also Chapter 6). This transport mechanism is apparently responsible for up to 50% of Na<sup>+</sup>-Cl<sup>-</sup> transport in mineralocorticoid-stimulated rat CCD,<sup>129,130</sup> allowing for ENaC-independent, electroneutral Na<sup>+</sup> absorption that will not directly affect K<sup>+</sup> secretion. The converse effect emerges after dietary K<sup>+</sup> loading, which increases the fraction of ENaC-independent, amiloride-resistant K<sup>+</sup> excretion to ~50%.<sup>84</sup>

NCC, the thiazide-sensitive Na<sup>+</sup>-Cl<sup>-</sup> cotransporter in the DCT, plays a key role in K<sup>+</sup> homeostasis. Selective increases in DCT and NCC activity, as seen in FHHt, reduce Na<sup>+</sup> delivery to principal cells in the downstream CNT and CCD,

leading to hyperkalemia.<sup>69</sup> The DCT also clearly functions as a “potassium sensor,” directly responding to changes in circulating potassium. Reduction in potassium intake and/or hypokalemia thus leads to reduced basolateral [K<sup>+</sup>] in the DCT; the subsequent hyperpolarization is dependent on basolateral Kir4.1-containing K<sup>+</sup> channels.<sup>4</sup> Hyperpolarization leads to chloride exit, via basolateral CLC-NKB chloride channels, and a reduction in intracellular chloride; the reduction in intracellular chloride activates the WNK cascade, resulting in phosphorylation of NCC and activation of the transporter.<sup>4</sup> Angiotensin-II also activates NCC via WNK-dependent activation of the SPAK kinase and phosphorylation of the transporter protein,<sup>134,135</sup> reducing delivery of Na<sup>+</sup> to the CNT and limiting K<sup>+</sup> secretion. In contrast, angiotensin-II inhibits ROMK activity via several mechanisms including downstream activation of c-src tyrosine kinases (see earlier).<sup>126-128</sup> Whereas K<sup>+</sup> restriction induces renin and circulating angiotensin-II (see “Consequences of Hypokalemia”), increases in dietary K<sup>+</sup> are suppressive.<sup>128,136</sup> A high-K<sup>+</sup> diet also inactivates NCC<sup>137</sup> due to the associated decrease in angiotensin-II, in addition to the increase in the ratio of WNK1-S to WNK1-L isoforms that occurs with increased K<sup>+</sup> intake<sup>115,116,118</sup>; WNK1-S antagonizes the effect of WNK1-L on NCC, leading to inhibition of NCC in conditions with a relative excess of WNK1-S.<sup>132</sup> Selective deletion of WNK1-S indicates that it plays a key role in the K<sup>+</sup> switch pathway in the DCT.<sup>10</sup> Nedd4-2 also negatively regulates NCC and WNK1, in addition to ENaC; mice with homozygous deletion of the Nedd4-2 gene develop severe hypokalemia during K<sup>+</sup> restriction, indicating a key role in renal adaptation.<sup>138</sup>

Finally, within principal cells, increases in aldosterone induce the SGK1 kinase, which phosphorylates WNK4 and attenuates the effect of WNK4 on ROMK<sup>139</sup> while activating ENaC.<sup>93,94,96</sup> However, when dietary K<sup>+</sup> intake is reduced, c-Src tyrosine kinase activity increases under the influence of increased angiotensin-II, causing inhibition of ROMK activity via tyrosine phosphorylation of the channel.<sup>122,140</sup> The increase in c-Src tyrosine kinase activity also abrogates the inhibitory effect of SGK1 on WNK4<sup>133</sup>; Src family tyrosine kinases also directly phosphorylate WNK4 and modulate its effects on ROMK.<sup>141</sup> Again, angiotensin-II appears to mediate part of its inhibitory effect on ROMK through activation of c-Src,<sup>128</sup> such that c-Src serves as an important component of the “switch” that regulates K<sup>+</sup> secretion in response to changes in dietary K<sup>+</sup>.

To summarize this important physiology, the differential effects of K<sup>+</sup> intake on angiotensin-II versus aldosterone appear to be critical in resolving the aldosterone paradox; so, too, are the differential effects of K<sup>+</sup> intake on NCC-dependent Na<sup>+</sup>-Cl<sup>-</sup> transport in the DCT and on secretory K<sup>+</sup> channels within the downstream CNT and CCD (see also Fig. 16.7). Under conditions of low-Na<sup>+</sup> intake but moderate K<sup>+</sup> intake, angiotensin-II and aldosterone are both strongly induced, leading to enhanced Na<sup>+</sup>-Cl<sup>-</sup> transport via NCC, increased ENaC activity, and decreased secretory K<sup>+</sup> channel activity. Although ENaC is activated, the relative inhibition of ROMK by the increased angiotensin-II prevents excessive kaliuresis. Angiotensin-II-dependent activation of c-Src kinases has direct inhibitory effects on ROMK trafficking and also abrogates the inhibitory effect of SGK1 on WNK4,<sup>133</sup> leading to unopposed inhibition of ROMK by



**Fig. 16.7 Integrated regulation of  $\text{Na}^+\text{-Cl}^-$  and  $\text{K}^+$  transport in the DCT, CNT, and CCD.** Green arrowheads (activating pathways), red blunt end (inhibitory pathway). The left panel shows the pathway in the setting of a low- $\text{Na}^+$  diet, wherein Ang-II and SGK-1 signaling leads to phosphorylation of WNK4. This stimulates phosphorylation of SPAK, which in turn phosphorylates and activates thiazide-sensitive  $\text{Na}^+\text{-Cl}^-$  cotransport in the DCT via NCC. Stimulation of unknown receptors is hypothesized to cause phosphorylation of L-WNK1, which can also stimulate SPAK phosphorylation. L-WNK1 has other functions: 1. It blocks the NCC-inhibitory form of WNK4, thus activating NCC; and 2. It inhibits secretion of  $\text{K}^+$  via ROMK channels. The right panel shows the pathway in the setting of high dietary  $\text{K}^+$  intake, wherein aldosterone is stimulated and Ang-II is low. In the absence of sufficient Ang-II,  $\text{AT}_1\text{R}$  cannot activate WNK4. This reduces SPAK activation and NCC phosphorylation. Dietary potassium loading also increases the level of KS-WNK1 isoform to suppress the activity of L-WNK1. In consequence, the inhibitory effect of WNK4 on NCC dominates, blocking traffic of NCC to the apical membrane and thereby reducing NCC activity. KS-WNK1 also blocks the effect of L-WNK1 on ROMK endocytosis, causing ROMK to increase at the apical membrane. The net effect is that  $\text{K}^+$  secretion in the DCT and CNT/CCD is maximized, whereas NCC is suppressed. Aldosterone stimulation of ENaC (not shown) offsets the decreased  $\text{Na}^+$  reabsorption by NCC, allowing robust potassium secretion without changes in sodium balance. The roles of WNK3, SGK1, and c-src cytoplasmic tyrosine kinases are not shown in the interest of clarity; see text for further details. NCC,  $\text{NaCl}$  cotransporter; ROMK, renal outer medullary K channel; SPAK, STE20/SPS1-related proline/alanine-rich kinase; WNK, With-No-K(lysine) kinase. (From Welling PA, Chang YP, Delpire E, Wade JB. Multigene kinase network, kidney transport, and salt in essential hypertension. *Kidney Int.* 2010;77:1063–1069.)

WN4. In addition, the aldosterone-dependent induction of electroneutral  $\text{Na}^+\text{-Cl}^-$  transport within the CCD<sup>129–131</sup> increases  $\text{Na}^+\text{-Cl}^-$  reabsorption but blunts the effect on the lumen-negative potential, thus limiting kaliuresis. When dietary  $\text{K}^+$  increases, circulating aldosterone is moderately induced but angiotensin-II is suppressed. This leads to inhibition of NCC and increased downstream delivery of  $\text{Na}^+$  to principal cells in the CNT and CCD, where ENaC activity is increased and ROMK and BK channels are significantly activated. ENaC-independent  $\text{K}^+$  secretion is also strongly induced by increased dietary  $\text{K}^+$  intake,<sup>74</sup> contributing significantly to the ability to excrete  $\text{K}^+$  in the urine.

#### REGULATION OF RENAL RENIN AND ADRENAL ALDOSTERONE

Modulation of the renin-angiotensin-aldosterone (RAAS) axis has profound clinical effects on  $\text{K}^+$  homeostasis. Although multiple tissues are capable of renin secretion, renin of renal origin has a dominant physiologic impact. Renin secretion by juxtaglomerular cells within the afferent arteriole is initiated in response to a signal from the macula densa,<sup>142</sup> specifically a decrease in luminal chloride<sup>143</sup> transported through the  $\text{Na}^+\text{-K}^+\text{-2Cl}^-$  cotransporter (NKCC2) at the apical membrane of macula densa cells.<sup>25</sup>

In addition to this macula densa signal, decreased renal perfusion pressure and renal sympathetic tone stimulate renal renin secretion.<sup>144</sup> The various inhibitors of renin release include angiotensin-II, endothelin,<sup>145</sup> adenosine,<sup>146</sup> atrial natriuretic peptide (ANP),<sup>147</sup> TNF- $\alpha$ ,<sup>148</sup> and active vitamin D.<sup>149</sup> The cGMP-dependent protein kinase type II (cGKII) tonically inhibits renin secretion, in that renin secretion in response to several stimuli is exaggerated in homozygous *cGKII* knockout mice.<sup>150</sup> Activation of cGKII by ANP and/or nitric oxide has a marked inhibitory effect on the release of renin from juxtaglomerular cells.<sup>147</sup> Local factors that stimulate renin release from juxtaglomerular cells include prostaglandins,<sup>151</sup> adrenomedullin,<sup>152</sup> catecholamines ( $\beta$ -1 receptors),<sup>153</sup> and succinate (GPR91 receptor).<sup>154</sup>

The relationship between renal renin release, the RAAS, and cyclooxygenase-2 (COX-2) is particularly complex.<sup>151</sup> COX-2 is heavily expressed in the macula densa,<sup>151</sup> with significant recruitment of COX-2<sup>(+)</sup> cells seen with salt restriction or furosemide treatment.<sup>49,151</sup> Reduced intracellular chloride in macula densa cells appears to stimulate COX-2 expression via p38 MAP kinase,<sup>155</sup> whereas both aldosterone and angiotensin-II reduce its expression.<sup>151</sup> Prostaglandins derived from COX-2 in the

macula densa play a dominant role in the stimulation of renal renin release by salt restriction, furosemide, renal artery occlusion, or angiotensin-converting enzyme (ACE) inhibition.<sup>49,156</sup> Specifically, COX-2-derived prostaglandins appear to play a role in tonic expression of renin in JG cells, via modulation of intracellular cyclic-AMP and calcium, rather than functioning in the acute regulation of renin release.<sup>151</sup> Prostaglandins generated by the macula densa also participate in the recruitment during salt restriction of CD44<sup>+</sup> mesenchymal stromal cells, which differentiate into renin-producing cells.<sup>157</sup>

Renin released from the kidney ultimately stimulates aldosterone release from the adrenal via angiotensin-II (AT-II). Hyperkalemia per se is also an independent and synergistic stimulus (see Fig. 16.8) for aldosterone release from the adrenal gland,<sup>49,158</sup> although dietary K<sup>+</sup> loading is less potent than dietary Na<sup>+</sup>-Cl<sup>-</sup> restriction in increasing circulating aldosterone.<sup>87</sup> The resting membrane potential of adrenal glomerulosa cells is hyperpolarized, due to the activity of the “leak” K<sup>+</sup> channels TASK-1 and TASK-3; combined deletion of genes encoding these channels leads to baseline depolarization of adrenal glomerulosa cells and an increase in serum aldosterone that is resistant to dietary sodium loading.<sup>159</sup> AT-II and K<sup>+</sup> both activate Ca<sup>2+</sup> entry in glomerulosa cells, via voltage-sensitive T-type Ca<sup>2+</sup> channels,<sup>49,160</sup> primarily Cav3.2.<sup>161</sup> Elevations in extracellular K<sup>+</sup> thus depolarize glomerulosa cells and activate these Ca<sup>2+</sup> channels, which are independently and synergistically activated by AT-II.<sup>160</sup> Calcium-dependent activation of calcium-calmodulin (CaM)-CaM-dependent protein kinase in turn activates the synthesis and release of aldosterone, via induction of aldosterone synthase.<sup>162</sup> K<sup>+</sup> and AT-II also enhance transcription of the Cav3.2 Ca<sup>2+</sup> channel, by abrogating repression of this gene by the neuron restrictive silencing factor (NRS); this ultimately amplifies the induction of aldosterone synthase.<sup>161</sup>

The role of adrenal K<sup>+</sup> sensing in aldosterone release has been dramatically underlined by the reports of both germline and somatic mutations, in aldosterone-producing

adenomas, of transport proteins that control membrane excitability of adrenal zona glomerulosa cells (see also “Hyperaldosteronism” later). For example, somatic mutations in the adrenal K<sup>+</sup> channel, KCNJ5 (GIRK4), can be detected in ~40% of aldosterone-producing adrenal adenomas<sup>163</sup>; these mutations endow the channel with a novel Na<sup>+</sup> conductance, leading to adrenal glomerulosa cell depolarization, Ca<sup>2+</sup> influx, and aldosterone release.

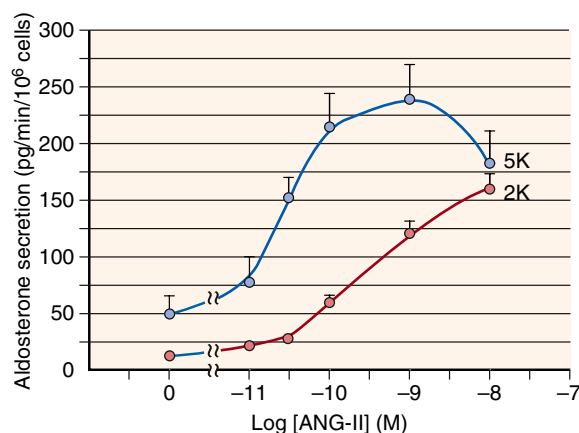
The adrenal release of aldosterone due to increased K<sup>+</sup> is dependent on an intact *adrenal* renin-angiotensin system,<sup>164</sup> particularly during Na<sup>+</sup> restriction. ACE-inhibitors and angiotensin-receptor blockers (ARBs) thus completely abrogate the effect of high K<sup>+</sup> on salt-restricted adrenals.<sup>165</sup> Direct, G-protein-dependent activation of the TASK-1 and/or TASK-3 K<sup>+</sup> channels by AT<sub>1A</sub> or AT<sub>1B</sub> receptors is thought to underlie the effect of AT-II on adrenal aldosterone release,<sup>159</sup> with abrogation of this effect by ARBs or ACE-inhibitors. Other clinically relevant activators of adrenal aldosterone release include prostaglandins<sup>166</sup> and catecholamines,<sup>167</sup> via increases in cyclic-AMP.<sup>168,169</sup> Finally, ANP exerts a potent negative effect on aldosterone release induced by K<sup>+</sup> and other stimuli,<sup>170</sup> at least in part by inhibiting early events in aldosterone synthesis.<sup>171</sup> ANP is therefore capable of inhibiting both renal renin release and adrenal aldosterone release, functions that may be central to the pathophysiology of hyporeninemic hypoaldosteronism.

## URINARY INDICES OF POTASSIUM EXCRETION

A bedside test to directly measure distal tubular K<sup>+</sup> secretion in humans would be ideal; however, for obvious reasons this is not technically feasible. A widely used surrogate is the “transtubular K<sup>+</sup> gradient” (TTKG), which is defined as follows:

$$\text{TTKG} = \frac{[\text{K}^+]_{\text{urine}} \times \text{Osm}_{\text{blood}}}{[\text{K}^+]_{\text{blood}} \times \text{Osm}_{\text{urine}}}$$

The expected values of the TTKG are largely based on historical data and are <3 to 4 in the presence of hypokalemia and >6 to 7 in the presence of hyperkalemia; the shifting opinions regarding the physiologically appropriate TTKG in hyperkalemia have been reviewed multiple times.<sup>172</sup> Clearly, water absorption in the CCD and medullary collecting duct is an important determinant of the absolute K<sup>+</sup> concentration in the final urine; hence the use of a ratio of urine-to-plasma osmolality. Indeed, water absorption may in large part determine the TTKG, such that it far exceeds the limiting K<sup>+</sup> gradient.<sup>173</sup> More recently, the originators of the TTKG have suggested that it fails to incorporate the putative effects of distal tubular urea reabsorption in K<sup>+</sup> excretion; however, neither urea transporter knockout mice nor rats treated with a urea transporter inhibitor<sup>174</sup> demonstrate abnormalities in K<sup>+</sup> homeostasis. The TTKG continues to play an important role in clinical investigation of potassium homeostasis.<sup>175</sup> The TTKG may be less useful in patients ingesting diets of changing K<sup>+</sup> and mineralocorticoid intake.<sup>176</sup> There is, however, a linear relation between plasma aldosterone and the TTKG, suggesting that it provides a rough approximation of the ability to respond to aldosterone with kaliuresis.<sup>177</sup> The response of the TTKG to mineralocorticoid administration, typically fludrocortisone, can thus be utilized in the diagnostic approach to hyperkalemia.<sup>172</sup> In hypokalemic patients, a TTKG of <2 to 3 separates patients with redistributive hypokalemia from those



**Fig. 16.8 Synergistic effect of increased extracellular K<sup>+</sup> and angiotensin-II (ANG-II) in inducing aldosterone release from bovine adrenal glomerulosa cells.** Dose-response curves for ANG-II were performed at extracellular K<sup>+</sup> of 2 mmol/L (○) and 5 mmol/L (●). (From Chen XL, Bayliss DA, Fern RJ, Barrett PQ. A role for T-type Ca<sup>2+</sup> channels in the synergistic control of aldosterone production by ANG II and K<sup>+</sup>. *Am J Physiol.* 1999;276:F674–683.)

with hypokalemia due to renal potassium wasting, who will have TTKG values that are  $>4$ .<sup>178</sup>

An alternative to the TTKG in hypokalemic patients is the measurement of urine  $K^+$ :creatinine ratio. The urine  $K^+$ :creatinine ratio is usually  $<13$  meq/g creatinine (1.5 meq/mmol creatinine) when hypokalemia is caused by poor dietary intake, transcellular potassium shifts, gastrointestinal losses, or previous use of diuretics.<sup>178</sup> Higher values are indicative of ongoing renal potassium wasting. The utility of the  $K^+$ :creatinine ratio was evaluated in a study of 43 patients with severe hypokalemia (ranging 1.5–2.6 mmol/L) associated with paralysis.<sup>178</sup> The urine  $K^+$ :creatinine ratio reliably distinguished between the 30 patients with hypokalemic periodic paralysis and the 13 patients with hypokalemia due mostly to renal potassium wasting. The  $K^+$ :creatinine ratio was thus significantly lower in the patients with periodic paralysis (11 vs. 36 meq/g creatinine, 1.3 vs. 4.1 meq/mmol creatinine). The cutoff value was approximately 22 meq/g creatinine (2.5 meq/mmol).

The determination of urinary electrolytes for calculation of the TTKG or urine  $K^+$ :creatinine ratio provides the opportunity for the measurement of urinary  $Na^+$ , which will determine whether significant prerenal stimuli are limiting distal  $Na^+$  delivery and thus  $K^+$  excretion (see also Fig. 16.5). Urinary electrolytes also afford the opportunity to calculate the urinary anion gap, an indirect index of urinary  $NH_4^+$  content and thus the ability to respond to acidemia.<sup>179</sup>

## CONSEQUENCES OF HYPERKALEMIA AND HYPOKALEMIA

### CONSEQUENCES OF HYPOKALEMIA

#### EXCITABLE TISSUES: MUSCLE AND HEART

Hypokalemia is a well-described risk factor for both ventricular and atrial arrhythmias.<sup>27,180</sup> For example, in patients undergoing cardiac surgery, a serum  $K^+$  of  $<3.5$  mmol/L is a predictor of serious intraoperative arrhythmia, perioperative arrhythmia, and postoperative atrial fibrillation.<sup>181</sup> Moderate hypokalemia does not, however, appear to increase the risk of serious arrhythmia during exercise stress testing.<sup>182</sup> Electrocardiographic changes in hypokalemia include broad flat T-waves, ST depression, and QT prolongation; these are most marked when serum  $K^+$  is  $<2.7$  mmol/L.<sup>183</sup> Hypokalemia, often accompanied by hypomagnesemia, is an important cause of the long QT syndrome (LQTS) and torsades de pointes, either alone<sup>184</sup> or in combination with drug toxicity<sup>185</sup> or with LQTS-associated mutations in cardiac  $K^+$  and  $Na^+$  channels.<sup>186</sup> Hypokalemia accelerates the clathrin-dependent internalization and degradation of the cardiac HERG (human ether-a-go-go)  $K^+$  channel protein.<sup>29</sup> HERG encodes pore-forming subunits of the cardiac rapidly activating delayed rectifier  $K^+$  channel ( $I_{Kr}$ );  $I_{Kr}$  is largely responsible for potassium efflux during phases 2 and 3 of the cardiac action potential.<sup>187</sup> Loss-of-function mutations in HERG reduce  $I_{Kr}$  and cause type II LQTS<sup>29</sup>; downregulation of HERG and  $I_{Kr}$  by hypokalemia provides an elegant explanation for the association with LQTS and torsades de pointes.

In accordance with the Nernst equation, the resting membrane potential is related to the ratio of the

intracellular to the extracellular potassium concentration. In skeletal muscle, a reduction in plasma  $K^+$  will increase this ratio and therefore hyperpolarize the cell membrane (i.e., make the resting potential more electronegative); this impairs the ability of the muscle to depolarize and contract, leading to weakness. However, in some human cardiac cells, particularly Purkinje fibers in the conducting system, hypokalemia results in a paradoxical depolarization<sup>188</sup>; this paradoxical depolarization plays an important role in the genesis of hypokalemic cardiac arrhythmias.<sup>188,189</sup> Resting membrane potential in excitable cells is determined mostly by a large family of “K2P1”  $K^+$  channels, so-named due to the presence of two pore-forming (P) loop domains in each subunit. Hypokalemia causes K2P1 channels, which are normally selective for potassium, to suddenly transport sodium into cells, causing paradoxical depolarization.<sup>190</sup> Notably, rodent cardiac cells respond to hypokalemia with a Nernst equation–predicted hyperpolarization and, unlike human cardiomyocytes, do not express the K2P1 channel TWIK-1; genetic manipulation indicates that TWIK-1 expression confers this paradoxical depolarization behavior on human and mouse cardiomyocytes.<sup>190</sup>

An additional mechanism for hypokalemia-induced arrhythmia involves downregulation of cardiac  $Na^+$ / $K^+$ -ATPase activity.<sup>191,192</sup> The resulting increase in intracellular  $Na^+$ , which impedes removal of intracellular  $Ca^{2+}$  by the  $Na^+$ - $Ca^{2+}$  exchanger and other mechanisms, leads to intracellular calcium overload. The ensuing increase in calmodulin kinase-II activity reduces repolarization reserve by activating late  $Na^+$  and  $Ca^{2+}$  currents.<sup>193</sup> This, in turn, predisposes the heart to early afterdepolarization-associated arrhythmias, such as torsades de pointes and polymorphic ventricular tachycardia.<sup>193</sup>

In skeletal muscle, hypokalemia causes hyperpolarization, thus impairing the capacity to depolarize and contract. Weakness and paralysis are therefore a not-infrequent consequence of hypokalemia of diverse etiologies.<sup>194,195</sup> On a historical note, the realization in 1946 that  $K^+$  replacement reversed the hypokalemic diaphragmatic paralysis induced by treatment of diabetic ketoacidosis (DKA) was a milestone in diabetes care.<sup>196</sup> Pathologically, muscle biopsies in hypokalemic myopathy demonstrate phagocytosis of degenerating muscle fibers, fiber regeneration, and atrophy of type 2 fibers.<sup>197</sup> Most patients with significant myopathy will have elevations in creatine kinase, and hypokalemia of diverse etiologies predisposes to rhabdomyolysis with acute renal failure.

### RENAL CONSEQUENCES

Hypokalemia causes a host of structural and functional changes in the kidney. In humans, the renal pathology includes a relatively specific proximal tubular vacuolization,<sup>27,198</sup> interstitial nephritis,<sup>199</sup> and renal cysts.<sup>200</sup> Hypokalemic nephropathy can cause ESRD, mostly in patients with longstanding hypokalemia due to eating disorders and/or laxative abuse<sup>201</sup>; acute renal failure with proximal tubular vasculopathy has also been described.<sup>202</sup> In animal models, hypokalemia increases susceptibility to acute renal failure induced by ischemia, gentamicin, and amphotericin.<sup>49</sup> Potassium restriction in rats induces cortical AT-II and medullary endothelin-1 expression, with an ischemic pattern of renal injury.<sup>203</sup> Hypokalemic

nephropathy in rats is associated with progressive capillary loss, with reduced angiogenesis due to reduced VEGF expression.<sup>204</sup>

The prominent functional changes in renal physiology that are induced by hypokalemia include  $\text{Na}^+\text{-Cl}^-$  retention, polyuria,<sup>205</sup> phosphaturia,<sup>27</sup> hypocitraturia,<sup>206</sup> and increased ammoniogenesis.  $\text{K}^+$  depletion in rats causes proximal tubular hyperabsorption of  $\text{Na}^+\text{-Cl}^-$ , in association with an upregulation of AT-II,<sup>203</sup>  $\text{AT}_1$  receptor,<sup>27</sup> and the  $\alpha_2$ -adrenergic receptor<sup>27</sup> in this nephron segment. NHE3, the dominant apical  $\text{Na}^+$  entry site in the proximal tubule, is massively (>700%) upregulated in  $\text{K}^+$ -deficient rats,<sup>207</sup> which is consistent with the observed hyperabsorption of both  $\text{Na}^+\text{-Cl}^-$  and bicarbonate.<sup>27</sup> Polyuria in hypokalemia is due to polydipsia<sup>208</sup> and a vasopressin-resistant defect in urinary concentrating ability. This renal concentrating defect is multifactorial, with evidence for both a reduced hydroosmotic response to vasopressin in the collecting duct and decreased  $\text{Na}^+\text{-Cl}^-$  absorption by the TAL.<sup>27</sup>  $\text{K}^+$  restriction has been shown to result in a rapid, reversible decrease in the expression of aquaporin-2 in the collecting duct,<sup>209</sup> beginning in the CCD and extending to the medullary collecting duct within the first 24 hours.<sup>210</sup> Downregulation of aquaporin-2 and several other proteins in hypokalemic nephrogenic diabetes insipidus appears to occur via autophagy.<sup>211,212</sup> In the TAL, the marked reductions seen during  $\text{K}^+$  restriction in both the apical  $\text{K}^+$  channel ROMK and the apical  $\text{Na}^+\text{-K}^+\text{-2Cl}^-$  cotransporter NKCC2<sup>207</sup> reduce  $\text{Na}^+\text{-Cl}^-$  absorption, and thus inhibit countercurrent multiplication and the driving force for water absorption by the collecting duct.

## CARDIOVASCULAR CONSEQUENCES

A large body of experimental and epidemiologic evidence implicates hypokalemia and/or reduced dietary  $\text{K}^+$  in the genesis or worsening of hypertension, heart failure, and stroke.<sup>213</sup>  $\text{K}^+$  depletion in young rats induces hypertension,<sup>214</sup> with a salt sensitivity that persists after  $\text{K}^+$  levels are normalized; presumably, this salt sensitivity is due to the significant tubulointerstitial injury induced by  $\text{K}^+$  restriction.<sup>203</sup> Hypokalemia has also been linked to vascular calcification and arterial stiffness, via induction of autophagy and promotion of vascular smooth muscle calcification.<sup>215</sup> Short-term  $\text{K}^+$  restriction in healthy humans and patients with essential hypertension also induces  $\text{Na}^+\text{-Cl}^-$  retention and hypertension,<sup>27</sup> and abundant epidemiologic data link dietary  $\text{K}^+$  deficiency and/or hypokalemia with hypertension.<sup>213,216</sup> Correction of hypokalemia is particularly important in hypertensive patients treated with diuretics; blood pressure in this setting is improved with the establishment of normokalemia,<sup>217</sup> and the cardiovascular benefits of diuretic agents are blunted by hypokalemia.<sup>27,218</sup> Hypokalemia reduces insulin secretion; this mechanism may play an important role in thiazide-associated diabetes.<sup>219</sup> Finally,  $\text{K}^+$  depletion may play important roles in the pathophysiology and progression of heart failure.<sup>213</sup>

## CONSEQUENCES OF HYPERKALEMIA

### EXCITABLE TISSUES: MUSCLE AND HEART

Hyperkalemia constitutes a medical emergency, primarily due to its effect on the heart. Hyperkalemia depolarizes

cardiac myocytes, reducing the membrane potential from  $-90$  mV to  $\sim -80$  mV. This brings the membrane potential closer to the threshold for generation of an action potential; mild and/or rapid onset hyperkalemia will initially increase cardiac excitability since a lesser depolarizing stimulus is required to generate an action potential. Mild increases in extracellular  $\text{K}^+$  also affect the repolarization phase of the cardiac action potential, via increases in  $I_{\text{Kr}}$ ; as discussed earlier (see “Consequences of Hypokalemia”),  $I_{\text{Kr}}$  is highly sensitive to changes in extracellular  $\text{K}^+$ .<sup>29</sup> This effect on repolarization is thought to underlie the “early” signs of hyperkalemia<sup>220</sup> including ST-T segment depression, peaked T-waves, and Q-T interval shortening.<sup>187</sup> Persistent and increasing depolarization inactivates cardiac sodium channels, thus reducing the rate of phase 0 of the action potential ( $V_{\text{max}}$ ); the decrease in  $V_{\text{max}}$  results in a reduction in myocardial conduction, with progressive prolongation of the P wave, PR interval, and QRS complex.<sup>187</sup> Severe hyperkalemia results in loss of the P wave and a progressive widening of the QRS complex; fusion with T-waves causes a “sine-wave” sinoventricular rhythm.

Cardiac arrhythmias associated with hyperkalemia include sinus bradycardia, sinus arrest, slow idioventricular rhythms, ventricular tachycardia, ventricular fibrillation, and asystole<sup>220,221</sup>; a multitude of mechanisms are involved.<sup>193</sup> The differential diagnosis and treatment of a wide-complex tachycardia in hyperkalemia can be particularly problematic; moreover, hyperkalemia potentiates the blocking effect of lidocaine on the cardiac  $\text{Na}^+$  channel, such that use of this agent may precipitate asystole or ventricular fibrillation in this setting.<sup>222</sup> Hyperkalemia can also cause a type I Brugada pattern in the electrocardiogram (ECG), with a pseudo-RBBB and persistent “coved” ST segment elevation in at least two precordial leads. This “hyperkalemic Brugada sign” occurs in critically ill patients with significant hyperkalemia (serum  $\text{K}^+ > 7$  mmol/L) and can be differentiated from genetic Brugada syndrome by an absence of P waves, marked QRS widening, and an abnormal QRS axis.<sup>223</sup>

Classically, the electrocardiographic manifestations in hyperkalemia progress as shown in Table 16.2. However, these changes are notoriously insensitive, such that only 55% of patients with serum  $\text{K}^+ > 6.8$  mmol/L in one case series manifested peaked T-waves.<sup>224</sup> There is large interpatient variability in the absolute potassium level leading to ECG changes and cardiac toxicity of hyperkalemia. Relevant variables include the rapidity of the onset of hyperkalemia<sup>225,226</sup> and the presence or absence of concomitant hypocalcemia, acidemia, and/or hyponatremia.<sup>227,228</sup> Hemodialysis patients<sup>228</sup> and patients with chronic renal failure<sup>229</sup> in particular may not demonstrate electrocardiographic changes. Care should also be taken to adequately distinguish the symmetrically peaked, “church steeple,” T-waves induced by hyperkalemia from T-wave changes due to other causes.<sup>230</sup> The ratio of precordial T-wave to R-wave amplitude (T:R ratio) may be a more specific sign of hyperkalemia than T-wave tenting.<sup>231</sup>

Hyperkalemia can also rarely present with ascending paralysis,<sup>49</sup> denoted “secondary hyperkalemic paralysis” to differentiate it from familial hyperkalemic periodic paralysis

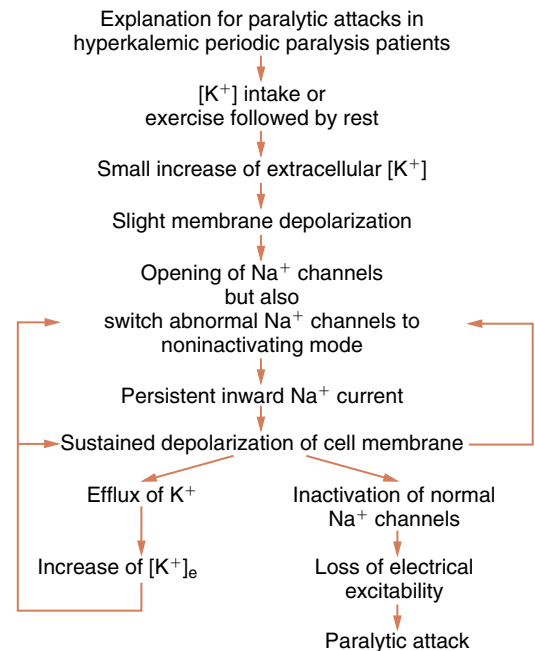
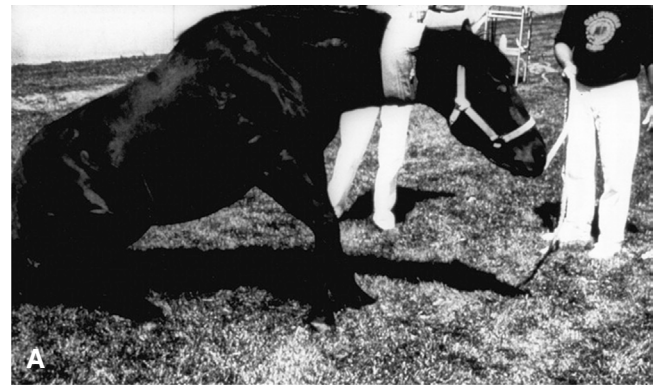
**Table 16.2** Approximate Relationship Between Hyperkalemic Electrocardiographic Changes and Serum K<sup>+</sup>

| Serum K <sup>+</sup>                    | Electrocardiogram Abnormality                                                                                                                                                                                                                                   |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mild hyperkalemia<br>5.5–6.5 mmol/L     | Tall peaked T-waves with narrow base, best seen in precordial leads                                                                                                                                                                                             |
| Moderate hyperkalemia<br>6.5–8.0 mmol/L | Peaked T-waves<br>Prolonged PR interval<br>Decreased amplitude of P waves                                                                                                                                                                                       |
| Moderate hyperkalemia<br>>8.0 mmol/L    | Widening of QRS complex<br>Absence of P wave<br>Intraventricular blocks, fascicular blocks, bundle branch blocks, QRS axis shift<br>Progressive widening of the QRS complex<br>“Sine-wave” pattern (sinoventricular rhythm), ventricular fibrillation, asystole |

From Mattu A, Brady WJ, Robinson DA. Electrocardiographic manifestations of hyperkalemia. *Am J Emerg Med.* 2000;18:721–729.

(HYPP). This presentation of hyperkalemia can mimic Guillain-Barré syndrome and may include diaphragmatic paralysis and respiratory failure.<sup>232</sup> Hyperkalemia from a diversity of causes can cause paralysis, as reviewed by Evers and colleagues.<sup>233</sup> The mechanism is not entirely clear; however, nerve conduction studies in one case suggest a neurogenic mechanism, rather than a direct effect on muscle excitability.<sup>233</sup>

In contrast to secondary hyperkalemic paralysis, HYPP is a primary myopathy. Patients with HYPP develop myopathic weakness during hyperkalemia induced by increased K<sup>+</sup> intake or rest after heavy exercise.<sup>234</sup> The hyperkalemic trigger in HYPP serves to differentiate this syndrome from *hypokalemic* periodic paralysis (HOKP); a further distinguishing feature is the presence of myotonia in HYPP.<sup>234</sup> Depolarization of skeletal muscle by hyperkalemia unmasks an inactivation defect in a tetrodotoxin-sensitive Na<sup>+</sup> channel in patients with HYPP, and autosomal dominant mutations in the *SCN4A* gene encoding this channel cause most forms of the disease.<sup>235</sup> Mild muscle depolarization (5–10 mV) in HYPP results in a persistent inward Na<sup>+</sup> current through the mutant channel; the normal, allelic *SCN4* channels quickly recover from inactivation and can then be reactivated, resulting in myotonia. When muscle depolarization is more marked (i.e., 20–30 mV), all of the Na<sup>+</sup> channels are inactivated, rendering the muscle inexcitable and causing weakness (Fig. 16.9). Related disorders due to mutations within the large *SCN4A* channel protein include HOKP type II,<sup>236</sup> paramyotonia congenita,<sup>235</sup> and K<sup>+</sup>-aggravated myopathy.<sup>235</sup> American Quarter Horses have historically had a high incidence (4.4%) of HYPP due to a mutation



B

**Fig. 16.9 Hyperkalemic periodic paralysis (HYPP) due to mutations in the voltage-gated Na<sup>+</sup> channel of skeletal muscle.** (A) This disorder is particularly common in American Quarter Horses; an affected horse is shown during a paralytic attack, triggered by rest after heavy exercise. (B) Mechanistic explanation for muscle paralysis in HYPP. (A, Courtesy Dr. Eric P. Hoffman. B, From Lehmann-Horn F, Jurkat-Rott K. Voltage-gated ion channels and hereditary disease. *Physiol Rev.* 1999;79:1317–1372.)

in equine *SCN4A* traced to the sire “Impressive,” but the horse industry has worked to eliminate it. Since 2007, a foal that tests positive for this genetic disease cannot be registered by the American Quarter Horse Association (see Fig. 16.9).<sup>235</sup> Finally, loss-of-function mutations in the muscle-specific K<sup>+</sup> channel subunit “MinK-related peptide 2” (MiRP2) have also been shown to cause HYPP; MiRP2 and the associated Kv3.4 K<sup>+</sup> channel play a role in setting the resting membrane potential of skeletal muscle.<sup>237</sup>

## RENAL CONSEQUENCES

Hyperkalemia has a significant effect on the ability to excrete acid urine due to interference with the urinary excretion of ammonium (NH<sub>4</sub><sup>+</sup>). Potassium loading in humans results in modest reduction in urinary NH<sub>4</sub><sup>+</sup> excretion and an impaired response to acid loading.<sup>238</sup> In rats, chronic potassium loading leads to hyperkalemia

and metabolic acidosis due to a 40% reduction in urinary  $\text{NH}_4^+$  excretion.<sup>239</sup> Proximal tubular ammonia generation falls but without a significant effect on proximal tubular secretion of  $\text{NH}_4^+$ .<sup>239</sup> The TAL absorbs  $\text{NH}_4^+$  from the tubular lumen, followed by countercurrent multiplication and ultimately excretion from the medullary interstitium<sup>240</sup>; hyperkalemia appears to inhibit renal acid excretion by competing with  $\text{NH}_4^+$  for reabsorption by the TAL.<sup>241</sup>

The  $\text{NH}_4^+$  ion has the same ionic radius as  $\text{K}^+$  and can be transported in lieu of  $\text{K}^+$  by  $\text{NKCC2}$ ,<sup>242</sup> the apical  $\text{Na}^+\text{-K}^+/\text{NH}_4^+\text{-2Cl}^-$  cotransporter of the TAL;  $\text{NH}_4^+$  exits the TAL via the basolateral  $\text{Na}^+/\text{H}^+$  exchanger  $\text{NHE4}$ .<sup>243</sup> As is the case for other cations, countercurrent multiplication of  $\text{NH}_4^+$  by the TAL greatly increases the concentration of  $\text{NH}_4^+/\text{NH}_3$  available for secretion in the collecting duct. The  $\text{NH}_4^+$  produced by the proximal tubule in response to acidosis is thus reabsorbed across the TAL, concentrated by countercurrent multiplication in the medullary interstitium, and secreted in the collecting duct. The capacity of the TAL to reabsorb  $\text{NH}_4^+$  is increased during acidosis, due to an induction of  $\text{NKCC2}$ <sup>242</sup> and  $\text{NHE4}$  expression.<sup>243</sup> Hyperkalemia induces acidosis in rats by reducing the  $\text{NH}_4^+$  between the vasa recta (surrogate for interstitial fluid) and collecting duct<sup>241</sup> due to interference with absorption of  $\text{NH}_4^+$  by the TAL.

More recently, in a mouse model for hyporeninemic hypoaldosteronism with prominent, treatable (by hydrochlorothiazide) hyperkalemic acidosis, hyperkalemia was correlated with reduced expression of ammonia-generating enzymes in the proximal tubule, combined with upregulation of the ammonia-recycling enzyme glutamine synthetase.<sup>244</sup> Hyperkalemia did not affect expression of  $\text{NKCC2}$  or  $\text{NHE3}$ . However, these mice demonstrated decreased expression of the ammonia transporter family member  $\text{Rhcg}$  and decreased apical polarization of  $\text{H}^+\text{-ATPase}$  in the inner stripe of the OMCD, further compromising urinary  $\text{NH}_4^+$  excretion. These various changes in enzyme and transporter expression were reversed by correcting the hyperkalemia.<sup>244</sup>

Clinically, patients with hyperkalemic acidosis due to hyporeninemic hypoaldosteronism demonstrate an increase in urinary  $\text{NH}_4^+$  excretion in response to normalization of plasma  $\text{K}^+$  with cation-exchange resins,<sup>245,246</sup> indicating a significant role for hyperkalemia in generation of the acidosis.

## CAUSES OF HYPOKALEMIA

### EPIDEMIOLOGY

Hypokalemia is a relatively common finding in both outpatients and inpatients, perhaps the most common electrolyte abnormality encountered in clinical practice.<sup>247</sup> When defined as a serum  $\text{K}^+$  of  $<3.6$  mmol/L, it is found in approximately 20% of hospitalized patients<sup>248,249</sup>; defined as a serum  $\text{K}^+$  of  $<3.4$  mmol/L, it occurs in 16.8% of first-time hospital admissions.<sup>250</sup> Hypokalemia is usually mild, with  $\text{K}^+$  levels in the 3.0 to 3.5 mmol/L range, but in up to 25% of hypokalemic patients, it can be moderate to severe ( $<3.0$  mmol/L).<sup>248,251</sup> The most common causative

factors in hospitalized patients with hypokalemia are gastrointestinal losses of potassium, diuretic therapy, and hypomagnesemia.<sup>252</sup> It is a particularly prominent problem in patients receiving thiazide diuretics for hypertension, with an incidence of up to 48% (average 15%–30%).<sup>27,253</sup> The thiazide-type diuretic, metolazone, is frequently utilized in the management of heart failure refractory to loop diuretics alone, causing moderate ( $\text{K}^+ \leq 3.0$  mmol/L) or severe ( $\text{K}^+ \leq 2.5$  mmol/L) hypokalemia in approximately 40% and 10% of patients, respectively.<sup>254</sup> Hypokalemia is also a common finding in patients receiving peritoneal dialysis, with 10% to 20% requiring potassium supplementation.<sup>255</sup> Hypokalemia per se can increase in-hospital mortality rate up to 10-fold,<sup>250,251</sup> likely due to the profound effects on arrhythmogenesis, blood pressure, and cardiovascular morbidity.<sup>213,256</sup>

### SPURIOUS HYPOKALEMIA

Delayed sample analysis is a well-recognized cause of spurious hypokalemia, due to increased cellular uptake; this may become clinically relevant if ambient temperature is increased.<sup>27,257,258</sup> Rarely, patients with profound leukocytosis due to acute leukemia present with artifactual hypokalemia caused by time-dependent uptake of  $\text{K}^+$  by the large white cell mass.<sup>257</sup> Such patients do not develop clinical or electrocardiographic complications of hypokalemia, and plasma  $\text{K}^+$  is normal if measured immediately after venipuncture.

### REDISTRIBUTION AND HYPOKALEMIA

Manipulation of the factors affecting internal distribution of  $\text{K}^+$  (see “Factors Affecting Internal Distribution of Potassium” earlier) can cause hypokalemia, due to redistribution of  $\text{K}^+$  between the extracellular and intracellular compartments. Endogenous insulin is rarely a cause of hypokalemia; however, administered insulin is a frequent cause of iatrogenic hypokalemia<sup>248</sup> and may be a factor in the “dead in bed syndrome” associated with aggressive glycemic control.<sup>259</sup> Insulin also may play a significant role in the hypokalemia associated with refeeding syndrome.<sup>260</sup> Alterations in the activity of the endogenous sympathetic nervous system can cause hypokalemia in several settings including alcohol withdrawal,<sup>261</sup> acute myocardial infarction,<sup>213,262</sup> and head injury.<sup>263,264</sup> Redistributive hypokalemia after severe head injury can be truly profound, with reported serum  $\text{K}^+$  of 1.2<sup>263</sup> and 1.9<sup>264</sup>, and marked rebound hyperkalemia after repletion.

Due to their ability to activate both  $\text{Na}^+/\text{K}^+\text{-ATPase}$ <sup>47</sup> and the  $\text{Na}^+\text{-K}^+\text{-2Cl}^-$  cotransporter  $\text{NKCC1}$ ,<sup>26,49</sup>  $\beta_2$  agonists are powerful activators of cellular  $\text{K}^+$  uptake. These agents are chiefly encountered in the therapy of asthma, but tocolytics such as ritodrine can induce hypokalemia and arrhythmias during maternal labor.<sup>265</sup> The long-acting  $\beta_2$  agonist, clenbuterol, which is not approved for medical use in the United States, has caused hypokalemia in poisonings including outbreaks of toxicity from clenbuterol-adulterated heroin.<sup>266</sup> Occult sources of sympathomimetics, such as pseudoephedrine and ephedrine in cough syrup<sup>195</sup> or dieting agents,<sup>267</sup> can be an overlooked cause of hypokalemia. Finally, downstream activation of cyclic-AMP by

xanthines, such as theophylline<sup>49,268</sup> and dietary caffeine,<sup>269</sup> may induce hypokalemia and may synergize in this respect with  $\beta_2$  agonists.<sup>270</sup>

Whereas  $\beta_2$  agonists activate  $K^+$  uptake via the  $Na^+/K^+$ -ATPase, one would expect that inhibition of passive  $K^+$  efflux would also lead to hypokalemia; this is accomplished by barium, a potent inhibitor of inward-rectifying  $K^+$  channels.<sup>271</sup> This rare cause of hypokalemia is usually due to ingestion of the rodenticide barium carbonate, either unintentionally or during a suicide attempt.<sup>272</sup> Suicidal ingestion of barium-containing shaving powder<sup>273</sup> and hair remover<sup>274</sup> has also been described. Barium salts are widely used in industry, and poisoning has been described by various mechanisms in industrial accidents.<sup>49,275</sup> Patients have a particularly prominent U wave, likely due to direct inhibition of cardiac inward-rectifying  $K^+$  channels.<sup>271</sup> Muscle paralysis can also occur<sup>271</sup> due to inhibition of muscle Kir channels. Treatment of barium poisoning with  $K^+$  serves to both increase plasma  $K^+$  and to displace barium from affected  $K^+$  channels<sup>272</sup>; hemodialysis is also an effective treatment.<sup>276</sup>

Cesium also inhibits multiple  $K^+$  channels and can provoke hypokalemia and associated arrhythmias, with inappropriate kaliuresis presumably due to inhibition of ROMK and BK secretory channels.<sup>277</sup> Hypokalemia is also common with chloroquine toxicity or overdose,<sup>278</sup> although the mechanism is not entirely clear.

## HYPOKALEMIC PERIODIC PARALYSIS

The periodic paralyses have both genetic and acquired causes and are further subdivided into hyperkalemic and hypokalemic forms.<sup>49,234-236</sup> The genetic and secondary forms of hyperkalemic paralysis are discussed earlier (see “Consequences of Hyperkalemia” earlier). Autosomal dominant mutations in the *CACNA1S* gene encoding the  $\alpha_1$  subunit of L-type calcium channels are the most common genetic cause of hypokalemic periodic paralysis (HOKP type I), whereas type II HOKP is due to mutations in the *SCN4A* gene encoding the skeletal  $Na^+$  channel.<sup>279</sup> In Andersen syndrome, autosomal dominant mutations in the *KCNJ2* gene encoding the inwardly rectifying  $K^+$  channel Kir2.1 cause periodic paralysis, cardiac arrhythmias, and dysmorphic features.<sup>280</sup> Paralysis in Andersen syndrome can be normokalemic, hypokalemic, or hyperkalemic; however, the symptomatic trigger is consistent within individual kindreds.<sup>280</sup>

The pathophysiology of HOKP is complex. Structurally, ~90% of the HOKP-associated mutations result in loss of positively charged arginine residues in the S4, voltage-sensor domains of L-type calcium channels, and the skeletal  $Na^+$  channel.<sup>236</sup> This generates a so-called “gating current,” generated by a cation leak through an aberrant pore; this abnormal cation leak may directly lead to  $K^+$ -dependent paradoxical depolarization and hypokalemic weakness.<sup>281</sup> Muscles of a  $Na^+$  channel knockin mutant mouse ( $Na_v1.4$ -R669H) also exhibit an anomalous inward current at hyperpolarized potentials, attributed to this gating pore current.<sup>282</sup>

Abnormalities in insulin-sensitive transport events may also contribute to the hypokalemic weakness in HOKP. Reversible attacks of paralysis with hypokalemia in HOKP

are typically precipitated by rest after exercise and/or meals rich in carbohydrates.<sup>236</sup> Although the induction of endogenous insulin by carbohydrate meals is thought to reduce plasma  $K^+$ , thus triggering weakness, insulin can precipitate paralysis in HOKP in the absence of significant hypokalemia.<sup>283</sup> The generation of action potentials and muscle contraction is reduced in type I and II HOKP muscle fibers exposed to insulin in vitro<sup>279,284</sup>; this effect is seen at an extracellular  $K^+$  of 4.0 mmol/L and is potentiated as  $K^+$  decreases.<sup>284</sup> Type I HOKP muscles have a reduced activity of ATP-sensitive, inward-rectifying  $K^+$  channels ( $K_{ATP}$ ),<sup>285,286</sup> which likely contributes to hypokalemia due to the resultant unopposed activity of muscle  $Na^+/K^+$ -ATPase.<sup>287</sup> Insulin inhibits the remaining  $K_{ATP}$  activity in muscle fibers of both type I HOKP patients<sup>284</sup> and hypokalemic rats,<sup>288</sup> resulting in a depolarizing shift toward the equilibrium potential for the  $Cl^-$  ion (approximately 50 mV); at this potential, voltage-dependent  $Na^+$  channels are largely inactivated, resulting in paralysis.

Paralysis is associated with multiple other causes of hypokalemia, both acquired and genetic.<sup>194,195,289</sup> Renal causes of hypokalemia with paralysis include Fanconi syndrome,<sup>290</sup> Gitelman syndrome,<sup>291</sup> and the various causes of hypokalemic distal renal tubular acidosis (RTA).<sup>27,292</sup> The activity and regulation of skeletal muscle  $K_{ATP}$  channels are aberrant in animal models of hypokalemia, suggesting a parallel muscle physiology to that of genetic HOKP (see earlier). However, the pathophysiology of thyrotoxic periodic paralysis (TPP), a particularly important cause of hypokalemic paralysis, is distinctly different from that of HOKP; for example, despite the clinical similarities between the two syndromes, thyroxine has no effect on HOKP.

TPP is classically seen in patients of Asian origin but also occurs at higher frequencies in Latin American patients<sup>293</sup>; this shared predisposition has been linked to genetic variation in the *KCNJ18* gene encoding Kir2.6, a muscle-specific, thyroid hormone-responsive  $K^+$  channel.<sup>28</sup> Genome-wide association studies have also implicated variation in the *KCNJ2* gene, which encodes a related muscle  $K^+$  channel, Kir 2.1, predisposition to TPP.<sup>294</sup> Patients typically present with weakness of the extremities and limb girdles, with attacks occurring most frequently between 1 and 6 A.M. As in HOKP, paralytic attacks in TPP may be precipitated by rest and/or carbohydrate-rich meals. Clinical signs and symptoms of hyperthyroidism are not invariably present.<sup>293,295</sup> Hypokalemia is profound, ranging between 1.1 and 3.4 mol/L, and is frequently accompanied by hypophosphatemia and hypomagnesemia<sup>293</sup>; all three abnormalities presumably contribute to the associated weakness. Diagnostically, a TTKG of <2 to 3 or urine K:creatinine ratio of 2.5 mmol/mmol separates patients with TPP from those with hypokalemia due to renal potassium wasting, who will have TTKG values that are >4.<sup>178</sup> This distinction is of considerable therapeutic relevance; patients with large potassium deficits require aggressive repletion with  $K^+$ - $Cl^-$ , whereas such an approach would incur significant risk of rebound hyperkalemia in TPP and related disorders.<sup>178,296</sup>

The hypokalemia in TPP is most likely due to both direct and indirect activation of the  $Na^+/K^+$ -ATPase, given the evidence for increased activity in erythrocytes and platelets in TPP patients.<sup>27,297</sup> Thyroid hormone clearly induces

expression of multiple subunits of the  $\text{Na}^+/\text{K}^+$ -ATPase in skeletal muscle.<sup>298</sup> Increases in  $\beta$ -adrenergic response due to hyperthyroidism also play an important role since high-dose propranolol (3 mg/kg) rapidly reverses the hypokalemia, hypophosphatemia, and paralysis seen in acute attacks.<sup>299,300</sup> Of particular importance, no rebound hyperkalemia is associated with this treatment, whereas aggressive  $\text{K}^+$  replacement in TPP is associated with an incidence of ~25%<sup>296</sup>; repletion-associated rebound hyperkalemia in TPP can be fatal.<sup>301</sup>

Outward-directed inward-rectifying  $\text{K}^+$  current, mediated by KIR channels (primarily Kir2.1 and Kir2.2 tetramers), is also reduced in skeletal muscles of patients with TPP,<sup>286</sup> providing an additional mechanism for hypokalemia. Together with increased  $\text{Na}^+/\text{K}^+$ -ATPase activity and increased circulating insulin, this reduced KIR current may trigger a “feed-forward” cycle of hypokalemia leading to inactivation of muscle  $\text{Na}^+$  channels, paradoxical depolarization, and paralysis.<sup>302</sup> The role of TPP-associated sequence variants in the *KCNJ18* gene that encodes Kir 2.6 is not entirely clear at this point, given that some genetic studies fail to find a convincing association.<sup>303</sup> As noted, the muscle Kir channel is largely formed by tetramers of Kir 2.1 with Kir 2.2 proteins, with robust expression of these channel proteins at the plasma membrane and T-tubules of skeletal muscle.<sup>304</sup> In contrast, wild-type Kir 2.6 protein appears to be restricted primarily to the ER, with a dominant negative effect on expression and function of Kir 2.1 and Kir 2.2 subunits.<sup>304</sup> It remains unclear how TPP-associated sequence variants, induced perhaps by high levels of thyroid hormone,<sup>28</sup> explain the proposed predisposition to TPP.

## NONRENAL POTASSIUM LOSS

The loss of  $\text{K}^+$  from skin is typically low, with the exception of extremes in physical exertion.<sup>14</sup> Direct gastric loss of  $\text{K}^+$  due to vomiting or nasogastric suctioning is also typically minimal, though the ensuing hypochloremic alkalosis results in persistent kaliuresis due to secondary hyperaldosteronism and bicarbonaturia.<sup>305,306</sup> Intestinal loss of  $\text{K}^+$  due to diarrhea is a quantitatively important cause of hypokalemia, given the worldwide prevalence of diarrheal disease, and may be associated with acute complications such as myopathy and flaccid paralysis.<sup>307</sup> The presence of a nonanion gap metabolic acidosis with a negative urinary anion gap<sup>179</sup> (consistent with an intact ability to increase  $\text{NH}_4^+$  excretion) should strongly suggest diarrhea as a cause of hypokalemia. Polyethylene glycol-based bowel preparation regimens for colonoscopy can also lead to hypokalemia in elderly patients<sup>308</sup> and/or patients on diuretics.<sup>309</sup> Noninfectious gastrointestinal processes such as celiac disease,<sup>310</sup> ileostomy, and chronic laxative abuse can present with acute hypokalemic syndromes or with chronic complications such as ESRD.<sup>49</sup>

Intestinal BK  $\text{K}^+$  channels play a major role in hypokalemia associated with intestinal disease. Three reports initially identified a novel association between colonic pseudoobstruction (Ogilvie syndrome) and hypokalemia due to secretory diarrhea with an abnormally high- $\text{K}^+$  content.<sup>311-313</sup> In one patient with concomitant ESRD, immunohistochemistry revealed massive upregulation of the apical BK channel throughout the surface-crypt

axes<sup>311</sup>; colonic BK channels may a significant role in intestinal  $\text{K}^+$  secretion in a variety of pathologies including ESRD.<sup>314</sup> Several hypotheses for the association between Ogilvie syndrome and enhanced intestinal  $\text{K}^+$  secretion have been postulated including active stimulation by catecholamines induced by colonic pseudoobstruction<sup>313</sup>; BK channels appear to mediate adrenaline-induced colonic  $\text{K}^+$  secretion.<sup>315</sup> Notably, aldosterone also induces colonic BK channel expression and activity<sup>316</sup>; consistent with a role for aldosterone, as a dramatic response to spironolactone in Ogilvie syndrome-associated diarrhea and hypokalemia has been reported.<sup>317</sup>

Increased fecal loss of  $\text{K}^+$  may play a broader role in hypokalemia associated with diarrhea.<sup>314</sup> Recruitment of colonic BK channels along intestinal crypts, similar to that seen in Ogilvie syndrome, has thus been demonstrated as a consistent feature of colonic biopsies in ulcerative colitis.<sup>318</sup> Direct enhancement of intestinal  $\text{K}^+$  excretion has also been demonstrated in a hypokalemic patient with Crohn disease, following treatment with budesonide.<sup>319</sup>

## RENAL POTASSIUM LOSS

### DRUGS

Diuretics are an especially important cause of hypokalemia, due to their ability to increase distal flow rate and distal delivery of  $\text{Na}^+$ . Thiazides generally cause more hypokalemia<sup>49,217,253</sup> than do loop diuretics, despite their lower natriuretic efficacy. One potential explanation is the differential effect of loop diuretics and thiazides on calcium excretion. Whereas thiazides and loss-of-function mutations in the  $\text{Na}^+\text{-Cl}^-$  cotransporter decrease  $\text{Ca}^{2+}$  excretion,<sup>320</sup> loop diuretics cause a significant calciuresis.<sup>321</sup> Increases in luminal  $\text{Ca}^{2+}$  in the distal nephron serve to reduce the lumen-negative driving force for  $\text{K}^+$  excretion,<sup>322</sup> perhaps by direct inhibition of ENaC in principal cells. A mechanistic explanation is provided by the presence of apical calcium-sensing receptor (CaSR) in the collecting duct<sup>323</sup>; analogous to the evident decrease in the apical trafficking of aquaporin-2 induced by luminal  $\text{Ca}^{2+}$ , tubular  $\text{Ca}^{2+}$  may stimulate endocytosis of ENaC via the CaSR and thus limit generation of the lumen-negative potential difference that is so critical for distal  $\text{K}^+$  excretion. Regardless of the underlying mechanism, the increase in distal delivery of  $\text{Ca}^{2+}$  induced by loop diuretics may serve to blunt kaliuresis; such a mechanism would not occur with thiazides, which reduce distal delivery of  $\text{Ca}^{2+}$ , with unopposed activity of ENaC and increased kaliuresis.

A substantial and growing body of evidence indicates a key role in  $\text{K}^+$  homeostasis for NCC, the thiazide-sensitive  $\text{Na}^+\text{-Cl}^-$  cotransporter in the DCT, such that it is not surprising that thiazide treatment has such potent effects on serum  $\text{K}^+$ . Selective increases in DCT and NCC activity, as seen in FHHt, reduce  $\text{Na}^+$  delivery to principal cells in the downstream CNT and CCD, leading to hyperkalemia.<sup>69</sup> The DCT also clearly functions as a “potassium sensor,” directly responding to changes in circulating potassium.<sup>4</sup> A high- $\text{K}^+$  diet also inactivates NCC, whereas NCC is activated in hypokalemia.<sup>137</sup>

Other drugs associated with hypokalemia due to kaliuresis include toxic levels of acetaminophen, which causes dose-dependent hypokalemia.<sup>324</sup> High doses of

penicillin-related antibiotics are another important cause of hypokalemia, increasing obligatory  $K^+$  excretion by acting as nonreabsorbable anions in the distal nephron; in addition to penicillin, implicated antibiotics include nafcillin, dicloxacillin, ticarcillin, oxacillin, and carbenicillin.<sup>325</sup> Increased distal delivery of other anions such as  $SO_4^{2-}$  and  $HCO_3^-$  also induces kaliuresis. The usual explanation is that  $K^+$  excretion increases so as to balance the negative charge of these nonreabsorbable anions. However, increased delivery of such anions will also increase the electrochemical gradient for  $K^+Cl^-$  exit via apical  $K^+Cl^-$  cotransport or parallel  $K^+H^+$  and  $Cl^-HCO_3^-$  exchange<sup>66,82,83</sup> (see also “Potassium Transport in the Distal Nephron”). Drugs are also an important cause of Fanconi syndrome,<sup>326</sup> which is often associated with significant hypokalemia (see “Renal Tubular Acidosis” later).

Several tubular toxins result in both  $K^+$  and magnesium wasting. These include gentamicin, which can cause tubular toxicity with hypokalemia that can masquerade as Bartter syndrome (BS).<sup>327</sup> Other drugs that can cause mixed magnesium and  $K^+$  wasting include amphotericin, foscarnet,<sup>328</sup> cisplatin,<sup>49,329</sup> and ifosfamide.<sup>330</sup> One intriguing cause of hypomagnesemia and hypokalemia is cetuximab, a humanized monoclonal antibody specific for the receptor for epidermal growth factor (EGF)<sup>331</sup>; paracrine EGF stimulates magnesium transport via the apical TRPM6 cation channel in the DCT, with magnesium wasting and hypomagnesemia in patients treated with cetuximab.<sup>332</sup> Aggressive replacement of magnesium is obligatory in the treatment of combined hypokalemia and hypomagnesemia since successful  $K^+$  replacement depends on treatment of the hypomagnesemia (see “Hypomagnesemia”).

A handful of drugs lead to hypokalemia through inhibition of  $11\beta$ -hydroxysteroid dehydrogenase-2 ( $11\beta$ HSD-2), resulting in unopposed mineralocorticoid activity of circulating cortisol. The classic  $11\beta$ HSD-2 inhibitor is licorice, through the active components glycyrrhetic/glycyrrhizinic acid and carbenoxolone. More recently, the antifungals itraconazole and posaconazole have been shown to inhibit  $11\beta$ HSD-2,<sup>333</sup> with several reports of hypertension and hypokalemia in patients treated with these agents<sup>334,335</sup> (see “Syndromes of Apparent Mineralocorticoid Excess” later).

Finally, the antiandrogen drug abiraterone, which is used to treat prostate cancer, can also cause apparent mineralocorticoid excess (AME) with hypokalemia and hypertension.<sup>336</sup> Abiraterone inhibits the CYP17A1 enzyme,  $17\alpha$ -hydroxylase; decreases cortisol; and increases plasma corticotropin (ACTH). Increased ACTH leads to the generation of excessive amounts of deoxycorticosterone, which has potent mineralocorticoid effects, causing hypokalemia and hypertension. This ACTH-dependent mineralocorticoid excess seen with abiraterone therapy can be minimized by coadministration of glucocorticoids, use of mineralocorticoid receptor (MR) antagonists, or salt restriction.<sup>337</sup>

## HYPERALDOSTERONISM

Increases in circulating aldosterone (hyperaldosteronism) may be primary or secondary. Increased levels of circulating renin in secondary forms of hyperaldosteronism lead to increased angiotensin-II (AT-II) and thus aldosterone and can be associated with hypokalemia; causes include renal

artery stenosis,<sup>338</sup> Page kidney (renal compression by a subcapsular mass or hematoma, with hyperreninemia),<sup>339</sup> a paraneoplastic process,<sup>340</sup> or renin-secreting renal tumors.<sup>341</sup> The incidence of hypokalemia in renal artery stenosis is thought to be <20%.<sup>338</sup> An unusual presentation of renal artery stenosis and renal ischemia is the “hyponatremic hypertensive syndrome,” in which concurrent hypokalemia may be profound.<sup>342</sup>

Primary hyperaldosteronism (PA) may have a hereditary, genetic cause. Hypertension and hypokalemia, generally attributed to increases in circulating  $11$ -deoxycorticosterone,<sup>343</sup> are seen in patients with congenital adrenal hyperplasia due to defects in either steroid  $11\beta$ -hydroxylase<sup>343</sup> or steroid  $17\alpha$ -hydroxylase<sup>344</sup>; deficient  $11\beta$ -hydroxylase results in virilization and other signs of androgen excess,<sup>343</sup> whereas reduced sex steroids in  $17\alpha$ -hydroxylase deficiency result in hypogonadism.<sup>344</sup>

The two major hereditary forms of isolated PA are denoted familial hyperaldosteronism type I (FH-I, also known as glucocorticoid-remediable hyperaldosteronism or GRA)<sup>345</sup> and familial hyperaldosteronism type II (FH-II), in which aldosterone production is not repressible by exogenous glucocorticoids. Patients with FH-II are clinically indistinguishable from sporadic forms of PA due to bilateral adrenal hyperplasia; a gene has been localized to chromosome 7p22 by linkage analysis but has yet to be characterized.<sup>346</sup> A third form of familial hyperaldosteronism (FH-III) was initially described in 2008, with hyporeninemia, hyperaldosteronism resistant to dexamethasone, and high levels of  $18$ -oxocortisol and  $18$ -hydroxycortisol.<sup>347</sup> FH-III is due to somatic mutations in the adrenal  $K^+$  channel KCNJ5, which endow the channel with a novel  $Na^+$  conductance and activate adrenal glomerulosa proliferation and aldosterone release<sup>2</sup>; somatic mutations in KCNJ5 are also found in spontaneous adrenal adenomas (see later). More recently, a fourth form of familial hyperaldosteronism (FH-IV) has been described; it is caused by autosomal dominant gain-of-function mutations in the *CACNA1H* gene, which encodes the CaV3.2 calcium channel.<sup>348,349</sup> The CaV3.2 calcium channel protein is expressed in human adrenal glomerulosa cells,<sup>348</sup> and the mutant channels demonstrate gain-of-function phenotypes, leading to increased induction of aldosterone synthase in cultured adrenal cells.<sup>349</sup> The clinical characteristics of patients with FH-IV have not been fully described, but the index patients developed PA in childhood and one subject had a multiplex developmental disorder.<sup>348,349</sup>

Patients with FH-I/GRA are generally hypertensive, typically presenting at an early age; the severity of hypertension is, however, variable, such that some affected individuals are normotensive.<sup>345</sup> Aldosterone levels are modestly elevated and regulated solely by ACTH. The diagnosis can be biochemically confirmed by a dexamethasone suppression test, with suppression of aldosterone to <4 ng/dL consistent with the diagnosis.<sup>350</sup> Patients also have high levels of abnormal “hybrid”  $18$ -hydroxylated steroids, generated by transformation of steroids typically formed in the zona fasciculata by aldosterone synthase, an enzyme that is normally expressed in the zona glomerulosa.<sup>351</sup> FH-I has been shown to be caused by a chimeric gene duplication between the homologous  $11\beta$ -hydroxylase (*CYP11B1*) and aldosterone synthase (*CYP11B2*) genes,

fusing the ACTH-responsive 11 $\beta$ -hydroxylase promoter to the coding region of aldosterone synthase; this chimeric gene is thus under the control of ACTH and expressed in a glucocorticoid-repressible fashion.<sup>351</sup> Ectopic expression of the hybrid *CYP11B1-CYP11B2* gene in the zona fasciculata has been reported in a single case where adrenal tissue became available for molecular analysis.<sup>352</sup> Direct genetic testing for the hybrid *CYP11B1-CYP11B2* has largely supplanted biochemical screening for FH-I; genetic testing for FH-I should be pursued in patients with PA and a family history of PA and/or strokes at a young age, or in young patients with PA (<20 years of age).<sup>353</sup> For reasons unclear, patients with FH-I have a high frequency of hemorrhagic stroke due to cerebral aneurysms, roughly equivalent to the frequency in autosomal polycystic kidney disease.<sup>354</sup>

Although the initial patients reported with FH-I were hypokalemic, the majority are, in fact, normokalemic,<sup>355,356</sup> albeit perhaps with a propensity to develop hypokalemia while on thiazide diuretics.<sup>356</sup> Patients with FH-I are able to appropriately increase K<sup>+</sup> excretion in response to K<sup>+</sup> loading or fludrocortisone but fail to increase plasma aldosterone in response to hyperkalemia.<sup>357</sup> This may reflect the ectopic expression of the chimeric aldosterone synthase in the adrenal fasciculata, which likely lacks the appropriate constellation of ion channels to respond to increases in extracellular K<sup>+</sup> with an increase in aldosterone secretion.

Acquired causes of PA classically include aldosterone-producing adenomas (APA, 35% of cases), primary or unilateral adrenal hyperplasia (PAH, 2% of cases), idiopathic hyperaldosteronism (IHA) due to bilateral adrenal hyperplasia (60% of cases), and adrenal carcinoma (<1% of cases).<sup>358</sup> A rare case involving paraneoplastic overexpression of aldosterone synthase in lymphoma has also been described.<sup>359</sup>

The molecular characterization of adrenal adenomas with whole-genome sequencing and related techniques has been remarkably fruitful. In particular, acquired mutations in the adrenal K<sup>+</sup> channel KCNJ5 can be detected in ~40% of aldosterone-producing adrenal adenomas.<sup>163</sup> As in FH-III (see earlier), these somatic mutations endow the channel with a novel Na<sup>+</sup> conductance, leading to adrenal glomerulosa cell depolarization, Ca<sup>2+</sup> influx, and aldosterone release. Clinically, patients with adrenal KCNJ5 mutations have higher preoperative aldosterone level<sup>163</sup> and higher lateralization index in adrenal vein sampling,<sup>360</sup> without affecting the surgical response to adrenalectomy.<sup>163</sup> Less frequently, somatic mutations in adrenal adenomas can be detected in the calcium channel CACNA1D<sup>361</sup> or in a subunit (ATP2B3) of the Ca<sup>2+</sup>-ATPase pump,<sup>362</sup> predicted to also lead to increased Ca<sup>2+</sup> influx and aldosterone release.<sup>361</sup> Acquired mutations in the ATP1A  $\alpha$ 1 subunit of the Na<sup>+</sup>/K<sup>+</sup>-ATPase are, in turn, thought to generate chronic depolarization, leading also to exaggerated aldosterone release.<sup>362</sup> A common consequence of the mutations in these “aldosterone-driver genes” is that they ultimately increase aldosterone production through increased expression of aldosterone synthase (*CYP11B2*).<sup>363</sup>

More recently, other pathways separate from aldosterone signaling have been implicated in the genesis of APAs. Mutations in the *CTNNB1* gene encoding  $\beta$ -catenin have also been described in APAs.<sup>364,365</sup> The mutations described appear to stabilize  $\beta$ -catenin, suggesting a role in WNT

signaling, which has previously been implicated in adrenal development and adrenal adenomas.<sup>364</sup>

Landmark studies with monoclonal antibodies to *CYP11B2* (aldosterone synthase) and *CYP11B1* (steroid 11 $\beta$ -hydroxylase, which catalyzes the terminal step in cortisol production) have recently provided a molecular definition of the zonation of human adrenal cortices.<sup>363,366</sup> Whereas *CYP11B1* expression is uniformly expressed throughout the zona fasciculata, there is only sporadic expression of *CYP11B2* in the zona glomerulosa of normal human adrenal glands.<sup>366</sup> However, a subcortical population of nonneoplastic clusters of cells can also be detected; given the coexpression of other steroidogenic enzymes that are required for aldosterone synthesis, these *CYP11B2*+ clusters have been designated “aldosterone-producing cell clusters” (APCCs).<sup>363,366,367</sup> APCCs have also been demonstrated in adrenal tissues adjacent to APAs, suggesting that APCCs are a precursor to APA.<sup>363,366</sup> Consistent with this hypothesis, mutations in the known “aldosterone-driver genes” can be seen in ~35% of APCCs in normal adrenal glands.<sup>367</sup>

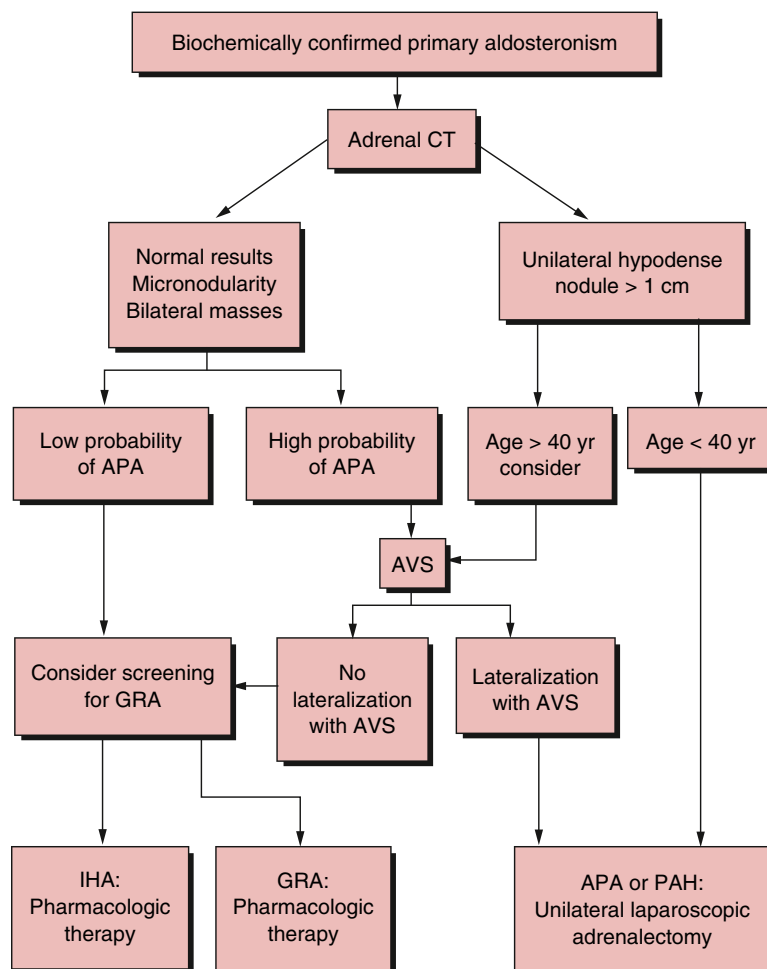
Immunochemical characterization of APAs also indicates expression of cortisol-producing enzymes *CYP17A1* and *CYP11B1*,<sup>366,368</sup> providing a molecular explanation for the hypercortisolism associated with hyperaldosteronism.<sup>368,369</sup> In hyperaldosteronism due to both BAH and APAs, there is significant urinary excretion of cortisol and other glucocorticoid metabolites, exceeded only in patients with clinically overt Cushing syndrome.<sup>369</sup> Glucocorticoid excretion in primary aldosteronism appears to correlate better with metabolic risk (type 2 diabetes, osteoporosis, metabolic syndrome) than does aldosterone excretion.<sup>369</sup> To the extent that hypercortisolism isn't affected by mineralocorticoid antagonists, these findings tend to underscore the preference for adrenalectomy in APAs. Increasing utilization of the plasma aldosterone (PAC)/plasma renin activity (PRA) ratio in hypertension clinics has led to reports of a much higher incidence of PA than previously appreciated, with incidence rates in hypertension ranging from zero to 72%<sup>370</sup>; however, the prevalence was 3.2% in a large, multicenter study of patients with mild to moderate hypertension *without* hypokalemia.<sup>371</sup> It should be noted that new studies have revealed that there is a continuum in renin-independent hyperaldosteronism, among normotensive individuals with suppressed renin and inappropriately “normal” or high aldosterone levels, individuals with unrecognized but biochemically overt PA, and individuals with severe hypertensive PA.<sup>372,373</sup> The presumed pathophysiology behind this continuum of increasing autonomous aldosterone secretion is the age-dependent development of APCCs, the evident precursor to APAs.<sup>363,366,367</sup> Regardless, the PAC:PRA ratio is a screening tool, which classically must be confirmed by aldosterone suppression testing, measuring PAC or aldosterone secretion after loading with salt or intravenous saline.<sup>374</sup> After controlling hypertension and hypokalemia (serum K<sup>+</sup>  $\geq$ 4.0 mEq/L),<sup>374</sup> oral salt loading over 3 days is followed by measurement of 24-hour urine aldosterone, sodium, and creatinine excretion; the 24-hour sodium excretion should exceed 200 mmol/day for adequate suppression, and a urinary aldosterone of >33 nmol/day (12 micrograms/day) is consistent with PA. Alternatively, in the saline infusion test, recumbent

patients are infused with 2 L of isotonic saline over 4 hours, followed by measurement of PAC. In patients without PA, the measured PAC after saline infusion should decrease to  $<139$  pmol/L (5 ng/dL). The measured PAC in patients with PA usually does not suppress to  $<277$  pmol/L (10 ng/dL); indeterminate values between 139 and 277 pmol/L (5–10 ng/dL) can be seen in patients with IHA.<sup>358</sup> Patients with high probability features (hypokalemia, hypertension, high PAC:PRA ratio, abnormalities) may not necessarily require confirmatory testing, proceeding instead directly to imaging and adrenal venous sampling.<sup>375</sup>

Since surgery can be curative in APA, adequate differentiation of APA from IHA is critical; this requires both adrenal imaging and adrenal venous sampling (see Fig. 16.10). Contemporary reports and recommendations have thus emphasized the continued importance of adrenal vein sampling in subtype differentiation.<sup>374</sup> Laparoscopic adrenalectomy is increasingly the preferred surgical management in APA or PAH.<sup>374</sup> Radiofrequency ablation appears to be an effective alternative in patients judged to be inappropriate for surgery.<sup>376</sup> MR antagonists are indicated for medical therapy of PA, with carefully monitored use

of glucocorticoid to suppress ACTH in some patients with FH-I/GRA.<sup>374</sup>

The true incidence of hypokalemia in patients with acquired forms of PA remains difficult to evaluate, due to a variety of factors. First, historically, patients have only been screened for hyperaldosteronism when hypokalemia is present, so even recent case series from clinics with such a referral pattern may suffer from a selection bias; other series have concentrated on hypertensive patients, also a selection bias. Second, the incidence of hypokalemia is higher in adrenal adenomas than in IHA, likely due to higher average levels of aldosterone.<sup>377</sup> Third, since increased kaliuresis in hyperaldosteronism can be induced by dietary  $\text{Na}^+\text{-Cl}^-$  loading or diuretics, dietary factors and/or medications may play a role in the incidence of hypokalemia at presentation. Fourth, as noted earlier, there is a continuum in renin-independent hyperaldosteronism, among normotensive individuals with suppressed renin and inappropriately “normal” or high aldosterone levels, individuals with unrecognized but biochemically overt PA, and individuals with severe hypertensive PA.<sup>372,373</sup> Regardless, it is clear that hypokalemia is not a universal feature of PA; this is perhaps



**Fig. 16.10 Diagnostic algorithm in patients with primary hyperaldosteronism.** Adrenal adenoma (APA) must be distinguished from glucocorticoid remediable hyperaldosteronism (FH-I or GRA), primary or unilateral adrenal hyperplasia (PAH), and idiopathic hyperaldosteronism (IHA). This requires computed axial tomography (CT), adrenal venous sampling (AVS), and the relevant diagnostic biochemical and hormonal assays (see text). (From Young WF Jr. Adrenalectomy for primary aldosteronism. *Ann Intern Med.* 2003;138(2):157–159.)

not unexpected, since aldosterone does not appear to affect the hypokalemic response of  $H^+/K^+$ -ATPase,<sup>378</sup> the major reabsorptive pathway for  $K^+$  in the distal nephron. A related issue is whether PA is underdiagnosed when hypokalemia is used as a criterion for further investigation.

Finally, hypokalemia may also occur with systemic increases in glucocorticoids.<sup>379,380</sup> In *bona fide* Cushing syndrome caused by increases in pituitary ACTH, the incidence of hypokalemia is only 10%,<sup>379</sup> whereas it is 57%<sup>380</sup> to 100%<sup>379</sup> in patients with ectopic ACTH, despite a similar incidence of hypertension. Ectopic ACTH expression is associated primarily with neuroendocrine malignancies, most commonly bronchial carcinoid tumors, small lung cancer, and other neuroendocrine tumors.<sup>381</sup> Indirect evidence suggests that the activity of renal  $11\beta$ -hydroxysteroid dehydrogenase-2 ( $11\beta$ HSD-2) is reduced in patients with ectopic ACTH compared with Cushing syndrome,<sup>382</sup> resulting in a syndrome of apparent mineralocorticoid excess (see later). Whether this reflects a greater degree of saturation of the enzyme by circulating cortisol or direct inhibition of  $11\beta$ HSD-2 by ACTH is not entirely clear, and there is evidence for both mechanisms<sup>380</sup>; however, indirect indices of  $11\beta$ HSD-2 activity in patients with ectopic ACTH expression correlate with hypokalemia and other measures of mineralocorticoid activity.<sup>383</sup> Similar mechanisms likely underlie the severe hypokalemia reported in patients with familial glucocorticoid resistance, in which loss-of-function mutations in the glucocorticoid receptor result in marked hypercortisolism without Cushingoid features, accompanied by high ACTH levels.<sup>384</sup>

### SYNDROMES OF APPARENT MINERALOCORTICOID EXCESS

The syndromes of “apparent mineralocorticoid excess” (AME) have a self-explanatory label. In the classic form of AME, recessive loss-of-function mutations in the  $11\beta$ -hydroxysteroid dehydrogenase-2 ( $11\beta$ HSD-2) gene cause a defect in the peripheral conversion of cortisol to the inactive glucocorticoid cortisone; the resulting increase in the half-life of cortisol is associated with a marked decrease in synthesis, such that plasma levels of cortisol are normal and patients are not Cushingoid.<sup>385</sup> The  $11\beta$ HSD-2 protein is expressed in epithelial cells that are targets for aldosterone; in the kidney, these include cells of the DCT, connecting segment (CNT), and CCD.<sup>386</sup> Since the MR has equivalent affinity for aldosterone and cortisol, generation of cortisone by  $11\beta$ HSD-2 serves to protect mineralocorticoid-responsive cells from illicit activation by cortisol.<sup>387</sup> In patients with AME, the unregulated mineralocorticoid effect of glucocorticoids results in hypertension, hypokalemia, and metabolic alkalosis, with suppressed PRA and aldosterone.<sup>385</sup> Biochemical diagnosis entails measuring the urinary-free cortisol to urinary-free cortisone ratio on a 24-hour urine collection. Biochemical studies of mutant enzymes usually indicate a complete loss of function; lesser enzymatic defects in patients with AME are associated with altered ratios of urinary cortisone/cortisol metabolites,<sup>388</sup> lesser impairment in the peripheral conversion of cortisol to cortisone,<sup>389</sup> and/or older age at presentation.<sup>390</sup>

Mice with a homozygous targeted deletion of  $11\beta$ HSD-2 exhibit hypertension, hypokalemia, and polyuria; the polyuria is likely secondary to the hypokalemia (see

“Renal Consequences of Hypokalemia”), which reaches 2.4 mmol/L in  $11\beta$ HSD-2-null mice.<sup>391</sup> As expected, both PRA and plasma aldosterone in the  $11\beta$ HSD-2-null mice are profoundly suppressed, with a decreased urinary  $Na^+/K^+$  ratio that is increased by dexamethasone (given to suppress endogenous cortisol). These knockout mice have significant nephromegaly due to massive hypertrophy and hyperplasia of DCTs. The relative effect of genotype on the morphology of cells in the DCT, CNT, and CCD was not determined by the appropriate phenotypic studies<sup>392</sup>; however, it is known that both the DCT and CCD are target cells for aldosterone<sup>393,394</sup> and both cell types express  $11\beta$ HSD-2. The induction of ENaC activity by unregulated glucocorticoid likely causes the  $Na^+$  retention and the marked increase in  $K^+$  excretion in  $11\beta$ HSD-2-null mice; distal tubular micropuncture studies in rats treated with a systemic inhibitor of  $11\beta$ HSD-2 are consistent with such a mechanism.<sup>395</sup> In addition, the cellular “gain-of-function” in the DCT would be expected to be associated with hypercalciuria, given the phenotype of pseudohypoaldosteronism type II and Gitelman syndrome (see “Hereditary Tubular Causes of Hyperkalemia” and “Gitelman Syndrome” later); indeed, patients with AME are reported to exhibit nephrocalcinosis.<sup>385</sup>

Pharmacologic inhibition of  $11\beta$ HSD-2 is also associated with hypokalemia and AME. The most infamous offender is licorice, in its multiple guises (licorice root, tea, candies, herbal remedies, etc.). The early observations that licorice required small amounts of cortisol to exert its kaliuretic effect, in the Addisonian absence of endogenous glucocorticoid,<sup>396</sup> presaged the observations that its active ingredients (glycyrrhentic/glycyrrhizinic acid and carbenoxolone) inhibit  $11\beta$ HSD-2 and related enzymes.<sup>385</sup> Licorice intake remains considerable in European countries, particularly Iceland, Netherlands, and Scandinavia<sup>397</sup>; Pontefract cakes, eaten both as sweets and as a laxative, are a continued source of licorice in the United Kingdom,<sup>397</sup> whereas it is an ingredient in several popular sweeteners and preservatives in Malaysia.<sup>398</sup> Glycyrrhizinic acid is also a component of Chinese herbal remedies, prescribed for disorders such as for allergic rhinitis.<sup>399</sup> Pharmacologic inhibition of  $11\beta$ HSD-2 has also been tested in patients with ESRD, as a novel mechanism to control hyperkalemia (see “Management of Hyperkalemia”).<sup>400</sup> Carbenoxolone is in turn used in some countries in the management of peptic ulcer disease.<sup>385</sup> More recently, the antifungals itraconazole and posaconazole have been shown to inhibit  $11\beta$ HSD-2,<sup>333</sup> with several reports of hypertension and hypokalemia in patients treated with these agents.<sup>334,335</sup>

Finally, a mechanistically distinct form of AME has been reported, due to a gain-of-function mutation in the MR.<sup>401</sup> A single kindred was thus described with autosomal dominant inheritance of severe hypertension and hypokalemia; the causative mutation involves a serine residue that is conserved in the MR from multiple species yet differs in other nuclear steroid receptors. This mutation results in constitutive activation of the MR in the absence of ligand and induces significant affinity for progesterone.<sup>401</sup> The MR is thus constitutively “on” in these patients, with a marked stimulation by progesterone; of interest, pregnancies in the affected female members of the family have all

been complicated by severe hypertension, due to marked increases in plasma progesterone induced by the gravid state.<sup>401</sup>

### LIDDLE SYNDROME

Liddle syndrome constitutes an autosomal dominant gain-in-function of ENaC, the amiloride-sensitive Na<sup>+</sup> channel of the CNT and CCD.<sup>402</sup> Patients manifest severe hypertension with hypokalemia, unresponsive to spironolactone yet sensitive to triamterene and amiloride. Liddle syndrome could therefore also be classified as a syndrome of AME. Both hypertension and hypokalemia are variable aspects of the Liddle phenotype; consistent features include a blunted aldosterone response to ACTH and reduced urinary aldosterone excretion.<sup>98,403</sup> The differential diagnosis for Liddle syndrome, as a cause of hereditary hypertension with hypokalemia and suppressed aldosterone, includes AME due to deficient 11 $\beta$ HSD-2; however, whereas the Liddle syndrome phenotype is resistant to blockade of the MR with spironolactone and sensitive to amiloride, AME patients are sensitive to both drugs. Commercial genetic testing for both syndromes is available in the United States.

The vast majority of mutations target the C-terminus of either the  $\beta$ - or  $\gamma$ -ENaC subunit, with a small minority of mutations in  $\alpha$ -ENaC. ENaC channels containing Liddle syndrome mutations are constitutively overexpressed at the cell membrane<sup>99,404</sup>; unlike wild-type ENaC channels, they are not sensitive to inhibition by intracellular Na<sup>+</sup><sup>405</sup>, an important regulator of endogenous channel activity in the CCD.<sup>406</sup> The mechanism whereby mutations in the C-terminus of ENaC subunits lead to this channel phenotype are discussed earlier in this chapter (see Fig. 16.6 and “Control of Potassium Secretion: Aldosterone” earlier). In addition to effects on interaction with Nedd4-2-dependent retrieval from the plasma membrane, Liddle-associated mutations increase proteolytic cleavage of ENaC at the cell membrane<sup>407</sup>; aldosterone-induced “channel-activating proteases” activate ENaC channels at the plasma membrane. This important result provides a mechanistic explanation for the longstanding observation that Liddle-associated mutations in ENaC appear to have a dual activating effect, on both the open probability of the channel (i.e., on channel activity) and expression at the cell membrane.<sup>99</sup>

Given the overlapping and synergistic mechanisms that regulate ENaC activity, it stands to reason that mutations in ENaC that give rise to Liddle syndrome might do so by a variety of means. Indeed, mutation of a residue within the extracellular domain of  $\gamma$ -ENaC, versus the usual C-terminal site of mutations in Liddle syndrome, increases open probability of the channel without changing surface expression; the patient with this mutation has a typical Liddle syndrome phenotype.<sup>408</sup> More recently, a missense mutation in the extracellular domain of  $\alpha$ -ENaC has also been described in a Liddle kindred, increasing open probability of the channel without increasing susceptibility to activation by proteases.<sup>409</sup> Extensive searches for more common mutations and polymorphisms in ENaC subunits that correlate with blood pressure in the general population have essentially been negative, despite characterization

of specific gain-of-function variants.<sup>410</sup> However, there are a handful of genetic studies that correlate specific variants in ENaC subunits with biochemical evidence of greater in vivo activity of the channel (i.e., a suppressed PRA and aldosterone and/or increased ratios of urinary K<sup>+</sup>: aldosterone/PRA).<sup>411,412</sup>

### FAMILIAL HYPOKALEMIC ALKALOSIS: BARTTER SYNDROME

Bartter and Gitelman syndromes are the two major variants of familial hypokalemic alkalosis; Gitelman syndrome is a much more common cause of hypokalemia than is BS.<sup>413</sup> Whereas a clinical subdivision of these syndromes has been used in the past, a genetic classification is increasingly in use, due in part to phenotypic overlap. Patients with “classic” BS typically suffer from polyuria and polydipsia and manifest a hypokalemic, hypochloremic alkalosis. They may have an increase in urinary calcium (Ca<sup>2+</sup>) excretion, and 20% are hypomagnesemic.<sup>414</sup> Other features include marked elevation of plasma angiotensin-II, plasma aldosterone, and plasma renin. Patients with “antenatal” BS present earlier in life with a severe systemic disorder characterized by marked electrolyte wasting, polyhydramnios, and significant hypercalciuria with nephrocalcinosis. Prostaglandin synthesis and excretion are significantly increased and may account for much of the systemic symptoms. Decreasing prostaglandin synthesis by cyclooxygenase inhibition can improve polyuria in patients with BS, by reducing the amplifying inhibition of urinary concentrating mechanisms by prostaglandins. Indomethacin also increases plasma K<sup>+</sup> and decreases PRA but does not correct the basic tubular defect; it does, however, appear to help increase the growth of BS patients.<sup>415</sup> Of interest, COX-2 immunoreactivity is increased in the TAL and macula densa of patients with BS,<sup>416</sup> and reports indicate a clinical benefit of COX-2 inhibitors.<sup>417</sup>

Early studies in BS suggested that these patients had a defect in the function of the TAL.<sup>418</sup> Many of the clinical features are mimicked by the administration of loop diuretics, to which at least a subset of patients with antenatal BS do not respond.<sup>419</sup> The apical Na<sup>+</sup>-K<sup>+</sup>-2Cl<sup>-</sup> cotransporter (NKCC2, SLC12A1) of the mammalian TAL<sup>25</sup> (see Fig. 16.2) was thus an early candidate gene. In 1996, disease-associated mutations were found in the human *NKCC2* gene in four kindreds with antenatal BS<sup>420</sup>; in the genetic classification of BS, these patients are considered to have BS type I. Although the functional consequences of disease-associated *NKCC2* mutations have not been comprehensively studied, the first<sup>420</sup> and subsequent reports<sup>27</sup> include patients with frameshift mutations and premature stop codons that predict the absence of a functional NKCC2 protein.

BS is a genetically heterogeneous disease. Given the role of apical K<sup>+</sup> permeability in the TAL, encoded at least in part by ROMK,<sup>77,421</sup> this K<sup>+</sup> channel was another early candidate gene. K<sup>+</sup> recycling via the Na<sup>+</sup>-K<sup>+</sup>-2Cl<sup>-</sup> cotransporter and apical K<sup>+</sup> channels generates a lumen-positive potential difference in the TAL, which drives the paracellular transport of Na<sup>+</sup> and other cations<sup>422</sup> (see also Fig. 16.2). Multiple disease-associated mutations in ROMK have been reported in patients with BS type II, most of whom exhibit the antenatal phenotype.<sup>415,423</sup> Finally, mutations in BS type III have been reported in the

chloride channel *CLC-NKB*,<sup>424</sup> which is expressed at the basolateral membrane of at least the TAL and DCT.<sup>425</sup> In a significant fraction of patients with BS the *NKCC2*, *ROMK*, and *CLC-NKB* genes are not involved.<sup>424</sup> For example, a subset of patients with associated sensorineural deafness exhibit linkage to chromosome 1p31<sup>27</sup>; the gene for this syndrome, denoted Barttin, is an obligatory subunit for the *CLC-NKB* chloride channel.<sup>426</sup> The occurrence of deafness in these patients suggests that Barttin functions in the regulation or function of Cl<sup>-</sup> channels in the inner ear. Notably, the *CLC-NKB* gene is immediately adjacent that for another epithelial Cl<sup>-</sup> channel, denoted *CLC-NKA*; digenic inactivation was described in two siblings with deafness and BS,<sup>427</sup> suggesting that *CLC-NKA* plays an important role in Barttin-dependent Cl<sup>-</sup> transport in the inner ear.

Patients with activating mutations in the calcium-sensing receptor (CaSR) have been described with autosomal dominant hypocalcemia and hypokalemic alkalosis.<sup>428,429</sup> The CaSR is heavily expressed at the basolateral membrane of the TAL,<sup>430</sup> where it is thought to play an important inhibitory role in regulating the transcellular transport of both Na<sup>+</sup>-Cl<sup>-</sup> and Ca<sup>2+</sup>. For example, activation of the basolateral CaSR in the TAL is known to reduce apical K<sup>+</sup> channel activity,<sup>431</sup> which would induce a Bartter-like syndrome (see Fig. 16.2). Coexpression of *NKCC2* with a BS gain-of-function mutant in the CaSR reveals reduced phosphorylation and reduced activity of *NKCC2*, dependent on the generation of inhibitory arachidonic acid-derived metabolites known to inhibit TAL function.<sup>432</sup> Genetic activation of the CaSR by these mutations is also expected to increase urinary Ca<sup>2+</sup> excretion, by inhibiting generation of the lumen-positive potential difference that drives paracellular Ca<sup>2+</sup> transport in the TAL. In addition, the “set-point” of the CaSR response to Ca<sup>2+</sup> in the parathyroid is shifted to the left, inhibiting PTH secretion by this gland. No doubt the positional cloning of other BS genes will have a considerable impact on mechanistic understanding of the TAL.

Despite the reasonable correlation between the disease gene involved and the associated subtype of familial alkalosis, there is significant phenotypic overlap and phenotypic variability in hereditary hypokalemic alkalosis. For example, patients with mutations in *CLC-NKB* most frequently (44.5%) exhibit classic BS but can present with a more severe antenatal phenotype (29.5%), or even with a phenotype similar to Gitelman syndrome (26.0%).<sup>27,433,434</sup> With respect to BS due to mutations in *NKCC2*, a number of patients have been described with variant presentations, including an absence of hypokalemia.<sup>27</sup> Two brothers were described with a late onset of mild BS; these patients were found to be compound heterozygotes for a mutant form of *NKCC2* that exhibits partial function, with a loss-of-function mutation on the other *NKCC2* allele.<sup>435</sup> Further genetic heterogeneity has also emerged, with the description of patients with hypokalemic alkalosis due to mutations in claudin 10, which presumably disrupt paracellular cation transport by the TAL.<sup>436</sup> Notably, the genetic heterogeneity in BS, GS, and related disorders is arguably not a major clinical issue in contemporary clinical practice; most commercial genetic analysis of patients with hereditary,

hypokalemic alkalosis now involves testing an extensive panel of relevant genes, rather than testing single genes.

BS type II is particularly relevant to K<sup>+</sup> homeostasis, given that ROMK is the SK secretory channel of the CNT and CCD (see “Potassium Secretion in the Distal Nephron”). Patients with BS type II typically have slightly higher serum K<sup>+</sup> than the other genetic forms of BS<sup>423,433</sup>; patients with severe (9.0 mmol/L), transient, neonatal hyperkalemia have also been described.<sup>437</sup> It is likely that this reflects a transient, developmental deficit in the other K<sup>+</sup> channels involved in distal K<sup>+</sup> secretion, including the apical maxi-K channel responsible for flow-dependent K<sup>+</sup> secretion in the distal nephron.<sup>76,438</sup> Distal K<sup>+</sup> secretion in ROMK knockout mice is primarily mediated by maxi-K/BK channel activity,<sup>439</sup> such that developmental deficits in this channel would indeed lead to hyperkalemia in BS type II. The mammalian TAL has two major apical K<sup>+</sup> conductances, the 30 pS channel corresponding to ROMK and a 70 pS channel<sup>440</sup>; both are thought to play a role in transepithelial salt transport by the TAL. ROMK is evidently a subunit of the 70 pS channel, given the absence of this conductance in TAL segments of ROMK knockout mice.<sup>441</sup> More recently, a distinctive form of BS has emerged due to X-linked mutations in the *MAGED2* gene. These kindreds are characterized by polyhydramnios with prematurity and a severe but transient form of antenatal BS. *MAGE-D2* affects the expression and function of *NKCC2* and *NCC*.<sup>442</sup>

Finally, BS must be clinically differentiated from various causes of “pseudo-Bartter” syndrome; these commonly include laxative abuse, furosemide abuse, and bulimia (see “The Clinical Approach to Hypokalemia” later). Other reported causes include gentamicin nephrotoxicity,<sup>327</sup> Sjögren syndrome,<sup>27</sup> and cystic fibrosis (CF).<sup>27,443</sup> Fixed loss of Na<sup>+</sup>-Cl<sup>-</sup> in sweat is likely the dominant predisposing factor for hypokalemic alkalosis in patients with CF; patients with this presentation generally respond promptly to intravenous fluids and electrolyte replacement. However, the CFTR protein coassociates with ROMK in the TAL and confers sensitivity to both ATP and glybenclamide to apical K<sup>+</sup> channels in this nephron segment.<sup>444</sup> Lu and colleagues<sup>444</sup> have proposed that this interaction serves to modulate the response of ROMK to cAMP and vasopressin, such that K<sup>+</sup> excretion in CFTR deficiency would not be appropriately reduced during water diuresis, thus predisposing such patients to the development of hypokalemic alkalosis.

## FAMILIAL HYPOKALEMIC ALKALOSIS: GITELMAN SYNDROME

A major advance in the understanding of hereditary alkaloses was the realization that a subset of patients exhibits marked hypocalciuria, rather than the hypercalciuria typically seen in BS; patients in this hypocalciuric subset are universally hypomagnesemic.<sup>320</sup> Such patients are now clinically classified as suffering from Gitelman syndrome.<sup>434</sup> Although PRA may be increased, renal prostaglandin excretion is not elevated in these hypocalciuric patients,<sup>445</sup> another distinguishing feature between Bartter and Gitelman syndromes. Gitelman syndrome is a milder disorder than BS; however, patients do report significant morbidity, mostly related to muscular symptoms and fatigue.<sup>434,446</sup> The QT interval is frequently prolonged in Gitelman syndrome, suggesting an increased risk of cardiac arrhythmia<sup>447</sup>;

however, a more exhaustive cardiac evaluation of a large group of patients failed to detect significant abnormalities of cardiac structure or rhythm.<sup>448</sup> Presyncope and/or ventricular tachycardia has been observed in at least two patients with Gitelman syndrome,<sup>27,186</sup> one with concomitant LQTS due to a mutation in the cardiac KCNQ1 K<sup>+</sup> channel.<sup>186</sup>

The hypocalciuria detected in Gitelman syndrome is an expected consequence of inactivating the thiazide-sensitive Na<sup>+</sup>-Cl<sup>-</sup> cotransporter NCC (SLC12A2), and loss-of-function mutations in the human gene have been reported<sup>449</sup>; many of these mutations lead to a defect in cellular trafficking when introduced into the human NCC protein. Gitelman syndrome is genetically homogeneous, except for patients with mutations in CLC-NKB and an overlapping phenotype.<sup>27,186,433,434</sup> More recently, loss-of-function mutations in the *KCNJ10* encoding the basolateral Kir 4.1 K<sup>+</sup> channel in the DCT have been implicated in a related syndrome of hypokalemic metabolic alkalosis with hypomagnesemia, accompanied by seizures, sensorineural deafness, ataxia, and mental retardation (SeSAME or EAST syndrome).<sup>30,450</sup> In Kir 4.1-deficient, *KCNJ10* knockout mice, the loss of basolateral K<sup>+</sup> conductance reduces basolateral Cl<sup>-</sup> conductance, leading to reduced expression of SPAK kinase and reduced apical NCC expression.<sup>141</sup> Notably, the genetic heterogeneity in Gitelman syndrome and related disorders is arguably not a major clinical issue in contemporary clinical practice; most commercial genetic analysis of patients with hereditary, hypokalemic alkalosis now involves testing an extensive panel of relevant genes, rather than testing for mutations in single candidate genes.

The NCC protein has been localized to the apical membrane of epithelial cells in the DCT and connecting segment. A mouse strain with targeted deletion of the *Slc12a2* gene encoding NCC exhibits hypocalciuria and hypomagnesemia, with a mild alkalosis and marked increase in circulating aldosterone.<sup>451</sup> These knockout mice exhibit marked morphologic defects in the early DCT,<sup>451</sup> with both a reduction in absolute number of DCT cells and changes in ultrastructural appearance. That Gitelman syndrome is a disorder of cellular development and/or cellular apoptosis should perhaps not be a surprise, given the observation that thiazide treatment promotes marked apoptosis of this nephron segment.<sup>452</sup> This cellular deficit leads to downregulation of the DCT magnesium channel TRPM6,<sup>453</sup> resulting in the magnesium wasting and hypomagnesemia seen in Gitelman syndrome. The downstream CNT tubules are hypertrophied in NCC-deficient mice,<sup>451</sup> reminiscent of the hypertrophic DCT and CNT segments seen in furosemide-treated animals.<sup>27</sup> These CNT cells also exhibit an increased expression of ENaC at their apical membranes versus littermate controls<sup>451</sup>; this is likely due to activation of SGK1-dependent trafficking of ENaC by the increase in circulating aldosterone (see “Control of Potassium Secretion: Aldosterone” earlier).

Hypokalemia does not occur in NCC null mice on standard rodent diet, but emerges on a K<sup>+</sup>-restricted diet; plasma K<sup>+</sup> of these mice is ~1 mM lower than K<sup>+</sup>-restricted littermate controls.<sup>454</sup> Several mechanisms account for the hypokalemia seen in GS and NCC <sup>-/-</sup> mice. The distal delivery of both Na<sup>+</sup> and fluid is decreased in NCC null mice, at least on a normal diet; however, the increased

circulating aldosterone and CNT hypertrophy likely compensate, leading to increased kaliuresis. As discussed earlier for thiazides, decreased luminal Ca<sup>2+</sup> in NCC-deficiency may augment baseline ENaC activity,<sup>322</sup> further exacerbating the kaliuresis. Of particular interest, NCC-deficient mice develop considerable polydipsia and polyuria on a K<sup>+</sup>-restricted diet<sup>454</sup>; this is reminiscent perhaps of the polydipsia that has been implicated in thiazide-associated hyponatremia.<sup>454</sup>

Hypocalciuria in Gitelman syndrome is not accompanied by changes in plasma calcium, phosphate, vitamin D, or PTH, suggesting a direct effect on renal calcium transport. The late DCT is morphologically intact in NCC-deficient mice, with preserved expression of the epithelial calcium channel (TRPV5) and the basolateral Na<sup>+</sup>-Ca<sup>2+</sup> exchanger.<sup>451</sup> Furthermore, the hypocalciuric effect of thiazides persists in mice deficient in TRPV5,<sup>453</sup> arguing against the putative effects of this drug on distal Ca<sup>2+</sup> absorption. Rather, several lines of evidence argue that the hypocalciuria of Gitelman syndrome and thiazide treatment is due to increased absorption of Na<sup>+</sup> by the proximal tubule,<sup>451,453</sup> with secondary increases in proximal Ca<sup>2+</sup> absorption. An interesting association has repeatedly been described between chondrocalcinosis, the abnormal deposition of calcium pyrophosphate dihydrate (CPPD) in joint cartilage, and Gitelman syndrome.<sup>456</sup> Patients have also been reported with ocular choroidal calcification.<sup>457</sup>

Treatment of Gitelman syndrome encompasses liberalization of salt intake and lifelong supplementation with oral magnesium and potassium. Reasonable targets for serum potassium and serum magnesium are 3.0 mEq/L and 0.6 mmol/L (1.46 mg/dL).<sup>458</sup> Organic anion salts of magnesium are preferred over magnesium chloride/hydroxide/oxide. Patients with refractory hypokalemia may require treatment with potassium-sparing diuretics, RAAS inhibitors, and/or nonsteroidal inflammatory drugs.<sup>458</sup>

Finally, as in BS, there are reports of acquired tubular defects that mimic Gitelman syndrome. These include patients with hypokalemic alkalosis, hypomagnesemia, and hypocalciuria after chemotherapy with cisplatin.<sup>459</sup> Patients have also been described with acquired Gitelman syndrome due to Sjögren syndrome and tubulointerstitial nephritis,<sup>7,199</sup> with a documented absence of coding sequence mutations in NCC.<sup>199</sup>

## RENAL TUBULAR ACIDOSIS

Renaltubularacidosis (RTA) and related tubular defects can be associated with hypokalemia. Proximal RTA is characterized by a reduction in proximal bicarbonate absorption, with a reduced plasma bicarbonate concentration. Isolated proximal RTA is quite rare; genetic causes include loss-of-function by mutations in the basolateral Na<sup>+</sup>-bicarbonate transporter. More commonly, proximal RTA occurs in the context of multiple proximal tubular transport defects, encompassing the Fanconi syndrome (FS).<sup>236</sup> The cardinal features of FS include hyperaminoaciduria, glycosuria with a normal plasma glucose concentration, and phosphate wasting; associated defects include the proximal RTA, hypouricemia, hypercalciuria, hypokalemia, salt wasting, and increased excretion of low-molecular-weight proteins. FS is most commonly drug associated; important contemporary

causes include aristolochic acid, ifosfamide, and the acyclic nucleoside phosphonates (tenofovir, cidofovir, and adefovir).<sup>236</sup> Before treatment with bicarbonate, patients with a proximal RTA will typically demonstrate mild hypokalemia, due primarily to baseline hyperaldosteronism<sup>460</sup>; patients have, however, been described with profound hypokalemia on presentation, before treatment.<sup>461</sup> Regardless, treatment with oral sodium bicarbonate will markedly increase distal tubular  $\text{Na}^+$  and  $\text{HCO}_3^-$  delivery, causing a marked increase in renal potassium wasting.<sup>462</sup> Patients will often require mixed base replacement with oral citrate and bicarbonate, in addition to aggressive  $\text{K}^+\text{-Cl}^-$  supplementation.

Hypokalemia is also associated with distal RTA, the so-called “type I” RTA. Hypokalemic distal RTA is most commonly due to a “secretory” defect, with reduced  $\text{H}^+\text{-ATPase}$  activity and decreased ability to acidify the urine. For example, hereditary defects in subunits of the  $\text{H}^+\text{-ATPase}$  are associated with profound hypokalemia, in addition to acidosis and hypercalciuria.<sup>463</sup> Pathophysiology of the associated hypokalemia is multifactorial, due to the loss of electrogenic  $\text{H}^+$  secretion (with enhanced  $\text{K}^+$  secretion to maintain electroneutrality in the distal nephron), loss of  $\text{H}^+\text{-ATPase}$  activity, and increases in aldosterone.<sup>462,464</sup> Notably, in hereditary distal RTA, hypokalemia is more common in those with  $\text{H}^+\text{-ATPase}$  defects than in patients with mutations in the basolateral  $\text{Cl}^-\text{-HCO}_3^-$  exchanger.<sup>465</sup> Sjögren syndrome is perhaps the most common cause of hypokalemic distal RTA in adults; the associated hypokalemia can be truly profound, often resulting in marked weakness and respiratory arrest.<sup>464</sup>

## MAGNESIUM DEFICIENCY

Magnesium deficiency results in refractory hypokalemia, particularly if the plasma  $\text{Mg}^{2+}$  is  $<0.5$  mmol/L<sup>248</sup>; hypomagnesemic patients are thus refractory to  $\text{K}^+$  replacement in the absence of  $\text{Mg}^{2+}$  repletion.<sup>466,467</sup> Magnesium deficiency is also a common concomitant of hypokalemia, in part because associated tubular disorders (e.g., aminoglycoside nephrotoxicity) may cause both kaliuresis and magnesium wasting. Plasma  $\text{Mg}^{2+}$  levels must thus be checked on a routine basis in hypokalemia.<sup>247,468</sup>

Several mechanisms appear to contribute to the effect of magnesium depletion on plasma  $\text{K}^+$ . Magnesium depletion has inhibitory effects on muscle  $\text{Na}^+\text{-ATPase}$  activity,<sup>469</sup> resulting in significant efflux from muscle and a secondary kaliuresis. Distal  $\text{K}^+$  secretion also appears to be enhanced, due to a reduction in the normal physiologic inward rectification of ROMK secretory  $\text{K}^+$  channels, with a subsequent increase in outward conductance.<sup>470</sup> ROMK and other KIR channels are inward-rectifying (i.e.,  $\text{K}^+$  flows inward more readily than outward); even though outward conductance is usually less than inward conductance,  $\text{K}^+$  efflux predominates in the CNT and CCD since the membrane potential is more positive than the equilibrium potential for  $\text{K}^+$ . Intracellular  $\text{Mg}^{2+}$  plays a key role in inward rectification, binding, and blocking the pore of the channel from the cytoplasmic side.<sup>470</sup> The hypomagnesemia-associated reduction in cytoplasmic  $\text{Mg}^{2+}$  in principal cells reduces inward rectification of ROMK, increasing outward conductance and increasing  $\text{K}^+$  secretion; this has been confirmed in vivo.<sup>471</sup> Finally, it has been suggested that the repletion of intracellular  $\text{K}^+$  is impaired

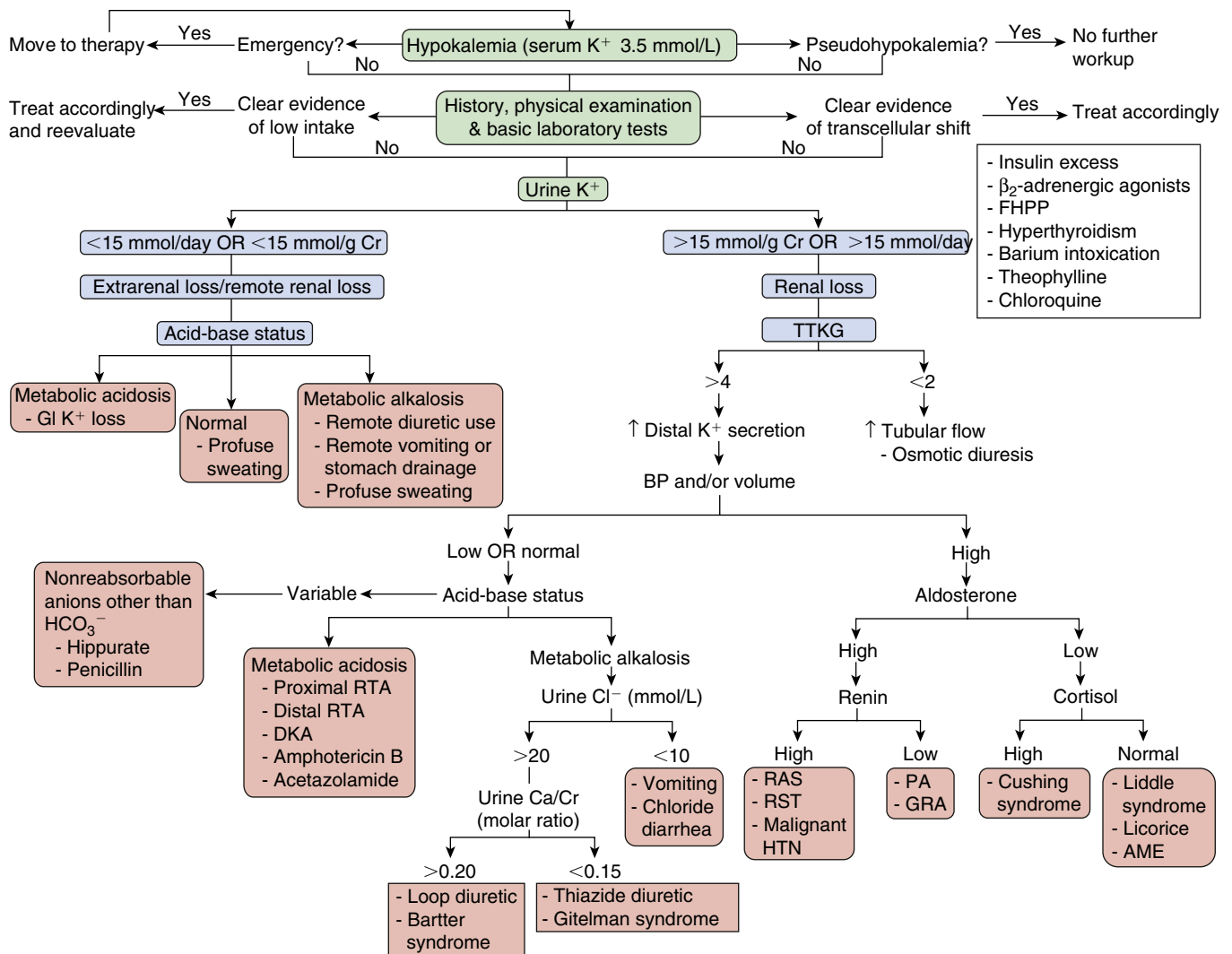
in hypomagnesemia, even in normokalemic patients.<sup>468</sup> Decreased intracellular  $\text{Mg}^{2+}$  enhances  $\text{K}^+$  efflux from the cytoplasm of cardiac and perhaps skeletal myocytes, likely due to reduced intracellular blockade of inward-rectifying  $\text{K}^+$  channels (increased efflux) and inhibition of the  $\text{Na}^+\text{-ATPase}$  (decreased influx); plasma  $\text{K}^+$  levels thus remain normal at the expense of intracellular  $\text{K}^+$ .<sup>49,468,472</sup> This phenomenon is particularly important in patients with cardiac disease taking both diuretics and digoxin. In such patients, hypokalemia and arrhythmias will respond to correction of magnesium deficiency and potassium supplementation.<sup>49,468</sup>

## THE CLINICAL APPROACH TO HYPOKALEMIA

The initial priority in the evaluation of hypokalemia is an assessment for signs and/or symptoms (muscle weakness, ECG changes, etc.) suggestive of an impending emergency that requires immediate treatment. The cause of hypokalemia is usually obvious from history, physical examination, and/or basic laboratory tests. However, persistent hypokalemia despite appropriate initial intervention requires a more rigorous workup; in most cases, a systematic approach reveals the underlying cause (Fig. 16.11).

The history should focus on medications (e.g., diuretics, laxatives, antibiotics, and herbal medications), diet and dietary supplements (e.g., licorice), and associated symptoms (e.g., diarrhea). During the physical examination, particular attention should be paid to blood pressure, volume status, and signs suggestive of specific disorders associated with hypokalemia (hyperthyroidism, Cushing syndrome, etc.). Initial laboratory tests should include electrolytes, blood urea nitrogen, creatinine, plasma osmolality,  $\text{Mg}^{2+}$ , and  $\text{Ca}^{2+}$ ; a complete blood count; and urinary pH, osmolality, creatinine, and electrolytes. Plasma and urine osmolality and urine creatinine are required for calculation of the transtubular  $\text{K}^+$  gradient (TTKG)<sup>172</sup> and urinary  $\text{K}^+:\text{creatinine}$  ratio (see “Urinary Indices of Potassium Excretion” earlier). A TTKG of  $<2$  to  $3$  separates patients with redistributive or extrarenal hypokalemia from those with hypokalemia due to renal potassium wasting, who will have TTKG values that are  $>4$ .<sup>178</sup> Further tests such as urinary  $\text{Mg}^{2+}$  and  $\text{Ca}^{2+}$  and plasma renin and aldosterone levels may be necessary in specific cases (see Fig. 16.11). The timing and evolution of hypokalemia are also helpful in differentiating the cause, particularly in hospitalized patients; for example, hypokalemia due to transcellular shift usually occurs in a matter of hours.<sup>473</sup>

The most common causes of chronic, diagnosis-resistant hypokalemia are Gitelman syndrome, surreptitious vomiting, and diuretic abuse.<sup>474</sup> Alternatively, an associated acidosis would suggest the diagnosis of hypokalemic distal or proximal RTA. Hypokalemia occurred in 5.5% of patients with eating disorders in a U.S. study from the mid-1990s,<sup>475</sup> mostly in those with surreptitious vomiting (bulimia) or laxative abuse (the purging<sup>306</sup> subtype of anorexia nervosa). These patients may have a constellation of associated symptoms and signs, including dental erosion and depression.<sup>476</sup> Hypokalemic patients with bulimia will have an associated metabolic alkalosis, with an obligatory natriuresis accompanying the loss of bicarbonate; urinary  $\text{Cl}^-$  is typically  $<10$  mmol/L and this clue can often yield the diagnosis.<sup>474,477</sup> Urinary electrolytes are,



**Fig. 16.11 The clinical approach to hypokalemia.** See text for details. AME, Apparent mineralocorticoid excess; BP, blood pressure; CCD, cortical collecting duct; DKA, diabetic ketoacidosis; FHPP, familial hypokalemic periodic paralysis; GI, gastrointestinal; GRA, glucocorticoid-remediable aldosteronism; HTN, hypertension; PA, primary aldosteronism; RAS, renal artery stenosis; RST, renin-secreting tumor; RTA, renal tubular acidosis; TTKG, transtubular potassium gradient.

however, generally unremarkable in unselected, mostly normokalemic patients with bulimia.<sup>476</sup> Urinary excretion of  $\text{Na}^+$ ,  $\text{K}^+$ , and  $\text{Cl}^-$  is high in patients who abuse diuretics, albeit not to the levels seen in Gitelman syndrome. Marked variability in urinary electrolytes is an important clue for diuretic abuse, which can be verified with urinary drug screens. Clinically, nephrocalcinosis is very common in furosemide abuse, due to the increase in urinary calcium excretion.<sup>478</sup> Differentiation of Gitelman syndrome from BS requires a 24-hour urine to assess calcium excretion since hypocalciuria is a distinguishing feature for the former<sup>320</sup>; patients with Gitelman syndrome are also invariably hypomagnesemic. BS must be differentiated from “pseudo-Bartter” syndrome due to gentamicin toxicity,<sup>327,479</sup> mutations in CFTR, the cystic fibrosis gene,<sup>443,480</sup> or Sjögren syndrome with tubulointerstitial nephritis.<sup>481</sup> Acquired forms of Gitelman syndrome have in turn been reported after cisplatin therapy<sup>459</sup> and in patients with Sjögren syndrome.<sup>27,289</sup> Finally, although laxative abuse is perhaps a less common cause of chronic

hypokalemia, an accompanying metabolic acidosis with a negative urinary anion gap should raise the diagnostic suspicion of this cause.<sup>179</sup>

## TREATMENT OF HYPOKALEMIA

The goals of therapy in hypokalemia are to prevent life-threatening conditions (diaphragmatic weakness, rhabdomyolysis, and cardiac arrhythmias), to replace any  $\text{K}^+$  deficit, and to diagnose and correct the underlying cause. The urgency of therapy depends on the severity of hypokalemia, associated conditions and settings (e.g., a patient with heart failure on digoxin, or a patient with hepatic encephalopathy), and the rate of decline in plasma  $\text{K}^+$ . A rapid drop to  $<2.5$  mmol/L poses a high risk of cardiac arrhythmias and calls for urgent replacement. Although replacement is usually limited to patients with a true deficit, it should be considered in patients with hypokalemia due to redistribution (e.g., hypokalemic periodic paralysis)

when serious complications such as muscle weakness, rhabdomyolysis, and cardiac arrhythmias are present or imminent.<sup>482</sup> The risk of arrhythmia from hypokalemia is highest in older patients, patients with evidence of organic heart disease, and patients on digoxin or antiarrhythmic drugs.<sup>247</sup> In these high-risk patients, an increased incidence of arrhythmias may occur at even mild to modest degrees of hypokalemia. The American Heart Association guidelines on the use of hospital telemetry recommend monitoring in patients with hypokalemia and a prolonged QT interval.<sup>483</sup>

It is also crucial to diagnose and eliminate the underlying cause, so as to tailor therapy to the pathophysiology involved. For example, the risk of overcorrection or rebound hyperkalemia in hypokalemia caused by redistribution is particularly high, with the potential for fatal hyperkalemic arrhythmias.<sup>248,263,301,482,484</sup> When increased sympathetic tone or increased sympathetic response is thought to play a dominant role, the use of nonspecific  $\beta$ -adrenergic blockade with propranolol generally avoids this complication and should be considered; the relevant causes of hypokalemia include thyrotoxic periodic paralysis,<sup>209</sup> theophylline overdose,<sup>485</sup> and acute head injury.<sup>263</sup>

$K^+$  replacement is the mainstay of therapy in hypokalemia. However, hypomagnesemic patients can be refractory to  $K^+$  replacement alone,<sup>467</sup> such that concomitant  $Mg^{2+}$  deficiency should always be addressed with oral or parenteral repletion. To prevent hyperkalemia due to excessive supplementation, the deficit and the rate of correction should be estimated as accurately as possible. Renal function, medications, and comorbid conditions such as diabetes (with a risk of both insulinopenia and autonomic neuropathy) should also be considered to gauge the risk of overcorrection. Adjustments in the dose of administered  $K^+$ - $Cl^-$  replacement based on estimated GFR can potentially reduce the risk of hyperkalemia.<sup>486</sup> The goal is to raise the plasma  $K^+$  to a safe range rapidly and then replace the remaining deficit at a slower rate over days to weeks.<sup>247,248,482</sup> In the absence of abnormal  $K^+$  redistribution, the total deficit correlates with serum  $K^+$ <sup>248,482,487</sup>, such that serum  $K^+$  drops by approximately 0.27 mmol/L for every 100-mmol reduction in total-body stores. Loss of 400 to 800 mmol of body  $K^+$  results in a reduction in serum  $K^+$  by approximately 2.0 mmol/L<sup>487</sup>; these parameters can be used to estimate replacement goals. However, such estimates are just an approximation of the amount of  $K^+$  replacement required to normalize plasma  $K^+$ , with as much as a one in six risk of overreplacement<sup>252</sup>; serum  $K^+$  should also be closely monitored during replacement, withdrawing, or adjusting  $K^+$  replacement if necessary.

Although the treatment of asymptomatic patients with borderline or low normal serum  $K^+$  remains controversial, supplementation is recommended in patients with serum  $K^+$  lower than 3 mmol/L. In high-risk patients (i.e., those with heart failure, cardiac arrhythmias, myocardial infarction, ischemic heart disease, or taking digoxin), serum  $K^+$  should be maintained at  $\geq 4.0$  mmol/L<sup>247</sup> or even  $\geq 4.5$  mmol/L.<sup>256</sup> Patients with severe hepatic disease may not be able to tolerate mild-to-moderate hypokalemia due to the associated augmentation in ammoniogenesis, and thus serum  $K^+$  should be maintained at approximately 4.0 mmol/L.<sup>488,489</sup> In asymptomatic patients with mild-to-moderate hypertension, an attempt should be made to

maintain serum  $K^+$  above 4.0 mmol/L<sup>247</sup> and potassium supplementation should be considered when serum  $K^+$  falls below 3.5 mmol/L.<sup>247</sup> Notably, prospective studies have shown an inverse relationship between dietary potassium intake and both fatal and nonfatal stroke, independent of the associated antihypertensive effect.<sup>247,490,493</sup>

Potassium is available in the form of potassium chloride, phosphate, potassium bicarbonate, or its precursors (potassium citrate, potassium acetate) and potassium gluconate.<sup>247,248,482</sup> Potassium phosphate is indicated when phosphate deficit accompanies  $K^+$  depletion (e.g., in diabetic ketoacidosis).<sup>482</sup> Potassium bicarbonate (or its precursors) should be considered in patients with hypokalemia and metabolic acidosis.<sup>247,482</sup> Potassium chloride should otherwise be the default salt of choice in most patients, for several reasons. First, metabolic alkalosis typically accompanies chloride loss from renal (e.g., diuretics) or upper gastrointestinal routes (e.g., vomiting) and contributes significantly to renal  $K^+$  wasting.<sup>248</sup> In this setting, replacing chloride along with  $K^+$  is essential in treating the alkalosis and preventing further kaliuresis; because dietary  $K^+$  is mainly in the form of potassium phosphate or potassium citrate, it usually does not suffice. Second, potassium bicarbonate may offset the benefits of  $K^+$  administration by aggravating concomitant alkalosis. Third, potassium chloride raises serum  $K^+$  at a faster rate than potassium bicarbonate, a factor that is crucial in patients with marked hypokalemia and related symptoms. In all likelihood, this faster rise in plasma  $K^+$  occurs because  $Cl^-$  is mainly an extracellular fluid anion that does not enter cells to the same extent as bicarbonate, keeping the  $K^+$  in the extracellular fluid compartment.<sup>492</sup> Finally, elegant animal physiology has revealed that a  $K^+$ -rich diet with  $HCO_3^-$  as the counterion enhances electrogenic  $K^+$  excretion via activation of ENaC and ROMK when compared with a high  $K^+$ - $Cl^-$  diet.<sup>9</sup>

Parenteral (intravenous)  $K^+$  administration should be limited to patients unable to utilize the enteral route or when the patient is experiencing associated signs and symptoms. However, rapid correction of hypokalemia through oral supplementation is possible and may be faster than intravenous  $K^+$  supplementation, due to limitations in the rapidity of intravenous  $K^+$  infusion. For example, serum  $K^+$  can be increased by 1 to 1.4 mmol/L in 60 to 90 minutes, following the oral intake of 75 mmol of  $K^+$ <sup>493</sup>; the ingestion of approximately 125 to 165 mmol of  $K^+$  as a single oral dose can increase serum  $K^+$  by approximately 2.5 to 3.5 mmol/L in 60 to 120 minutes.<sup>494</sup> The oral route is thus both effective and appropriate in patients with asymptomatic severe hypokalemia. If the patient is experiencing life-threatening signs and symptoms of hypokalemia, however, the maximum possible IV infusion of  $K^+$  should be administered acutely for symptom control, followed by rapid oral supplementation.

The usual intravenous dose is 20 to 40 mmol of  $K^+$ - $Cl^-$  in a liter of vehicle solution.<sup>482</sup> The vehicle solution should be dextrose free to prevent a transient reduction in serum  $K^+$  level of 0.2 to 1.4 mmol/L, due to an enhanced endogenous insulin secretion induced by the dextrose.<sup>495</sup> Higher concentrations of  $K^+$ - $Cl^-$  (up to 400 mmol/L, as 40 mmol in 100 mL normal saline) have been used in life-threatening conditions.<sup>496,497</sup> In these cases, the amount of  $K^+$  per intravenous bag should be limited (e.g., 20 mmol in 100 mL saline solution) to prevent inadvertent infusion of a

large dose.<sup>497</sup> These solutions are best given through a large central vein. Femoral veins are preferable since infusion through upper body central lines can acutely increase the local concentration of  $K^+$  with deleterious effects on cardiac conduction.<sup>497</sup> As a general rule and to avoid venous pain, irritation, and sclerosis, concentrations of more than 60 mmol/L should not be given through a peripheral vein.<sup>482</sup> Although the recommended rate of administration is 10 to 20 mmol/hour, rates of 40 to 100 mmol/hour or even higher (for a short period) have been used in patients with life-threatening conditions.<sup>27,496,498</sup> However, a rapid increase in serum  $K^+$  associated with electrocardiographic changes may occur with higher rates of infusion (e.g.,  $\geq 80$  mmol/hour).<sup>499</sup> Intravenous administration of  $K^+$  at a rate of more than 10 mmol/hour requires continuous ECG monitoring.<sup>482</sup> In patients receiving such high infusion rates, close monitoring of the appropriate physiologic consequences of hypokalemia is essential; after these effects have abated, the rate of infusion should be decreased to the standard dose of 10 to 20 mmol/hour. It is important to remember that volume expansion in patients with moderate-to-severe hypokalemia and  $Cl^-$ -responsive metabolic alkalosis should be performed cautiously and with close follow-up of serum  $K^+$ , since bicarbonaturia associated with volume expansion may aggravate renal  $K^+$  wasting and hypokalemia. In patients with combined severe hypokalemia and hypophosphatemia (e.g., diabetic ketoacidosis), intravenous  $K^+$  phosphate can be used. However, this solution should be infused at a rate of  $<50$  mmol over 8 hours to prevent the risk of hypocalcemia and metastatic calcification.<sup>27</sup> A combination of potassium phosphate and potassium chloride may be necessary to correct hypokalemia effectively in these patients.

The easiest and most straightforward method of oral  $K^+$  supplementation is to increase dietary intake of potassium-rich foods<sup>248</sup> (Table 16.3). One study compared the effectiveness of “diet versus medication supplementation” in cardiac surgery patients receiving diuretics in hospital and found no difference between the two groups in respect to maintenance of serum  $K^+$ . However, limitations of this study include a small number of subjects, relatively short duration, and lack of information on acid-base status, making it less than conclusive and not generalizable.<sup>500</sup> Regardless, dietary  $K^+$  is mainly in the form of potassium phosphate or potassium citrate and is inadequate in the majority of patients who have concomitant  $K^+$  and  $Cl^-$  deficiency.<sup>501</sup> For example, bananas contain only 2 mEq of potassium per inch, with a dominance of nonchloride anions.<sup>501</sup> Most patients will therefore need to combine a high- $K^+$  diet with a prescribed dose of  $K^+-Cl^-$ .<sup>248</sup> Salt substitutes are an inexpensive and potent source of  $K^+-Cl^-$ ; each gram contains 10 to 13 mmol of  $K^+$ <sup>502</sup>; however, patients, particularly those with an impaired ability to excrete potassium, need to be counseled regarding the appropriate amount and the potential for hyperkalemia.<sup>503</sup> Potassium chloride is also available in either liquid or tablet form.<sup>247</sup> In general, the available preparations are well absorbed.<sup>248</sup> Liquid forms are less expensive but are less well tolerated. Slow-release forms are more palatable and better tolerated; however, they have been associated with gastrointestinal ulceration and bleeding, ascribed to local accumulation of high concentrations of  $K^+$ .<sup>248</sup> Notably, this risk is rather low, and lower still with the microencapsulated forms.<sup>248</sup> The chance of overdose and hyperkalemia is higher with

**Table 16.3 Foods With High Potassium Content**

|                                                        |
|--------------------------------------------------------|
| Highest content ( $>1000$ mg [ $20$ mmol]/ $100$ g)    |
| Dried figs                                             |
| Molasses                                               |
| Seaweed                                                |
| Very high content ( $>500$ mg [ $12.5$ mmol]/ $100$ g) |
| Dried fruits (dates, prunes)                           |
| Nuts                                                   |
| Avocados                                               |
| Bran cereals                                           |
| Wheat germ                                             |
| Lima beans                                             |
| High content ( $>250$ mg [ $6.2$ mmol]/ $100$ g)       |
| Vegetables                                             |
| Spinach                                                |
| Tomatoes                                               |
| Broccoli                                               |
| Winter squash                                          |
| Beets                                                  |
| Carrots                                                |
| Cauliflower                                            |
| Potatoes                                               |
| Fruits                                                 |
| Bananas                                                |
| Cantaloupe                                             |
| Kiwis                                                  |
| Oranges                                                |
| Mangos                                                 |
| Meats                                                  |
| Ground beef                                            |
| Steak                                                  |
| Pork                                                   |
| Veal                                                   |
| Lamb                                                   |

From Gennari FJ. Hypokalemia. *N Engl J Med*. 1998;339:451–458.

slow-release formulations; unlike the immediate-release forms, these tablets are less irritating to the stomach and less likely to induce vomiting.<sup>504</sup> The usual dose is 40 to 100 mmol of  $K^+$  (as  $K^+-Cl^-$ ) per day, divided in 2 to 3 doses, in patients taking diuretics.<sup>248</sup> ( $K^+-Cl^-$  can be toxic in doses of  $>2$  mmol/kg<sup>504</sup>). This dose is effective in maintaining serum  $K^+$  in up to 90%; however, in the 10% of patients who remain to hypokalemic, increasing the oral dose or adding a  $K^+$ -sparing diuretic is an appropriate choice.<sup>248</sup>

In addition to potassium supplementation, strategies to minimize  $K^+$  losses should be considered. These measures may include minimizing the dose of non- $K^+$ -sparing diuretics, restricting  $Na^+$  intake, and using a combination of non- $K^+$ -sparing and  $K^+$ -sparing medications (e.g., angiotensin-converting enzyme inhibitors, angiotensin-receptor blockers,  $K^+$ -sparing diuretics, and  $\beta$ -blockers).<sup>217,247</sup> The use of a  $K^+$ -sparing diuretic is of particular importance in hypokalemia resulting from primary hyperaldosteronism and related disorders, such as Liddle syndrome and AME;  $K^+$  supplementation alone may be ineffective in these settings.<sup>505–507</sup> In patients with hypokalemia due to loss through upper gastrointestinal secretion (continuous nasogastric tube suction, continuous or self-induced vomiting), proton-pump

inhibitors are reportedly useful in helping to correct metabolic alkalosis and reduce hypokalemia.<sup>508</sup>

## HYPERKALEMIA

### EPIDEMIOLOGY

Hyperkalemia is usually defined as a potassium level of 5.5 mmol/L or higher,<sup>509,510</sup> although in some studies 5.0 to 5.4 mmol/L qualifies for the diagnosis.<sup>511</sup> Hyperkalemia has been reported in 1.1% to 10% of all hospitalized patients,<sup>224,509-512</sup> with approximately 1.0% of patients (8%–10% of hyperkalemic patients) having significant hyperkalemia ( $\geq 6.0$  mmol/L).<sup>509</sup> Hyperkalemia has been associated with a higher mortality rate (14.3%–41%),<sup>27,509,510</sup> accounting for approximately 1 death per 1000 patients in one case series from the mid-1980s.<sup>513</sup> In most hospitalized patients, the pathophysiology of hyperkalemia is multifactorial, with reduced renal function, medications, older age ( $\geq 60$  years), and hyperglycemia being the most common contributing factors.<sup>224,509,510</sup>

In patients with ESRD, the prevalence of hyperkalemia is 5% to 10%.<sup>514-516</sup> The prevalence increases from 2% to 42% as GFR decreases from 60 to 90 mL/min per 1.73 m<sup>2</sup>.<sup>517</sup> The risk of hyperkalemia is increased in males with chronic kidney disease (CKD) and tripled by treatment with ACE-inhibitors or ARBs.<sup>517</sup> Hyperkalemia accounts for or contributes to 1.9% to 5% of deaths among patients with ESRD.<sup>224,516</sup> Notably, however, the risk of death from hyperkalemia is reduced as CKD progresses, presumably due to as-yet-uncharacterized cardiac adaptation to chronic hyperkalemia.<sup>518</sup> However, multiple studies implicate hyperkalemia with increased risk of death in ESRD, as managed by both hemodialysis<sup>519</sup> and peritoneal dialysis.<sup>520</sup> Hyperkalemia is the reason for emergency hemodialysis in 24% of patients with ESRD on hemodialysis,<sup>516</sup> and renal failure is the most common cause of hyperkalemia diagnosed in the emergency department.<sup>514</sup> The prevalence of marked hyperkalemia ( $K^+ \geq 5.8$  mmol/L) is approximately 1% in a general medicine outpatient setting. Alarming, the management of outpatient hyperkalemia is often suboptimal, with approximately 25% of the patients lacking any follow-up, ECGs performed in only 36% of cases, and frequent delays in repeating serum  $K^+$ .<sup>521</sup>

### PSEUDOHYPERKALEMIA

Factitious or pseudohyperkalemia is an artifactual increase in serum  $K^+$  due to the release of  $K^+$  during or after venipuncture. There are several potential causes for pseudohyperkalemia.<sup>522</sup> First, forearm contraction, fist clenching,<sup>49</sup> or tourniquet use<sup>522</sup> may increase  $K^+$  efflux from local muscle and thus raise the measured serum  $K^+$ . Second, thrombocytosis,<sup>523</sup> leukocytosis,<sup>524</sup> and/or erythrocytosis<sup>525</sup> may cause pseudohyperkalemia due to release from these cellular elements. Third, acute anxiety during venipuncture may provoke respiratory alkalosis and hyperkalemia due to redistribution.<sup>52,53,55,62</sup> Fourth, sample contamination with  $K^+$ -EDTA, used as a sample anticoagulant for some laboratory assays, can cause spurious hyperkalemia.<sup>526</sup> There are several mechanisms

for sample contamination with  $K^+$ -EDTA during blood draws or sample handling.<sup>526</sup> Gross contamination with  $K^+$ -EDTA usually results in spurious hypocalcemia and hypomagnesemia; lesser contamination is less obvious, leading to the practice in some laboratories to perform EDTA assays on samples with a plasma  $K^+$  of  $>6$  mmol/L with  $K^+$ -EDTA. Finally, mechanical and physical factors may induce pseudohyperkalemia after blood has been drawn. For example, pneumatic tube transport has been shown to induce pseudohyperkalemia in one patient with leukemia and massive leukocytosis.<sup>527</sup> Cooling of blood before the separation of cells from serum or plasma is also a well-recognized cause of artifactual hyperkalemia.<sup>528</sup> The converse is the risk of increased uptake of  $K^+$  by cells at high ambient temperatures, leading to normal values for hyperkalemic patients and/or to spurious hypokalemia in patients who are normokalemic.<sup>27,258</sup> This issue is particularly important for outpatient primary practice samples that are transported off-site and analyzed at a central facility<sup>27</sup>; this phenomenon leads to “seasonal pseudohyperkalemia and hypokalemia,”<sup>27,529</sup> with fluctuations of outpatient samples as a function of season and ambient temperature.

Finally, there are several hereditary subtypes of pseudohyperkalemia, caused by increase in passive  $K^+$  permeability of erythrocytes. Abnormal red cell morphology, varying degrees of hemolysis, and/or perinatal edema can accompany hereditary pseudohyperkalemia, whereas in many kindreds there are no overt hematologic consequences. Serum  $K^+$  increases in pseudohyperkalemia patient samples that have been left at room temperature, due to abnormal  $K^+$  permeability of erythrocytes. Several subtypes have been defined, based on differences in the temperature-dependence curve of this red cell leak pathway.<sup>27,530</sup> The disorder is genetically heterogeneous, with characterized genes on chromosome 17q21 (*SLC4A1*), 16q23-ter (*PIEZO1*), and 2q35-36 (*ABCB6*).<sup>27,531</sup> Initially, 11 pedigrees of patients with autosomal dominant hemolysis, pseudohyperkalemia, and temperature-dependent loss of red cell  $K^+$  were found to have heterozygous mutations in the *SLC4A1* gene on chromosome 17q21, which encodes the band 3 anion exchanger, AE1.<sup>530</sup> The mutations that were detected all cluster within exon 17 of the gene,<sup>530</sup> between transmembrane domains 8 and 10 of the AE1 protein. These mutations reduce anion transport in both red cells and *Xenopus* oocytes injected with AE1, with the novel acquisition of a nonselective transport pathway for both  $Na^+$  and  $K^+$ . Pseudohyperkalemia in these patients thus results from a genetic event that endows AE1 with the ability to transport  $K^+$ ; single-point mutations can convert an anion exchanger to a nonselective cation channel serves to underline the narrow boundaries that separate exchangers and transporters from ion channels.<sup>530</sup> Notably, however, newer reports have suggested that these mutations in *SLC4A1* induce cation fluxes that are independent of the AE1 anion exchanger protein (i.e., mediated by other transport pathways).<sup>532,533</sup>

Mutations in the red cell Rh-associated glycoprotein (RhAG) have been linked to the monovalent cation leak associated with overhydrated hereditary stomatocytosis.<sup>534</sup> These mutations cause an exaggerated cation leak in the RhAG, thought to function as an  $NH_3$  or  $NH_4^+$

transporter RhAG. Exaggerated red cell cation leaks are also implicated in stomatocytosis due to mutations in the mechanically activated cation channel PIEZO1<sup>535</sup>; disease-associated mutations give rise to mechanically activated currents that inactivate more slowly than the wild-type channel. Physiologically, calcium influx via the PIEZO1 channel collaborates with the calcium-activated “Gardos” K<sup>+</sup> channel (encoded by *KCNN4*) in controlling red cell volume.<sup>531</sup> Not surprisingly, mutations in *KCNN4* that increase calcium sensitivity of the Gardos channel are associated with stomatocytosis.<sup>536</sup> Finally, mutations in the gene encoding the ATP-binding cassette family member 6 (*ABCB6*) transporter have also been associated with familial pseudohyperkalemia.<sup>531</sup>

## EXCESS INTAKE OF POTASSIUM AND TISSUE NECROSIS

Increased intake of even small amounts of K<sup>+</sup> may provoke severe hyperkalemia in patients with predisposing factors. For example, oral administration of 32 millimoles to a diabetic patient with hyporeninemic hypoaldosteronism resulted in an increase in serum K<sup>+</sup> from 4.9 mmol/L to a peak of 7.3 mmol/L, within 3 hours.<sup>537</sup> Increased intake or changes in intake of dietary sources rich in K<sup>+</sup> (see Table 16.3) may also provoke hyperkalemia in susceptible patients. Rarely, marked intake of K<sup>+</sup> (e.g., in sports beverages)<sup>538</sup> may provoke severe hyperkalemia in individuals free of predisposing factors. Other occult sources of K<sup>+</sup> must also be considered including salt substitutes,<sup>502</sup> alternative medicines,<sup>539</sup> and alternative diets.<sup>540</sup> Geophagia with ingestion of K<sup>+</sup>-rich clay<sup>27</sup> and cautoypreophagia<sup>541</sup> (ingestion of burnt matchsticks) are two forms of pica that have been reported to cause hyperkalemia in dialysis patients. Sustained-release K<sup>+</sup>-Cl<sup>-</sup> tablets can cause hyperkalemia in suicidal overdoses.<sup>504</sup> Such pills are radiopaque and may thus be seen on radiographs; whole-bowel irrigation should be used for gastrointestinal decontamination.<sup>504</sup> Iatrogenic causes include simple overreplacement with K<sup>+</sup>-Cl<sup>-</sup>, as can occur commonly in hypokalemic patients,<sup>252</sup> or administration of a potassium-containing medication, such as K<sup>+</sup>-penicillin to a susceptible patient.

Red cell transfusion is a well-described cause of hyperkalemia, typically seen in children or in massive transfusions. Risk factors for transfusion-related hyperkalemia include the rate and volume of the transfusion, the use of a central venous infusion and/or pressure pumping, the use of irradiated blood, and the age of the blood infused<sup>49,542</sup>; whereas 7-day-old blood has a free K<sup>+</sup> concentration of ~23 mmol/L, this rises to the 50 mmol/L range in 42-day-old blood.<sup>543</sup> Hyperkalemia is a common occurrence in patients with severe trauma, with a period prevalence of 29% in massively traumatized patients at a U.S. military combat support hospital in Iraq.<sup>544</sup> Although red cell and/or blood product transfusion plays an important role, this and other studies indicate a complex pathophysiology for resuscitative hyperkalemia, with low cardiac output, acidosis, hypocalcemia, and other factors contributing to the risk of hyperkalemia in patients with severe trauma.<sup>542,544</sup>

Tissue necrosis is an important cause of hyperkalemia. Hyperkalemia due to rhabdomyolysis is particularly common due to the enormous store of K<sup>+</sup> in muscle (see

Fig. 16.1). In many cases, volume depletion, medications (statins in particular), and metabolic predisposition contribute to the genesis of rhabdomyolysis. Hypokalemia is an important metabolic predisposing factor in rhabdomyolysis (see “Consequences of Hypokalemia”); others include hypophosphatemia, hypernatremia and hyponatremia, and hyperglycemia. Those patients with hypokalemia-associated rhabdomyolysis in whom redistribution is the cause of hypokalemia are at particular risk of subsequent hyperkalemia, as rhabdomyolysis evolves and renal function worsens.<sup>49,274</sup> Finally, massive release of K<sup>+</sup> and other intracellular contents may occur as a result of acute tumor lysis.<sup>537</sup>

## REDISTRIBUTION AND HYPERKALEMIA

Several different mechanisms can induce an efflux of intracellular K<sup>+</sup>, resulting in hyperkalemia. The infusion of hypertonic mannitol or saline, but not hypertonic bicarbonate, generates an increase in serum K<sup>+</sup>.<sup>545</sup> Potential mechanisms include a dilutional acidosis with subsequent shift in K<sup>+</sup>, increased passive exit of K<sup>+</sup> due to an increase in intracellular K<sup>+</sup> activity from intracellular water loss, acute hemolysis, and a “solvent drag” effect as water exits cells.<sup>546,547</sup> Regardless, severe hyperkalemia, typically with acute dilutional hyponatremia, is a well-described complication of mannitol for the treatment or prevention of cerebral edema.<sup>547-549</sup> Diabetic patients are prone to severe hyperkalemia in response to intravenous hypertonic glucose in the absence of adequate coadministered insulin, due to a similar osmotic effect.<sup>550,551</sup> Finally, a retrospective report documented considerable increases in serum K<sup>+</sup> after IV contrast dye in five patients with CKD, four on dialysis, and one with stage IV CKD<sup>552</sup>; again, the acute osmolar load was the likely cause of the acute hyperkalemia in these patients. The implications of this provocative study are not entirely clear; however, one would expect the development or worsening of hyperkalemia in dialysis patients exposed to large volumes of hyperosmolar contrast dye.

Several reports have appeared regarding the risk of hyperkalemia with epsilon-aminocaproic acid (Amicar),<sup>553-555</sup> a cationic amino acid that is structurally similar to lysine and arginine. Cationic but not anionic amino acids induce efflux of K<sup>+</sup> from cells, although the transport pathways involved are unknown.<sup>49</sup> A resurgence in the use of cationic amino acid infusions during endoradiotherapy has served to emphasize this risk, with the development of significant hyperkalemic acidosis.<sup>556</sup>

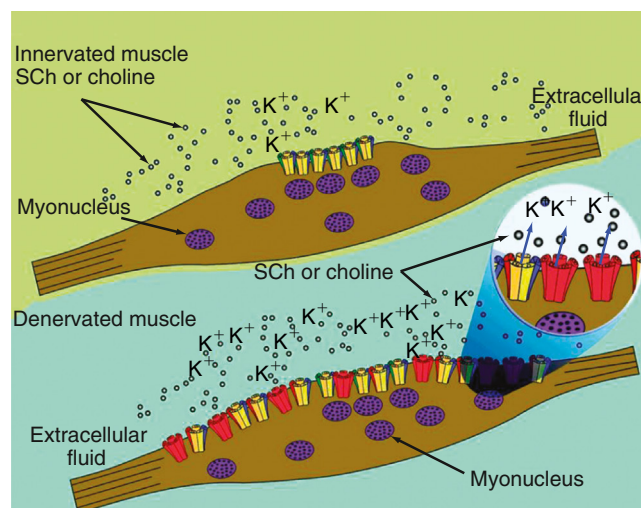
Muscle plays a dominant role in extrarenal K<sup>+</sup> homeostasis, primarily via regulated uptake by the Na<sup>+</sup>/K<sup>+</sup>-ATPase. Although exercise is a well-described cause of acute hyperkalemia, this effect is usually transient and clinical relevance is difficult to judge. ESRD patients on dialysis do not have an exaggerated increase in plasma K<sup>+</sup> with maximal exercise, perhaps due to greater insulin, catecholamine, and aldosterone responses to exercise and/or to their preexisting hyperkalemia.<sup>557</sup> The results and design of this and other studies of exercise-associated hyperkalemia in ESRD have been criticized by a report that linked abnormal extrarenal K<sup>+</sup> homeostasis to increased fatigue in ESRD.<sup>558</sup> Regardless, exercise-associated hyperkalemia is not a major clinical cause of hyperkalemia. Dialysis patients are,

however, susceptible to modest increases in plasma  $K^+$  after prolonged fasting, due to the relative insulinopenia in this setting.<sup>559</sup> This may be clinically relevant in preoperative ESRD patients, for whom intravenous glucose infusions  $\pm$  insulin are appropriate preventive measures for the development of hyperkalemia.<sup>559</sup>

Insulin stimulates the uptake of  $K^+$  by several tissues, primarily via stimulation of  $Na^+/K^+$ -ATPase activity.<sup>27,39,41,44</sup> Reduction in circulating insulin is thus an important factor or cofactor in the generation of hyperkalemia in diabetic patients. Patients with DKA typically present with serum  $K^+$  levels that are within normal limits or moderately elevated but with profound whole-body potassium deficits. However, significant hyperkalemia (serum  $K^+ > 6$ – $6.5$  mmol/L) is not uncommon in DKA<sup>560,561</sup> due to a variety of potential factors: insulinopenia, renal dysfunction, and the hyperosmolar effect of severe hyperglycemia.<sup>560-562</sup> Inhibition of insulin secretion by the somatostatin agonist octreotide can also cause significant hyperkalemia, in both anephric patients<sup>37</sup> and patients with normal renal function.<sup>38</sup>

Digoxin inhibits  $Na^+/K^+$ -ATPase and thus impairs the uptake of  $K^+$  by skeletal muscle (see “Factors Affecting Internal Distribution of Potassium” earlier), such that digoxin overdose can result in hyperkalemia. The skin and venom gland of the cane toad *Bufo marinus* contains high concentrations of bufadienolide, a structurally similar glycoside. The direct ingestion of such toads<sup>563</sup> or of toad extracts can result in fatal hyperkalemia. In particular, certain herbal aphrodisiac pills contain appreciable amounts of toad venom and have led to several case reports in the United States.<sup>49,564</sup> Patients may have detectable plasma levels using standard digoxin assays since bufadienolide is immunologically similar to digoxin. Moreover, treatment with digoxin-specific Fab fragment, indicated for treatment of digoxin overdoses, may be effective and life-saving in bufadienolide toxicity.<sup>49,564</sup> Finally, fluoride ions also inhibit  $Na^+/K^+$ -ATPase, such that fluoride poisoning is typically associated with hyperkalemia.<sup>565</sup>

Succinylcholine depolarizes muscle cells, resulting in the efflux of  $K^+$  through acetylcholine receptors (AChRs) and a rapid, but usually, transient hyperkalemia. The use of this agent is contraindicated in patients who have sustained thermal trauma, neuromuscular injury (upper or lower motor neuron), disuse atrophy, mucositis, or prolonged immobilization in an intensive care unit setting; the efflux of  $K^+$  induced by succinylcholine is enhanced in these patients and can result in significant hyperkalemia.<sup>566</sup> These disorders share a 2- to 100-fold upregulation of AChRs at the plasma membrane of muscle cells, with loss of the normal clustering at the neuromuscular junction.<sup>566</sup> Depolarization of these upregulated AChRs by succinylcholine results in an exaggerated efflux of  $K^+$  through the receptor-associated cation channels that are spread throughout the muscle cell membrane (see Fig. 16.12). Concomitant upregulation of the neuronal  $\alpha 7$  AChR subunit has also been observed in denervated muscle; the  $\alpha 7$ -containing AChR is a homomeric, pentameric channel that depolarizes in response to both succinylcholine and choline, its metabolite.<sup>566</sup> Depolarization  $\alpha 7$ -AChRs in response to choline is furthermore not subject to desensitization and may explain in part the hyperkalemic effect that persists in some patients well after the paralytic



**Fig. 16.12 Succinylcholine-induced efflux of potassium is increased in denervated muscle.** In innervated muscle, succinylcholine interacts with the entire plasma membrane but depolarizes only the junctional ( $\alpha 1$ ,  $\beta 1$ ,  $\delta$ , and  $\epsilon$ —multicolored) acetylcholine receptors (AChRs); this leads to a modest, transient hyperkalemia. With denervation, there is a considerable upregulation of muscle AChRs, with increased extrajunctional AChRs ( $\alpha 1$ ,  $\beta 1$ ,  $\delta$ , and  $\gamma$ —multicolored) and acquisition of homomeric, neuronal-type  $\alpha 7$ -AChRs. Depolarization of denervated muscle leads to an exaggerated  $K^+$  efflux, due to the upregulation and redistribution of these AChRs. In addition, choline generated from metabolism of succinylcholine maintains the depolarization mediated via  $\alpha 7$ -AChRs, thus enhancing and prolonging the  $K^+$  efflux after paralysis has subsided. (From Martyn JA, Richtsfeld M. Succinylcholine-induced hyperkalemia in acquired pathologic states: etiologic factors and molecular mechanisms. *Anesthesiology*. 2006;104:158–169.)

effect of succinylcholine has subsided.<sup>566</sup> Consistent perhaps with this neuromuscular pathophysiology, patients with renal failure do not appear to have an increased risk of succinylcholine-associated hyperkalemia.<sup>567</sup>

A report of three patients suggested the possibility that drugs that share the ability to open  $K_{ATP}$  channels may have an underappreciated propensity to cause hyperkalemia in critically ill patients. The implicated drugs included cyclosporine, isoflurane, and nicorandil.<sup>568</sup> These patients exhibited hyperkalemia that resisted usual therapies (insulin/dextrose  $\pm$  hemofiltration), with a temporal hypokalemic response to the  $K_{ATP}$  inhibitor, glibenclamide (glyburide). The daring, off-label use of glibenclamide was presumably instigated by the senior author's observation that cyclosporine activates  $K_{ATP}$  channels in vascular smooth muscle.<sup>569</sup>  $K_{ATP}$  channels are widely distributed including in skeletal muscle, such that activation of such channels is indeed a plausible cause of acute hyperkalemia. Other case reports have emerged of nicorandil-associated hyperkalemia.<sup>570,571</sup>

Finally,  $\beta$ -blockers cause hyperkalemia in part by inhibiting cellular uptake, but also through hyporeninemic hypoaldosteronism induced by effect of these drugs on both renal renin release and adrenal aldosterone release (see “Regulation of Renal Renin and Adrenal Aldosterone Release” earlier). Labetalol, a broadly reactive sympathetic blocker, is a particularly common cause of hyperkalemia in susceptible patients.<sup>49,572</sup> However, both nonspecific and

cardiospecific  $\beta$ -blockers have been shown to reduce PRA, angiotensin-II, and aldosterone,<sup>573</sup> such that  $\beta$ -blockade in general will increase susceptibility to hyperkalemia.

## REDUCED RENAL EXCRETION OF POTASSIUM

### HYPOALDOSTERONISM

Aldosterone promotes kaliuresis by activating apical amiloride-sensitive  $\text{Na}^+$  currents in the CNT and CCD and thus increasing the lumen-negative driving force for  $\text{K}^+$  excretion (see “Control of Potassium Excretion: Aldosterone”). Aldosterone release from the adrenal may be reduced by hyporeninemic hypoaldosteronism and its multiple causes, by medications, or due to isolated deficiency of ACTH. The isolated loss in pituitary secretion of ACTH leads to a deficit in circulating cortisol; variable defects in other pituitary hormones are likely secondary to this reduction in cortisol.<sup>574</sup> Concomitant hyporeninemic hypoaldosteronism is frequent<sup>27</sup>; however, hyperkalemia is perhaps less common in secondary hypoaldosteronism than in Addison disease.<sup>574</sup>

Primary hypoaldosteronism may be genetic or acquired. The X-linked disorder adrenal hypoplasia congenita (AHC) is caused by loss-of-function mutations in the transcriptional repressor Dax-1. Patients with AHC present with primary adrenal failure and hyperkalemia either shortly after birth or much later in childhood.<sup>575</sup> This bimodal presentation pattern does not appear to be influenced by *Dax-1* genotype; rather, if patients survive the early neonatal period, they will then miss being diagnosed until much later in life, presenting either with delayed puberty (see later) or an adrenal crisis. The steroidogenic factor-1 (SF-1), a functional partner for Dax-1, is also required for adrenal development in both mouse and man. Both genes are involved in gonadal development, with *Dax-1* deficiency leading to hypogonadotropic hypogonadism<sup>575</sup> and *SF-1* deficiency causing male-to-female sex reversal, in addition to adrenal insufficiency.

Reduced steroidogenesis causes two other important forms of primary hypoaldosteronism. Congenital lipoid adrenal hyperplasia (lipoid CAH) is a severe autosomal recessive syndrome characterized by impaired synthesis of mineralocorticoids, glucocorticoids, and gonadal steroids.<sup>49</sup> Patients present in early infancy with adrenal crisis including severe hyperkalemia.<sup>576</sup> Genotypically male 46,XY patients with lipoid CAH have female external genitalia due to the developmental absence of testosterone. Lipoid CAH is caused by loss-of-function mutations in steroidogenic acute regulatory protein, a small mitochondrial protein that helps shuttle cholesterol from the outer to the inner mitochondrial membrane, thus initiating steroidogenesis<sup>577</sup>; some patients may alternatively have mutations in the side-chain cleavage P450 enzyme.<sup>578</sup> The classic, salt-wasting form of congenital adrenal hyperplasia due to 21-hydroxylase deficiency is associated with marked reductions in both cortisol and aldosterone, leading to adrenal insufficiency.<sup>579</sup> Concomitant overproduction of androgenic steroids results in virilization in female patients with this form of CAH.

Isolated deficits in aldosterone synthesis with hyperreninemia are caused by loss-of-function mutations in aldosterone synthase, although genetic heterogeneity has been reported.<sup>580</sup> Patients typically present in childhood

with volume depletion and hyperkalemia.<sup>581</sup> Much like pseudohypoaldosteronism due to loss-of-function mutations in the MR (see later), patients tend to become asymptomatic in adulthood. Acquired hyperreninemic hypoaldosteronism has been described in critical illness,<sup>25</sup> type II diabetes,<sup>582</sup> amyloidosis due to familial Mediterranean fever,<sup>583</sup> and after metastasis of carcinoma to the adrenal gland.<sup>25</sup> Finally, aldosterone synthesis is selectively reduced by heparin, with a 7% incidence of hyperkalemia associated with heparin therapy.<sup>584</sup> Both unfractionated<sup>584</sup> and low-molecular-weight<sup>49,585</sup> heparin can cause hyperkalemia. Hyperkalemia due to prophylactic subcutaneous unfractionated heparin (5000 units twice daily) has also been reported.<sup>586</sup> Heparin reduces the adrenal aldosterone response to both angiotensin-II and hyperkalemia, resulting in hyperreninemic hypoaldosteronism. Histologic findings in experimental animals include a marked diminution in size of the zona glomerulosa and an attenuated hyperplastic response to salt depletion.<sup>584</sup>

Most primary adrenal insufficiency is due to autoimmunity in either Addison disease or the context of a polyglandular endocrinopathy.<sup>587</sup> Adrenal insufficiency can be seen following adrenalectomy for primary hyperaldosteronism, with 14% developing postoperative hyperkalemia and 5% developing long-term insufficiency requiring fludrocortisone.<sup>588</sup> The antiphospholipid syndrome may also cause bilateral adrenal hemorrhage and adrenal insufficiency.<sup>589</sup> Another renal syndrome in which there should be a high index of suspicion for adrenal insufficiency is renal amyloidosis.<sup>590</sup> Finally, HIV is a particularly important infectious cause of adrenal insufficiency. The most common cause of adrenalitis in HIV disease is cytomegalovirus; however, a long list of infectious, degenerative, and infiltrative processes may involve the adrenal glands in these patients.<sup>591</sup> Although the adrenal involvement in HIV is usually subclinical, adrenal insufficiency may be precipitated by stress, drugs such as ketoconazole that inhibit steroidogenesis, or the acute withdrawal of steroid agents such as megesterol.

Contemporary estimates of the risk of hyperkalemia with Addison disease are lacking, but the incidence is likely 50% to 60%.<sup>49</sup> The absence of hyperkalemia in such a high percentage of hypoadrenal patients underscores the importance of aldosterone-independent modulation of  $\text{K}^+$  excretion by the distal nephron. A high- $\text{K}^+$  diet and high peritubular  $\text{K}^+$  serve to increase apical  $\text{Na}^+$  reabsorption and  $\text{K}^+$  secretion in the CNT and CCD (see “Control of Potassium Excretion”); in most patients with reductions in circulating aldosterone, this homeostatic mechanism would appear to be sufficient to regulate plasma  $\text{K}^+$  to within normal limits.

### HYPORENINEMIC HYPOALDOSTERONISM

Hyporeninemic hypoaldosteronism<sup>592</sup> is a common predisposing factor in several large, overlapping subsets of hyperkalemic patients: diabetic patients,<sup>593</sup> the elderly,<sup>49,170,594</sup> and patients with renal insufficiency.<sup>49</sup> Hyporeninemic hypoaldosteronism has also been described in systemic lupus erythematosus (SLE),<sup>49,595</sup> multiple myeloma,<sup>27</sup> and acute glomerulonephritis.<sup>27</sup> Classically, patients should have suppressed PRA and aldosterone, which cannot be activated by typical maneuvers such as

furosemide or sodium restriction.<sup>592</sup> Approximately 50% have an associated acidosis, with reduced renal excretion of  $\text{NH}_4^+$ , a positive urinary anion gap, and urine pH  $<5.5$ .<sup>179,246</sup> Although the generation of this acidosis is clearly multifactorial,<sup>596</sup> strong clinical<sup>245,246,597</sup> and experimental<sup>241</sup> evidence suggests that hyperkalemia per se is the dominant factor, due to competitive inhibition of  $\text{NH}_4^+$  transport in the TAL and reduced distal excretion of  $\text{NH}_4^+$ <sup>598</sup> (see also “Consequences of Hyperkalemia” earlier).

Several factors account for the reduced PRA in diabetic patients with hyporeninemic hypoaldosteronism.<sup>593</sup> First, many patients have an associated autonomic neuropathy, with impaired release of renin during orthostatic challenges.<sup>49</sup> Failure to respond to isoproterenol with an increase in PRA, despite an adequate cardiovascular response, suggests a postreceptor defect in the ability of the juxtaglomerular apparatus to respond to  $\beta$ -adrenergic stimuli<sup>49</sup> (see also “Regulation of Renal Renin” earlier). Second, the conversion of prorenin to active renin is impaired in some diabetic patients,<sup>593</sup> despite adequate release of prorenin in response to furosemide<sup>49</sup>; this suggests a defect in the normal processing of prorenin. Third, as is the case with perhaps all patients with hyporeninemic hypoaldosteronism (see later), many diabetic patients appear to be volume expanded, with subsequent suppression of PRA.

The most attractive, unifying hypothesis for the suppression of PRA in hyporeninemic hypoaldosteronism is that primary volume expansion increases circulating ANP, which then exerts a negative effect on both renal renin release and adrenal aldosterone release (see also “Regulation of Renal Renin and Adrenal Aldosterone”). There is evidence that these patients are volume expanded, and many will respond to either  $\text{Na}^+\text{-Cl}^-$  restriction or to furosemide with an increased PRA (i.e., renin is physiologically rather than pathologically suppressed).<sup>597,599,600</sup> Patients with hyporeninemic hypoaldosteronism due to a diversity of underlying causes have elevated ANP levels,<sup>27,49,170,599,601</sup> which is also an indicator of their underlying volume expansion. Patients who respond to furosemide with an increase in PRA exhibit a concomitant decrease in ANP.<sup>599</sup> Furthermore, the infusion of exogenous ANP can suppress the adrenal aldosterone response to both hyperkalemia<sup>170</sup> and dietary  $\text{Na}^+\text{-Cl}^-$  depletion.<sup>602</sup>

#### ACQUIRED TUBULAR DEFECTS AND POTASSIUM EXCRETION

Unlike hyporeninemic hypoaldosteronism, hyperkalemic distal RTA is associated with a normal or increased aldosterone and/or PRA. Urine pH in these patients is  $>5.5$ , and they are unable to increase acid or  $\text{K}^+$  excretion in response to furosemide,  $\text{Na}^+\text{-SO}_4^{2-}$  or fludrocortisone.<sup>603-605</sup> Classic causes include SLE,<sup>603</sup> sickle cell anemia,<sup>49,605</sup> and amyloidosis.<sup>49</sup>

#### HEREDITARY TUBULAR DEFECTS AND POTASSIUM EXCRETION

Hereditary tubular causes of hyperkalemia have overlapping clinical features with hypoaldosteronism; hence the shared label “pseudohypoaldosteronism” (PHA). PHA-I has both an autosomal recessive and an autosomal dominant form.

The autosomal dominant form is due to loss-of-function mutations in the MR.<sup>606</sup> These patients require aggressive salt supplementation during early childhood; however, similar to the hypoaldosteronism due to mutations in aldosterone synthase, they typically become asymptomatic in adulthood.<sup>402</sup> Of interest, the lifelong increases in circulating aldosterone, angiotensin-II, and renin seen in this syndrome do not appear to have untoward cardiovascular consequences.<sup>606</sup>

The recessive form of PHA-I is caused by various combinations of mutations in all three subunits of ENaC, resulting in impairment in its channel activity.<sup>402</sup> Patients with this syndrome present with severe neonatal salt wasting, hypotension, and hyperkalemia; in contrast to the autosomal dominant form of PHA-I, the syndrome does not improve in adulthood.<sup>402</sup> One unexpected result in the physiologic characterization of ENaC was that mice with a targeted deletion of the  $\alpha$ -ENaC subunit were found to die within 40 hours of birth due to pulmonary edema.<sup>27</sup> Patients with recessive PHA-I may have pulmonary symptoms, which can occasionally be severe<sup>607</sup>; however, it appears that, unlike in ENaC-deficient mice, the modest residual activity associated with heteromeric PHA-I channels is generally sufficient to mediate pulmonary  $\text{Na}^+$  and fluid clearance in humans with loss-of-function mutations in ENaC.<sup>608</sup>

Pseudohypoaldosteronism type II (PHA-II) (also known as Gordon syndrome and more recently as FHHT [Familial Hyperkalemia with Hypertension]) is in every respect the “mirror image” of Gitelman syndrome; the clinical phenotype includes hypertension, hyperkalemia, hyperchloremic metabolic acidosis, suppressed PRA and aldosterone, hypercalciuria, and reduced bone density.<sup>609</sup> FHHT behaves like a gain-of-function in the thiazide-sensitive  $\text{Na}^+\text{-Cl}^-$  cotransporter NCC, and treatment with thiazides typically results in resolution of the entire clinical picture.<sup>609</sup> FHHT is an extreme form of hyporeninemic hypoaldosteronism due to volume expansion; aggressive salt restriction decreases ANP levels and increases PRA, with resolution of the hypertension, hyperkalemia, and metabolic acidosis.<sup>601</sup>

FHHT is an autosomal dominant syndrome, with four genetic loci. In an initial landmark paper, mutations in two related serine-threonine kinases were detected in various kindreds with FHHT.<sup>610</sup> The catalytic sites of these kinases lack specific catalytic lysines conserved in other kinases, hence the designation “WNK” (With No Lysine). Whereas FHHT mutations in *WNK4* affect the C-terminus of the coding sequence, large intronic deletions in the *WNK1* gene result in increased expression. Both kinases are expressed within the distal nephron, in both DCT and CCD cells; whereas *WNK1* localizes to the cytoplasm and basolateral membrane, *WNK4* protein is found at the apical tight junctions.<sup>610</sup> WNK-dependent phosphorylation and activation of the downstream SPAK (STE20/SPS1-related proline/alanine-rich kinase) and OSR1 (oxidative stress-responsive kinase 1) kinases leads to phosphorylation of a cluster of N-terminal threonines in NCC, resulting in an activation of  $\text{Na}^+\text{-Cl}^-$  cotransport<sup>132</sup> (see also Fig. 16.6). However, coexpression of *WNK4* with NCC reveals additional *inhibitory* influence of the kinase on NCC, effects which are blocked by FHHT-associated point mutations in the kinase.<sup>611</sup> In particular, the inhibitory effects of *WNK4*

appear to dominate in mouse models with overexpression of wild-type versus FHHt mutant WNK4.<sup>69</sup> Mutations in the *Cullin 3* and *kelch-like 3* (*KLHL3*) genes also cause FHHt; the proteins encoded by these genes are part of a ubiquitin-ligase complex that regulates the WNK1 and WNK4 kinase proteins. Autosomal dominant mutations in both *Cullin 3* and *KLHL3* act through dominant negative effects.<sup>612,613</sup>

A key insight from the mechanistic study of FHHt is that the activation of NCC in the DCT in this syndrome serves to reduce Na<sup>+</sup> delivery to principal cells in the downstream CNT and CCD, leading to hyperkalemia.<sup>69</sup> This and other effects of the WNK pathways on distal K<sup>+</sup> secretion are discussed earlier in this chapter (see “Control of Potassium Secretion: the Effect of Potassium Intake”).

## MEDICATION-RELATED HYPERKALEMIA

### CYCLO-OXYGENASE INHIBITORS

Hyperkalemia is a well-recognized complication of nonsteroidal anti-inflammatory drugs (NSAIDs) that inhibit cyclo-oxygenases. NSAIDs cause hyperkalemia by a variety of mechanisms, as would be predicted from the relevant physiology. By decreasing glomerular filtration rate and increasing sodium retention, they decrease distal delivery of Na<sup>+</sup> and reduce distal flow rate. Moreover, the flow-activated apical maxi-K channel in the CNT and CCD is activated by prostaglandins,<sup>614</sup> so NSAIDs will reduce its activity and the flow-dependent component of K<sup>+</sup> excretion.<sup>76,438</sup> NSAIDs are also a classic cause of hyporeninemic hypoaldosteronism.<sup>615,616</sup> The administration of indomethacin to normal volunteers thus attenuates furosemide-induced increases in PRA.<sup>153,617</sup> Finally, NSAIDs would not cause hyperkalemia with such regularity if they did not also blunt the adrenal response to hyperkalemia, which is at least partially dependent on prostaglandins acting through prostaglandin EP2 receptors and cyclic-AMP.<sup>169</sup>

The physiology reviewed earlier in this chapter (see “Regulation of Renal Renin and Adrenal Aldosterone”) would suggest that COX-2 inhibitors are equally likely to cause hyperkalemia. Indeed, COX-2 inhibitors can clearly cause sodium retention and a decrease in glomerular filtration rate,<sup>618,619</sup> suggesting NSAID-like effects on renal pathophysiology. COX-2-derived prostaglandins stimulate renal renin release,<sup>49</sup> and COX-2 inhibitors reduce PRA in both dogs<sup>27</sup> and humans.<sup>156</sup> Salt restriction potentiates the hyperkalemia seen in dogs treated with COX-2 inhibitors,<sup>27</sup> such that hypovolemic patients may be particularly prone to hyperkalemia in this setting. The COX-2 inhibitor, celecoxib, and the nonselective NSAID ibuprofen have equivalent negative effects on K<sup>+</sup> excretion after a defined oral load.<sup>620</sup> Not surprisingly, hyperkalemia and acute kidney injury are associated with COX-2 inhibitors.<sup>49,621,622</sup> Where the data have been reported, circulating PRA and/or aldosterone have been reduced in hyperkalemia associated with COX-2 inhibitors.<sup>27,622</sup>

### CYCLOSPORINE AND TACROLIMUS

Both cyclosporine<sup>623</sup> (CsA) and tacrolimus<sup>624</sup> cause hyperkalemia; the risk of sustained hyperkalemia may be higher in renal transplant patients treated with tacrolimus than in those treated with CsA.<sup>625</sup> CsA is perhaps the

most versatile of all drugs in the variety of mechanisms whereby it causes hyperkalemia. It causes hyporeninemic hypoaldosteronism<sup>626</sup> due in part to its inhibitory effect on COX-2 expression in the macula densa.<sup>627</sup> CsA inhibits apical SK secretory K<sup>+</sup> channels in the distal nephron,<sup>628</sup> in addition to basolateral Na<sup>+</sup>-K<sup>+</sup>-ATPase.<sup>49</sup> Finally, CsA causes redistribution of K<sup>+</sup> and hyperkalemia, particularly when used in combination with  $\beta$ -blockers.<sup>629</sup> A provocative but preliminary report has linked acute hyperkalemia secondary to CsA to indirect activation of K<sub>ATP</sub> channels (see also earlier).<sup>568</sup>

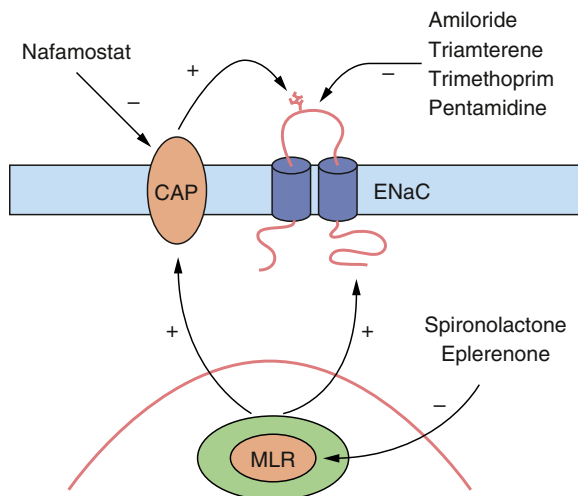
### ENaC INHIBITION

Inhibition of apical ENaC activity in the distal nephron by amiloride and other K<sup>+</sup>-sparing diuretics predictably results in hyperkalemia. Amiloride is structurally similar to the antibiotics trimethoprim (TMP) and pentamidine, which can also inhibit ENaC.<sup>630-632</sup> Trimethoprim thus inhibits Na<sup>+</sup> reabsorption and K<sup>+</sup> secretion in perfused CCDs.<sup>633</sup> Both TMP/SMX (Bactrim) and pentamidine were reported to cause hyperkalemia during high-dose treatment of *Pneumocystis* pneumonia in HIV patients,<sup>49,632</sup> who are otherwise predisposed to hyperkalemia. However, this side effect is not restricted to high-dose intravenous therapy; in a study of hospitalized patients treated with standard doses of trimethoprim, significant hyperkalemia occurred in >50%, with severe hyperkalemia (>5.5 mmol/L) in 21%.<sup>634</sup> Risk factors for hyperkalemia due to normal-dose TMP include renal insufficiency,<sup>634</sup> hyporeninemic hypoaldosteronism,<sup>57</sup> and concomitant use of ACE-inhibitors and ARBs.<sup>635</sup> This is not a trivial association, in that TMP/SMX administration increases the risk of sudden death in patients treated with ACE-inhibitors, angiotensin-receptor blockers, or spironolactone.<sup>636</sup>

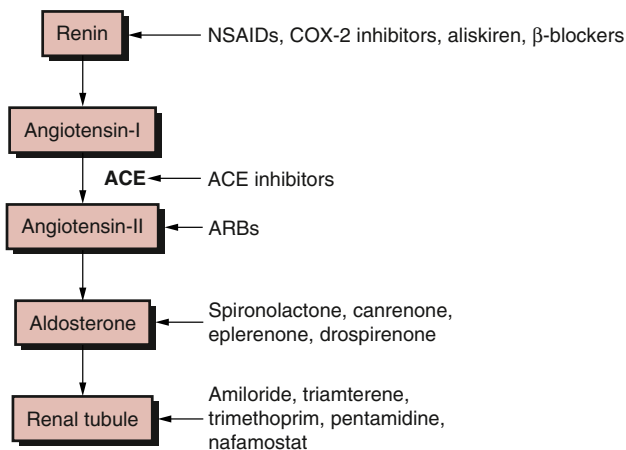
Whereas TMP and pentamidine directly inhibit ENaC, a novel, indirect mechanism for ENaC inhibition-associated hyperkalemia has also been reported.<sup>49,107</sup> Aldosterone induces expression of the membrane associated proteases CAP1-3 (see “Control of Potassium Excretion: Aldosterone”). Nafamostat, a protease inhibitor widely used in Japan for pancreatitis and other indications, is known to cause hyperkalemia<sup>107</sup>; indirect evidence suggests that the mechanism involves inhibition of amiloride-sensitive Na<sup>+</sup> channels in the CCD.<sup>49</sup> It has been evaluated as a treatment for COVID-19, with hyperkalemia reported in trials.<sup>637</sup> Treatment of rats with nafamostat was shown to reduce the urinary excretion of CAP1/prostasin, in contrast to the reported effect of aldosterone.<sup>101</sup> Thus inhibition of the protease activity of CAP1 by nafamostat appears to abrogate its activating effect on ENaC (see Fig. 16.13) and may reduce expression of the protein in the CCD.<sup>108</sup>

### ACE INHIBITORS, MINERALOCORTICOID, AND ANGIOTENSIN ANTAGONISTS

Hyperkalemia is a predictable and common effect of ACE inhibition, direct renin inhibition, and antagonism of the mineralocorticoid and angiotensin receptors<sup>638</sup> (see also Fig. 16.14). The oral contraceptive agent Yasmin-28 and related products contain the progestin drospirenone, which inhibits the MR<sup>639</sup> and can potentially cause hyperkalemia in susceptible patients. As with many other causes of



**Fig. 16.13 Pharmacologic inhibition of the epithelial Na<sup>+</sup> channel ENaC.** Whereas amiloride and related compounds directly inhibit the channel, the protease inhibitor nafamostat inhibits membrane-associated proteases such as CAP1, thus indirectly inhibiting the channel. Spironolactone and related drugs inhibit the mineralocorticoid receptor, thus reducing transcription of the  $\alpha$ -subunit of ENaC, the ENaC-activating kinase SGK, and several other target genes (see text for details).



**Fig. 16.14** Medications that target the renin-angiotensin-aldosterone axis are common causes of hyperkalemia, as are drugs that inhibit epithelial Na<sup>+</sup> channels (ENaC) in the renal tubule (CNT or CCD).

hyperkalemia, that induced by pharmacologic targeting of the RAAS axis depends on concomitant inhibition of the adrenal aldosterone release by hyperkalemia; the adrenal release of aldosterone due to increased K<sup>+</sup> is clearly dependent on an intact adrenal renal-angiotensin system, such that this response is abrogated by systemic ACE inhibitors and ARBs<sup>164</sup> (see “Regulation of Renal Renin and Adrenal Aldosterone Release”). Dual treatment with lisinopril and spironolactone in subjects with CKD is also associated with a reduction in extrarenal potassium disposition, given that reduced K<sup>+</sup> excretion alone does not explain the substantial increase in serum K<sup>+</sup> after a defined oral potassium load.<sup>640</sup> Similarly, the addition of spironolactone to losartan in treatment of diabetic nephropathy causes a significant increase in serum K<sup>+</sup> without significant change in urinary K<sup>+</sup> excretion.<sup>641</sup> ACE

inhibitors and ARBs have the additional potential to cause acute renal failure and acute hyperkalemia in patients with an angiotensin-dependent GFR; the renin inhibitor aliskiren has also been reported to cause acute renal failure with acute hyperkalemia, albeit in conjunction with spironolactone.<sup>642</sup>

RAAS inhibitors are an increasingly important cause of hyperkalemia, given the indications to combine spironolactone with ACE inhibitors and/or ARBs in renal and cardiac disease,<sup>643</sup> in addition to the emergence of MR antagonists with perhaps a greater potential for hyperkalemia.<sup>644</sup> Hyperkalemia can occur within a week of starting angiotensin-receptor blockade.<sup>645</sup> Heart failure, diabetes, and CKD increase the risk of hyperkalemia from these agents.<sup>638,646,647</sup> The prevalence of hyperkalemia associated with the combined use of MR antagonists and ACE inhibitors/ARBs appears to be much higher in clinical practice (~10%)<sup>648</sup> than what has been reported in large clinical trials,<sup>638</sup> in part due to the use of higher than recommended doses.<sup>49</sup> Notably, Juurlink and colleagues<sup>649</sup> studied the correlation between the rate of spironolactone prescription for Canadian patients with heart failure on ACE inhibitors, following the publication of The Randomized Aldactone Evaluation Study (RALES), with hyperkalemia and associated morbidity.<sup>650</sup> This provocative study found an abrupt increase in the rate of prescription for spironolactone after release of RALES, with a temporal correlation to increases in the rate of admissions with hyperkalemia<sup>650</sup>; the association remained statistically significant for admissions where hyperkalemia was the primary diagnosis.<sup>651</sup> However, a study from the United Kingdom found a similar increase in spironolactone use after the publication of RALES, but without an increase in hyperkalemia or hyperkalemia-associated admissions to hospital.<sup>652</sup> It should also be emphasized that the development of hyperkalemia, or for that matter the presence of predisposing factors for hyperkalemia, does not appear to mitigate the mortality benefits of eplerenone in heart failure.<sup>647</sup>

More recently, a new mineralocorticoid, finerenone, has emerged for clinical use, with greater selectivity and potency than spironolactone and eplerenone, combined with a shorter half-life.<sup>653</sup> Given the results of the FIDELIO and FIGARO trials, finerenone has emerged as a key “pillar” in management of diabetic kidney disease.<sup>653</sup> Although head-to-head comparison with spironolactone is lacking, comparative post-hoc analysis has indicated a lesser risk of hyperkalemia from finerenone when compared with spironolactone.<sup>654</sup> Another key advance is the availability of SGLT2 inhibitors, which clearly reduce the risk of serious hyperkalemia in type 2 diabetes without increasing the risk of hypokalemia.<sup>655</sup>

Given the mounting evidence supporting the combined use of ACE-inhibitors, ARBs, and/or MR antagonists, it is prudent to systematically adhere to measures that will minimize the chance of associated hyperkalemia, therefore allowing patients to benefit from the cardiovascular and renal effects of these agents. The patients at risk for the development of hyperkalemia in response to drugs that target the RAAS axis, singly or in combination therapy, are those in whom the ability of kidneys to excrete the potassium load is markedly diminished due to one or a combination of the following: 1. decreased

delivery of sodium to the cortical collecting duct (as in congestive heart failure, volume depletion, etc.); 2. decreased circulating aldosterone (hyporeninemic hypoaldosteronism, drugs such as heparin or ketoconazole, etc.); 3. inhibition of amiloride-sensitive  $\text{Na}^+$  channels in the CNT and CCD, by coadministration of TMP/SMX, pentamidine, or amiloride; 4. chronic tubulointerstitial disease, with associated dysfunction of the distal nephron; and 5. increased potassium intake (salt substitutes, diet, etc.). Overall, patients with diabetes, heart failure, and/or CKD are at particular risk for hyperkalemia from RAAS inhibition.<sup>638,646,647</sup> In these susceptible patients, the following approach is recommended to prevent or minimize the occurrence of hyperkalemia in response to medications that interfere with the RAAS system<sup>27,656</sup>:

- Estimate glomerular filtration rate using MDRD and/or related equations.
- Inquire about diet and dietary supplements (e.g., salt substitutes and licorice) and prescribe a low-potassium diet, preferably with the assistance of a nutritionist.
- Inquire about medications, particularly those that can interfere with renal  $\text{K}^+$  excretion (e.g., NSAIDs, COX-2 inhibitors, and  $\text{K}^+$ -sparing diuretics) and, if appropriate, discontinue these agents.
- Continue or initiate loop or thiazide-like diuretics, if otherwise appropriate (hypertension, edema).
- Correct acidosis with sodium bicarbonate.
- Initiate treatment with a low dose of only one of the agents (i.e., of ACE-inhibitors, ARB, or MR antagonists).
- Check serum  $\text{K}^+$  3 to 5 days after initiation of the therapy and each dose increment, at most within 1 week,<sup>645</sup> followed by another measurement 1 week later.
- If the serum  $\text{K}^+$  is  $>5.6$ , ACE inhibitors, ARBs, and/or MR blockers should be stopped and the patient should be treated for hyperkalemia.
- If serum  $\text{K}^+$  is increased but  $<5.6$  mmol/L, reduce the dose and reassess the possible contributing factors. If the patient is on a combination of ACE inhibitors, ARBs, and/or MR blockers, all but one should be stopped and potassium rechecked.
- If therapy with ACE inhibitors, ARBs, and/or MR blockers is considered particularly critical for management of the patient, consider coadministration of a  $\text{K}^+$  binder such as sodium zirconium cyclosilicate (SZC).
- A combination of a MR blocker and either an ACE inhibitor or an ARB should be used with extreme caution in patients with stage IV or V of CKD (eGFR  $<30$ ). If otherwise indicated in a diabetic patient, preference should be given to finerenone if available.
- The dose of spironolactone in combination with ACE inhibitors or ARBs should be no more than 25 mg/day. For diabetics, consider utilizing finerenone in preference to spironolactone.
- Consider therapy with an SGLT2 inhibitor in hyperkalemic patients, if otherwise indicated.

## THE CLINICAL APPROACH TO HYPERKALEMIA

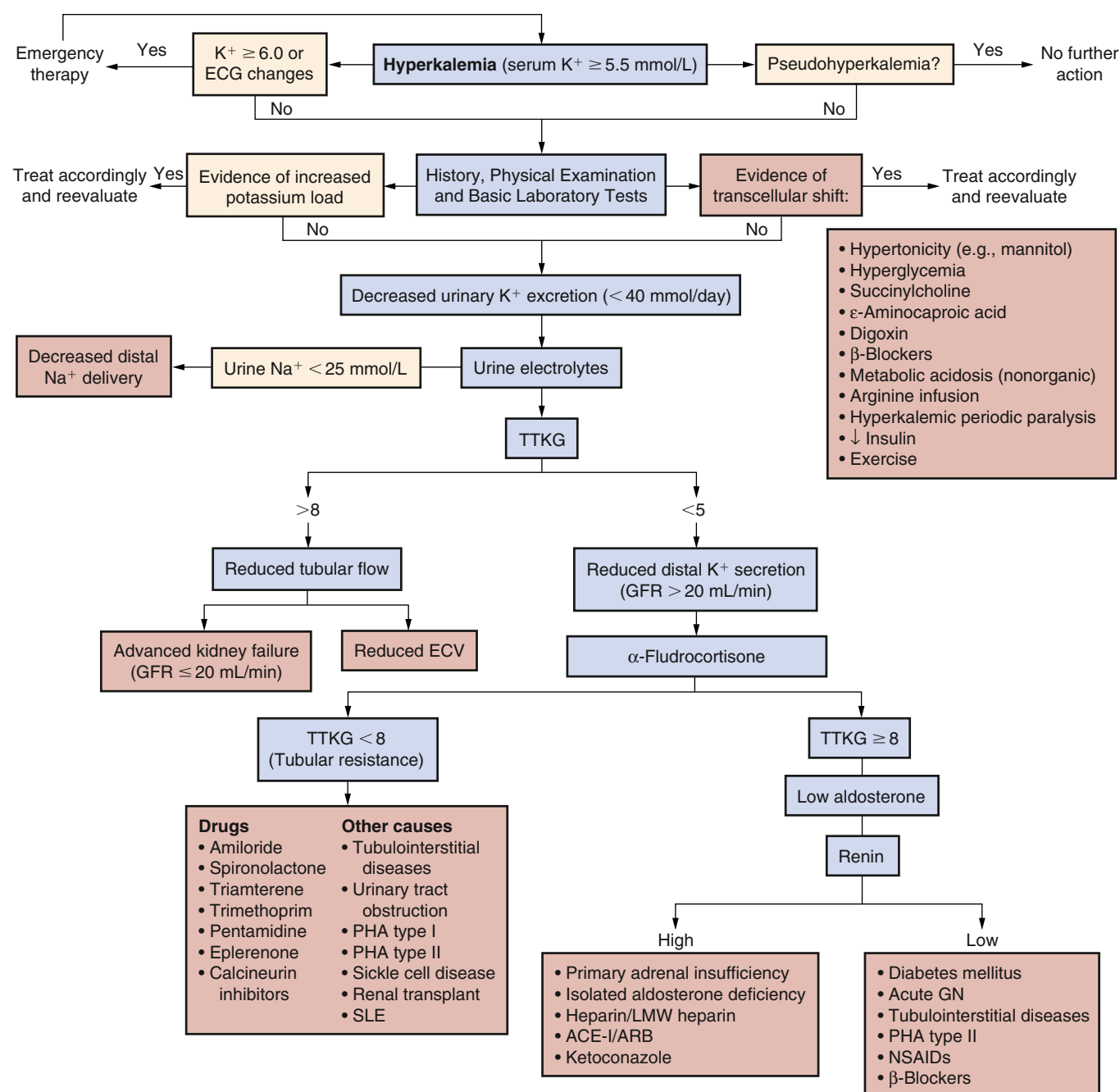
The first priority in the management of hyperkalemia is to assess the need for emergency treatment (ECG changes,  $\text{K}^+ \geq 6.5$  mmol/L). This should be followed by

a comprehensive workup to determine the cause (Fig. 16.15). History and physical examination should focus on medications (e.g., angiotensin-converting enzyme inhibitors, NSAIDs, trimethoprim/sulfamethoxazole), diet and dietary supplements (e.g., salt substitute), risk factors for kidney failure, reduction in urine output, blood pressure, and volume status. Initial laboratory tests should include electrolytes, BUN, creatinine, plasma osmolality,  $\text{Mg}^{2+}$ , and  $\text{Ca}^{2+}$ ; a complete blood count; and urinary pH, osmolality, creatinine, and electrolytes. Plasma and urine osmolality are required for calculation of the TTKG. The determination of urinary electrolytes for calculation of the TTKG or urine  $\text{K}^+:\text{creatinine}$  ratio also provides the opportunity for the measurement of urinary  $\text{Na}^+$ , which will determine whether significant prerenal stimuli are limiting distal  $\text{Na}^+$  delivery and thus  $\text{K}^+$  excretion (see “Urinary Indices of Potassium Excretion” earlier). PRA, plasma aldosterone, and the response in TTKG or urinary  $\text{K}^+:\text{creatinine}$  ratio to fludrocortisone may be necessary to determine the specific cause of an inappropriately low TTKG in hyperkalemia.

## MANAGEMENT OF HYPERKALEMIA

Indications for the hospitalization of patients with hyperkalemia are poorly defined, in part because there is no universally accepted definition for mild, moderate, or severe hyperkalemia. The clinical sequelae of hyperkalemia, which are primarily cardiac and neuromuscular, depend on many other variables (e.g., plasma calcium level, acid-base status, and the rate of change in plasma  $\text{K}^+$ <sup>225-228</sup>), in addition to the absolute value of the serum  $\text{K}^+$ <sup>512,657</sup>; these issues are likely to influence management decisions. Severe hyperkalemia (serum  $\text{K}^+ \geq 8.0$  mmol/L), ECG changes other than peaked T-waves, acute deterioration of renal function, and the existence of additional medical problems have been suggested as appropriate criteria for hospitalization.<sup>512</sup> However, hyperkalemia in patients with any ECG manifestation should be considered a true medical emergency and treated urgently<sup>221,511,658,659</sup>; adequate management and serial monitoring of serum  $\text{K}^+$  will generally require admission. Given the limitations of ECG changes as a predictor of cardiac toxicity (see “Consequences of Hyperkalemia” earlier), patients with severe hyperkalemia ( $\text{K}^+ \geq 6.5\text{--}7.0$  mmol/L) in the absence of ECG changes should be aggressively managed.<sup>187,221,482,511,659,660</sup>

Urgent management of hyperkalemia constitutes a 12-lead electrocardiogram, admission to the hospital, continuous ECG monitoring, and immediate treatment. The treatment of hyperkalemia is generally divided into three categories: 1. antagonism of the cardiac effects of hyperkalemia, 2. rapid reduction in  $\text{K}^+$  by redistribution into cells, and 3. removal of  $\text{K}^+$  from the body. The necessary measures to treat the underlying conditions causing hyperkalemia should be undertaken to minimize the factors that are contributing to hyperkalemia and to prevent future episodes.<sup>221</sup> Dietary restriction (usually 60 meq/day) with emphasis on  $\text{K}^+$  content of total parenteral nutrition solutions and enteral feeding products (typically 25–50 mmol/L) and adjustment of medications and intravenous fluids are necessary; hidden



**Fig. 16.15 Clinical approach to hyperkalemia.** See text for details. ACE-I, Angiotensin-converting enzyme inhibitor; ARB, angiotensin-II receptor blocker; CCD, cortical collecting duct; ECG, electrocardiogram; ECV, effective circulatory volume; GFR, glomerular filtration rate; GN, glomerulonephritis; HIV, human immunodeficiency virus; LMW, low-molecular-weight; NSAIDs, nonsteroidal anti-inflammatory drugs; PHA, pseudohypoaldosteronism; SLE, systemic lupus erythematosus; TTKG, transtubular potassium gradient.

sources of  $K^+$ , such as intravenous antibiotics formulated as potassium salts, should not be overlooked.

## ANTAGONISM OF CARDIAC EFFECTS

Intravenous calcium is the first-line drug in the emergency management of hyperkalemia, even in patients with normal calcium levels. The mutually antagonistic effects of calcium and  $K^+$  on the myocardium and the protective role of  $Ca^{2+}$  in hyperkalemia have long been known.<sup>661</sup> Calcium raises the action potential threshold to a less negative value, without

changing the resting membrane potential; by restoring the usual 15 mV difference between resting and threshold potentials, myocyte excitability is reduced.<sup>187</sup> Administration of calcium also alters the relationship between  $V_{max}$  and resting membrane potential, maintaining a more normal  $V_{max}$  at less negative resting membrane potentials and thus restoring myocardial conduction.<sup>187</sup>

Calcium is available as calcium chloride or calcium gluconate (10-mL ampules of 10% solutions) for intravenous infusion. Each milliliter of 10% calcium gluconate or calcium chloride has 8.9 mg (0.22 mmol) and

27.2 mg (0.68 mmol) of elemental calcium, respectively.<sup>662</sup> Calcium gluconate<sup>663</sup> is less irritating to the veins and can be used through a peripheral intravenous (IV) line; calcium chloride can cause tissue necrosis if it extravasates and requires a central line. The efficacy and onset of action are equivalent for calcium chloride and calcium gluconate, despite the higher molar calcium concentration of calcium chloride ampules.<sup>664</sup> A study of patients undergoing cardiac surgery with extracorporeal perfusion (with concomitant high gluconate infusion) suggested that the increase in the ionized calcium level is significantly lower with calcium gluconate.<sup>665</sup> This finding was attributed to a requirement for hepatic metabolism in the release of ionized calcium from calcium gluconate, such that less ionized calcium would be bioavailable in cases of liver failure or diminished hepatic perfusion.<sup>665</sup> However, further studies in vitro, in animals, in humans with normal hepatic function, and during the anhepatic stage of liver transplantation, have shown equal and rapid dissociation of ionized calcium from equal doses of calcium chloride and calcium gluconate, indicating that release of ionized calcium from calcium gluconate is independent of hepatic metabolism.<sup>27,664</sup>

The recommended dose is 10 mL of 10% calcium gluconate (3–4 mL of calcium chloride), infused intravenously over 2 to 3 minutes and under continuous ECG monitoring. The effect of the infusion starts in 1 to 3 minutes and lasts 30 to 60 minutes.<sup>516,660</sup> The dose should be repeated if there is no change in ECG findings or if they recur after initial improvement.<sup>516,660,664</sup> However, calcium should be used with extreme caution in patients taking digitalis because hypercalcemia potentiates the toxic effects of digitalis on the myocardium.<sup>663</sup> In this case, 10 mL of 10% calcium gluconate should be added to 100 mL of 5% dextrose in water and infused over 20 to 30 minutes to avoid hypercalcemia and to allow for an even distribution of calcium in the extracellular compartment.<sup>514,666</sup> To prevent the precipitation of calcium carbonate, calcium should not be administered in solutions containing bicarbonate.

## REDISTRIBUTION OF K<sup>+</sup> INTO CELLS

Sodium bicarbonate,  $\beta_2$  agonists, and insulin with glucose are all used in the treatment of hyperkalemia to induce a redistribution of K<sup>+</sup>. Of these treatments, insulin with glucose is the most constant and reliable, whereas bicarbonate is the most controversial. However, they are all temporary measures and should not be substituted for the definitive therapy of hyperkalemia, which is removal of K<sup>+</sup> from the body.

## INSULIN AND GLUCOSE

Insulin has the ability to lower plasma K<sup>+</sup> levels by shifting K<sup>+</sup> into cells, particularly into skeletal myocytes and hepatocytes (see “Factors Affecting Internal Distribution” earlier). This effect is reliable, reproducible, dose dependent,<sup>516</sup> and effective, even in patients with CKD and ESRD<sup>667-669</sup> and in the anhepatic stage of liver transplantation.<sup>670</sup> The effect of insulin on plasma K<sup>+</sup> levels is independent of age, of adrenergic activity,<sup>671</sup> and of its hypoglycemic effect, which in fact may be impaired in patients with CKD and/or ESRD.<sup>27</sup>

Insulin can be administered with glucose as a constant infusion or bolus injection.<sup>668,669</sup> One recommended dose for insulin with glucose infusion is 10 units of regular insulin in 500 mL of 10% dextrose, given over 60 minutes (there is no further drop in plasma K<sup>+</sup> after 90 minutes of insulin infusion<sup>671</sup>). However, a bolus injection is easier to administer, particularly under emergency conditions.<sup>221</sup> The recommended dose is 10 units of regular insulin administered intravenously followed immediately by 50 mL of 50% dextrose (25 g of glucose).<sup>658,668,672,673</sup> The effect of insulin on K<sup>+</sup> level begins in 10 to 20 minutes, peaks at 30 to 60 minutes, and lasts for 4 to 6 hours.<sup>516,666,668,674</sup> In almost all patients, the serum K<sup>+</sup> drops by 0.5 to 1.2 mmol/L after this treatment.<sup>669,670,673,674</sup> The dose can be repeated as needed.

Despite glucose administration, hypoglycemia may occur in up to 75% of patients treated with the bolus regimen described earlier, typically 1 hour after the infusion.<sup>668</sup> The likelihood of hypoglycemia is greater when the dose of glucose given is <30 g.<sup>514</sup> To prevent this, infusion of 10% dextrose at 50 to 75 mL/hour and close monitoring of the blood glucose levels is recommended.<sup>658,672</sup> Administration of glucose without insulin is not recommended, since the endogenous insulin release may be variable.<sup>559</sup> Glucose in the absence of insulin may, in fact, increase plasma K<sup>+</sup> by increasing plasma osmolality.<sup>550,551,672</sup> In hyperglycemic patients with glucose levels of  $\geq 200$  to 250 mg/dL, insulin should be administered without glucose and with close monitoring of plasma glucose. Combined treatment with  $\beta_2$  agonists, in addition to their synergism with insulin in lowering plasma K<sup>+</sup>, may reduce the level of hypoglycemia.<sup>668</sup> Of note, the combined regimen may increase the heart rate by  $15.1 \pm 6.0$  beats per minute.<sup>668</sup>

There are theoretical reasons to consider the use of newer short-acting forms of insulin, particularly in patients with renal insufficiency.<sup>675</sup> At present, however, in the absence of rigorous clinical trials that directly compare different regimens,<sup>676</sup> there is no perceived need to alter the “typical” recommended dose of 10 units of regular insulin administered intravenously, followed immediately by 50 mL of 50% dextrose (25 g of glucose).

## $\beta_2$ ADRENERGIC AGONISTS

$\beta_2$ -agonists are an important but underused group of agents for the acute management of hyperkalemia. They exert their effect by activating Na<sup>+</sup>/K<sup>+</sup>-ATPase and the NKCC1 Na<sup>+</sup>-K<sup>+</sup>-2Cl<sup>-</sup> cotransporter, shifting K<sup>+</sup> into hepatocytes and skeletal myocytes (see also “Factors Affecting Internal Distribution”). Albuterol (Salbutamol), a selective  $\beta_2$ -agonist, is the most widely studied and used. It is available in oral, inhaled, and intravenous forms; both the intravenous and inhaled or nebulized forms are effective.<sup>677</sup>

The recommended dose for intravenous administration, which is not available in the United States, is 0.5 mg of albuterol in 100 mL of 5% dextrose, given over 10 to 15 minutes.<sup>677,678</sup> Its K<sup>+</sup>-lowering effect starts in a few minutes and is maximal at about 30 to 40 minutes.<sup>677,678</sup> It lasts for 2 to 6 hours<sup>514</sup> and reduces serum K<sup>+</sup> levels by approximately 0.9 to 1.4 mmol/L.<sup>514</sup>

The recommended dose for inhaled albuterol is 10 to 20 mg of nebulized albuterol in 4 mL of normal saline, inhaled over 10 minutes<sup>668</sup> (nebulized levalbuterol is as

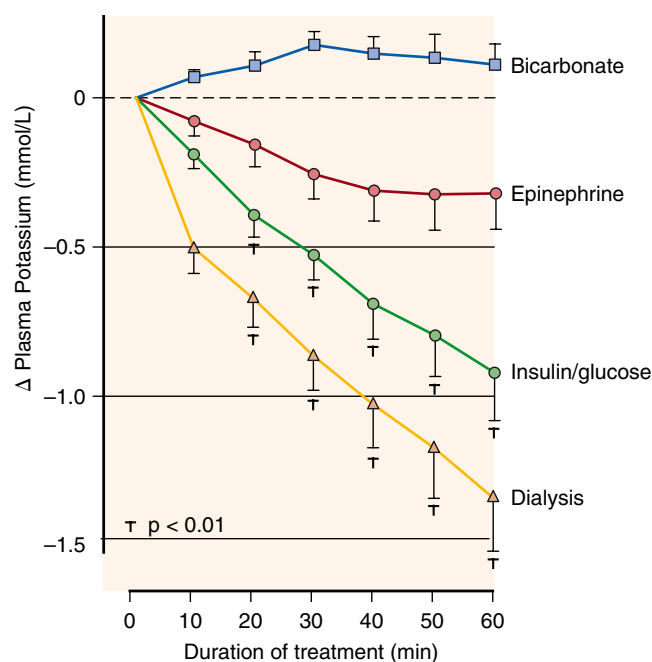
effective as albuterol<sup>679</sup>). Its kaliopenic effect starts at about 30 minutes, reaches its peak at about 90 minutes,<sup>668,677</sup> and lasts for 2 to 6 hours.<sup>514,677</sup> Inhaled albuterol reduces serum  $K^+$  levels by approximately 0.5 to 1.0 mmol/L<sup>514</sup>; albuterol administered by metered-dose inhaler with spacer reduced serum  $K^+$  level by approximately 0.4 mmol/L.<sup>680</sup> Albuterol (in inhaled or parenteral form) and insulin with glucose have an additive effect on reducing serum  $K^+$  levels, by approximately 1.2 to 1.5 mmol/L in total.<sup>514,668,674</sup> However, a subset of patients with ESRD (~20%–40%) are not responsive to  $K^+$ -lowering effect of albuterol ( $\Delta K \leq 0.4$  mmol/L); albuterol (or other  $\beta_2$ -agonists) should not be used as a single agent in the treatment of hyperkalemia.<sup>516,559</sup> In an attempt to reduce pharmacokinetic variability, one study tested the effects of “weight-based dosing” on serum  $K^+$  levels, using 7  $\mu\text{g}/\text{kg}$  of subcutaneous terbutaline (a  $\beta_2$ -agonist) in a group of ESRD patients.<sup>681</sup> The results showed a significant decline in serum  $K^+$  levels in almost all patients (mean  $1.31 \pm 0.5$  mmol/L, range 0.5 to 2.3 mmol/L) in 30 to 90 minutes; of note, heart rate increased by an average of  $25.8 \pm 10.5$  beats per minute (range 6.5–48).<sup>681</sup>

Treatment with albuterol may result in an increase in plasma glucose (~2–3 mmol/L) and heart rate. The increase in heart rate is more pronounced with the intravenous form (~20 beats per minute) than with inhaled form (~6–10 beats per minute).<sup>677,682</sup> There is no significant increase in systolic or diastolic blood pressure with nebulized or intravenous administration of albuterol.<sup>677</sup> However, it is prudent to use these agents with caution in patients with ischemic heart disease.<sup>514</sup>

## SODIUM BICARBONATE

Bicarbonate prevailed as a preferred treatment modality of hyperkalemia for decades. For example, in a survey of nephrology-training program directors in 1989, it was ranked as the second-line treatment, after  $\text{Ca}^{2+}$ .<sup>683</sup> Its use to treat acute hyperkalemia was mainly based on small older uncontrolled clinical studies with a limited number of patients,<sup>55,684,685</sup> in which bicarbonate was typically administered as a long infusion over many hours (contrary to IV push, which later became the routine).<sup>686</sup> One of these studies, which is frequently quoted, concluded that the  $K^+$ -lowering effect of bicarbonate is independent of changes in pH.<sup>685</sup> However, confounding variables included the duration of infusion, the use of glucose-containing solutions, and infrequent monitoring of serum  $K^+$ .<sup>685,687</sup>

The role of bicarbonate in the acute treatment of hyperkalemia has been challenged.<sup>669,686,688</sup> Blumberg and colleagues<sup>669</sup> compared different  $K^+$ -lowering modalities (Fig. 16.16) and showed that bicarbonate infusion (isotonic or hypertonic) for up to 60 minutes had no effect on serum  $K^+$  in their cohort of ESRD patients on hemodialysis; there is, however, an effect of isotonic bicarbonate at 4 to 6 hours (see later).<sup>686</sup> These observations were later confirmed by others, who failed to show any acute (60–120 minutes)  $K^+$ -lowering effects for bicarbonate.<sup>686–688</sup> A few studies have shown that metabolic acidosis may attenuate the physiologic responses to insulin and  $\beta_2$  agonists.<sup>516,688</sup> The combined effect of bicarbonate and insulin with glucose has been studied, with conflicting results.<sup>688</sup> In addition, bicarbonate



**Fig. 16.16** Changes in serum  $K^+$  during intravenous infusion of bicarbonate, epinephrine, or insulin in glucose, and during hemodialysis. (From Blumberg A, Weidmann P, Shaw S, Gnadinger M. Effect of various therapeutic approaches on plasma potassium and major regulating factors in terminal renal failure. *Am J Med.* 1988;85:507–512.)

and albuterol coadministration failed to show any additional benefit over albuterol alone.<sup>689</sup>

In summary, bicarbonate administration, especially as a single agent without dilution, has no role in the contemporary treatment of acute hyperkalemia. Prolonged infusion of isotonic bicarbonate in ESRD patients does, however, reduce serum  $K^+$  at 5 to 6 hours, by up to 0.7 mmol/L; approximately half of this effect can be attributed to volume expansion.<sup>686</sup> Regardless of the mechanism, bicarbonate infusion may thus have a limited role in the *subacute* control of hyperkalemia (e.g., in the nondialytic management of patients with severe hyperkalemia).<sup>690</sup> The acute effect of bicarbonate infusion on serum  $K^+$  in severely acidemic patients is not clear; however, it may be of some benefit in this setting.<sup>221,658</sup> Of note, the infusion of sodium bicarbonate may reduce serum ionized calcium levels and cause volume overload, issues of relevance in acidemic patients with renal failure.<sup>516,666</sup> When bicarbonate administration is utilized for hyperkalemia, we recommend isotonic infusion of sodium bicarbonate; although hypertonic sodium bicarbonate does not increase serum  $K^+$ ,<sup>545</sup> it has been reported to cause hyponatremia.<sup>669</sup>

## REMOVAL OF POTASSIUM

### DIURETICS

Diuretics have a relatively modest effect on urinary  $K^+$  excretion in patients with CKD, particularly in an acute setting.<sup>660</sup> However, diuretics are reasonable therapy for hyperkalemia in patients who have sufficient residual renal function to respond with an increase in urine

output, particularly if successful control of serum  $K^+$  may avoid the need for dialysis. In patients with impaired renal function, use of the following agents is recommended for acute control of serum  $K^+$ : 1. oral diuretics with the highest bioavailability (e.g., torsemide) and the least renal metabolism (e.g., torsemide and bumetanide) to minimize the chance of accumulation and toxicity; 2. intravenous agents (short-term treatment) with the least hepatic metabolism (e.g., furosemide rather than bumetanide); 3. combinations of loop and thiazide-like diuretics for better efficacy, although this may decrease glomerular filtration rate due to activation of tubuloglomerular feedback<sup>691</sup>; and 4. the maximal effective “ceiling” dose.<sup>691</sup> To avoid causing hypovolemia, it may be necessary to increase the rate of IV hydration with normal saline or isotonic bicarbonate. This will also serve to help increase distal tubular delivery of  $Na^+$ , augmenting  $K^+$  excretion.

For chronic management of hyperkalemia, diuretic therapy is not recommended in patients who are euvoletic or who otherwise lack indications for diuretic therapy, given the potential for inducing hypovolemia. However, with liberalization of fluid and salt intake, if appropriate, diuretics can be useful for correcting hyperkalemia in patients with the syndrome of hyporeninemic hypoaldosteronism<sup>692</sup> and selective renal  $K^+$  secretory problems (e.g., after transplantation or administration of trimethoprim).<sup>693,694</sup>

### MINERALOCORTICOIDS

Limited data are available on the role of mineralocorticoids in the management of acute hyperkalemia.<sup>27,695</sup> However, these agents have been utilized in treating chronic hyperkalemia in patients with hypoaldosteronism with or without hyporeninism, those with SLE,<sup>696</sup> kidney transplant patients on cyclosporine,<sup>697</sup> and ESRD patients on hemodialysis with interdialytic hyperkalemia.<sup>698,699</sup> The recommended dose is 0.1 to 0.3 mg/day of fludrocortisone, a synthetic glucocorticoid with potent mineralocorticoid activity and moderate glucocorticoid activity (0.3 mg of fludrocortisone is equal to 1 mg of prednisone with regard to glucocorticoid activity).<sup>27,697,699</sup> In patients with ESRD on hemodialysis, this regimen reduces serum  $K^+$  by 0.5 to 0.7 mmol/L and has not been associated with significant changes in blood pressure or weight (as a surrogate for fluid retention).<sup>698</sup> However, more recent studies of 0.1 mg/day of fludrocortisone in patients on chronic hemodialysis have resulted in statistically significant but clinically inconsequential effects on serum  $K^+$ .<sup>700,701</sup> The long-term safety of fludrocortisone in ESRD has not been established, and given the minimal effect on serum  $K^+$ , we do not recommend its use for the management of interdialytic hyperkalemia.

Pharmacologic inhibition of  $11\beta$ HSD-2 with glycyrrhetic acid (GA) has also been tested as a novel mechanism to control hyperkalemia in ESRD.<sup>400</sup> As in other aldosterone-sensitive epithelia, the  $11\beta$ HSD-2 enzyme protects colonic epithelial cells from illicit activation of the MR by cortisol. Hypothesizing that GA would activate extrarenal potassium secretion by the colon and other tissues, Farese and colleagues<sup>400</sup> tested the effect of the drug in a double-blind, placebo-controlled trial in 10 ESRD patients. Treatment with GA significantly increased the serum ratio

of cortisol:cortisone, consistent with successful inhibition of  $11\beta$ HSD-2. This effect was associated with a significant reduction in mean serum  $K^+$ , with 70% of predialysis values in the normal range (3.5–4.7 mmol/L) in the GA phase of the trial, versus 24% in the placebo phase.<sup>400</sup> PRA and aldosterone levels also dropped, perhaps due to the lower median plasma  $K^+$  level. Glycyrrhetic acid is thus a promising agent for long-term management of plasma  $K^+$  in ESRD; clearly, however, more extensive clinical testing is clearly required before widespread utilization.

### CATION-EXCHANGE RESINS

Ion-exchange resins are cross-linked polymers containing acidic or basic structural units that can exchange either anions or cations on contact with a solution. They are capable of binding to a variety of monovalent and divalent cations. Cation-exchange resins are classified on the basis of the cation (i.e., hydrogen, ammonium, sodium, potassium, or calcium) that is cycled during the synthesis of the resin to saturate sulfonic or carboxylic groups. The dominant resin clinically available in the United States is sodium polystyrene sulfonate (SPS).

SPS exchanges  $Na^+$  for  $K^+$  in the gastrointestinal tract, mainly in the colon, and has been shown to increase the fecal excretion of  $K^+$ .<sup>702</sup> Occasional constipation easily controlled with enema or cathartics has been reported with oral administration of SPS in water.<sup>702</sup> To prevent constipation and to facilitate the passage of the resin through the gastrointestinal tract, sorbitol has been coadministered with the resin. It has thus been routine to administer SPS with sorbitol, with approximately five million annual doses administered in the United States alone.<sup>703</sup> Notably, although we use SPS herein to denote sodium polystyrene sulfonate, SPS is, in fact, a brand name for sodium polystyrene sulfonate in sorbitol, illustrating the frequency in which these agents are administered together.<sup>659</sup>

The effect of ingested SPS on serum  $K^+$  is slow; it may take from 4 to 24 hours to see a significant effect on serum  $K^+$ .<sup>660,666,702</sup> The oral dose is usually 15 to 30 grams, which can be repeated every 4 to 6 hours. Each gram of resin binds 0.5 to 1.2 mEq of  $K^+$  in exchange for 2 to 3 mEq of  $Na^+$ . Earlier studies with SPS, mostly before the era of chronic hemodialysis, used multiple doses of the exchange resin orally or rectally as an enema and were associated with declines in serum  $K^+$  of 1 mmol/L and 0.8 mmol/L in 24 hours, respectively.<sup>702</sup> However, with the advent of routine hemodialysis, it has become common to order only a single dose of resin-cathartic in the management of acute hyperkalemia.

An increasing concern with SPS in sorbitol is intestinal necrosis due to the administration of this preparation; this is frequently a fatal complication.<sup>703-705</sup> Studies in experimental animals suggest that sorbitol is required for intestinal injury<sup>705</sup>; however, a new animal study indicated that SPS alone can cause intestinal necrosis, with no requirement for sorbitol.<sup>706</sup> Additionally, SPS crystals can often be detected in human pathologic specimens, adherent to the injured mucosa.<sup>704,707</sup> Several cases of colonic necrosis following oral SPS alone, without sorbitol, have also been reported, directly implicating SPS in intestinal injury. The risk of intestinal necrosis appears to be greatest when SPS is given with sorbitol within the first week after surgery. Notably, however, in a case series of SPS in

sorbitol-associated intestinal necrosis, only 2 of 11 confirmed cases occurred in the postoperative setting.<sup>704</sup> Although most cases of intestinal necrosis have occurred in patients receiving SPS in 70% sorbitol, this has also been reported in patients receiving SPS in 33% sorbitol.<sup>703</sup>

In response to these reports, the U.S. Food and Drug Administration (FDA) removed recommendations for concomitant or postdosing use of sorbitol from the labeling of SPS in 2005.<sup>703</sup> However, the FDA allowed continuous marketing of the most frequently used ready-made SPS in sorbitol suspension, given that it contained only 33% sorbitol. Subsequently, more cases of intestinal necrosis were reported, some reportedly associated with SPS in 33% sorbitol.<sup>703</sup> As a result, in September 2009 the FDA changed safety labeling for SPS powder, stating that concomitant administration of sorbitol is no longer recommended.<sup>703</sup> Given these serious concerns, clinicians must carefully consider whether emergency treatment with SPS is actually necessary for the treatment of hyperkalemia.<sup>659</sup> Additionally, patiomer and SZC, newly available alternatives, should be used in preference to SPS.

There are minimal data on the efficacy of SPS within the first 24 hours of administration for hyperkalemia<sup>703</sup>; at best, the effect occurs within 4 to 6 hours of administration.<sup>660,666,702</sup> This temporal limitation should be taken into consideration when deciding whether or not to administer SPS in acute hyperkalemia. In patients with intact renal function, alternative measures such as hydration, to increase distal tubular delivery of  $\text{Na}^+$  and distal tubular flow rate, and/or diuretics, are often sufficient for potassium removal. In patients with advanced renal failure, the use of SPS is reasonable as a temporizing maneuver while awaiting hemodialysis; but if hemodialysis is available within 1 to 4 hours, we question the need for SPS, given the delayed hypokalemic response and the risk of potentially fatal intestinal necrosis. Furthermore, if a patient has existing vascular access for hemodialysis, the risk of intestinal necrosis outweighs that of the dialysis procedure.

If SPS is administered, the preparation should ideally not contain sorbitol; SPS without sorbitol is typically available as a powder, which must be reconstituted with water. If a laxative other than sorbitol is coadministered, it should not contain potassium or other cations such as magnesium or calcium, which can compete with potassium for binding to the resin. In patients with renal insufficiency, the laxative should not contain phosphorus. Reasonable laxatives for this purpose include lactulose and some preparations of polyethylene glycol 3350. However, data demonstrating the efficacy and safety of these laxatives with SPS are not available.<sup>659</sup> We should also note that administering SPS without sorbitol might not eliminate the risk of intestinal necrosis, given the role of the SPS resin itself in this complication.

Finally, oral SPS with sorbitol can also injure the upper gastrointestinal tract, although the clinical significance of these findings is not known. Other potential complications include reduction of serum calcium, volume overload, interference with lithium absorption, and iatrogenic hypokalemia.<sup>27</sup>

#### PATROMER AND SODIUM ZIRCONIUM CYCLOSILICATE

Over the past decade, SPS has largely been supplanted in many countries by two alternative  $\text{K}^+$  binders, patiomer and SZC.

Patiomer is a nonabsorbed polymer, provided as a powder for suspension, which binds  $\text{K}^+$  in exchange for  $\text{Ca}^{2+}$ . In a study of 237 patients with CKD and hyperkalemia, the mean change in serum  $\text{K}^+$  after treatment with patiomer was  $-1.01 \pm 0.03$  mM.<sup>708</sup> Approximately 75% of patients achieved the target serum  $\text{K}^+$  of 3.8 to 5.0 meq/L. The 107 patients whose baseline serum potassium was 5.5 meq/L or greater and who achieved the target serum potassium during the initial 4-week treatment period were randomly assigned to patiomer or placebo for another 8 weeks. Serum  $\text{K}^+$  remained the same in patients continuing patiomer and increased by 0.7 meq/L in those assigned placebo. The incidence of hyperkalemia ( $\geq 5.5$  meq/L) was significantly higher in the placebo group (60% vs. 15%). Serious adverse events were rare. However, a serum magnesium level of  $<1.4$  mg/dL occurred in eight patients (3%) during the initial phase, and nine patients were initiated on magnesium replacement, suggesting that impaired intestinal absorption of magnesium may be a limitation in long-term use. Patiomer also binds to ingested medications, with characterized interactions with ciprofloxacin, thyroxine, and metformin; these three drugs need to be administered more than 3 hours before or after patiomer.<sup>709</sup>

SZC is an inorganic, nonabsorbable crystalline compound that exchanges both sodium and hydrogen ions for  $\text{K}^+$  and  $\text{NH}_4^+$  in the intestine. The binding of  $\text{K}^+$  to SZC has some similarity to the selectivity filter of  $\text{K}^+$  channels, with  $>25$ -fold selectivity for  $\text{K}^+$  over  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$ .<sup>710</sup> The efficacy of SZC in hyperkalemic outpatients was evaluated in two nearly identical phase III randomized placebo-controlled trials.

In the Hyperkalemia Randomized Intervention Multidose SZC Maintenance (HARMONIZE) study, 258 adult patients with persistent hyperkalemia entered a 48-hour open-label run-in, during which they received 10 g of SZC three times daily.<sup>711</sup> Of the 258 patients who entered the open-label run-in, 237 (92%) achieved a normal serum  $\text{K}^+$  (3.5–5.0 meq/L) at 48 hours and were then randomly assigned to placebo or 5 g, 10 g, or 15 g of SZC once daily for 4 weeks. During randomized therapy, the mean serum  $\text{K}^+$  was significantly lower with SZC (4.8, 4.5, and 4.4 meq/L with 5, 10, and 15 g dosing, respectively) as compared with placebo (5.1 meq/L). Similarly, the proportion of normokalemic patients at the end of the study was significantly greater with SZC (71%–85% as compared with 48% with placebo). Serious adverse events were uncommon and were not significantly increased with SZC. However, edema was more common with the 10 g and 15 g doses compared with placebo, as was hypokalemia.

In the second initial SZC trial,<sup>712</sup> 753 adult patients with a serum  $\text{K}^+$  of 5.0 to 6.5 mM were randomly assigned to receive 1.25, 2.5, 5, or 10 g of SZC three times daily for 48 hours. A normal serum  $\text{K}^+$  (3.5–4.9 meq/L) at 48 hours was attained by 543 patients (72%), and these individuals were then reassigned to receive placebo or 1.25, 2.5, 5, or 10 g of SZC once daily for 2 weeks. The serum potassium at 2 weeks was significantly lower in patients receiving 5 and 10 g doses of SZC as compared with placebo (by  $\approx 0.3$  and 0.5 meq/L, respectively), but not with 1.25 and 2.5 g doses. Adverse events were similar with placebo and SZC.

In these trials, the steepest decline in serum potassium with SZC occurred during the first 4 hours of therapy.<sup>711,712</sup>

This suggests an acute effect on intestinal potassium secretion, rather than simply a reduction in intestinal potassium absorption.

More recently, SZC has been shown to mitigate the risk of predialysis hyperkalemia in hemodialysis patients<sup>713</sup> and to reduce the risk of arrhythmia in hemodialysis patients managed with 3K dialysates, compared with 2K dialysates without SZC.<sup>714</sup> Additionally, perhaps due to reduced intestinal absorption of  $\text{NH}_4^+$  ions, which bind to SZC, treatment of hyperkalemia with this binder is associated with an increase in serum bicarbonate.<sup>715</sup>

## DIALYSIS

All modes of acute renal replacement therapies are effective in removing  $\text{K}^+$ . Continuous hemodiafiltration is increasingly used in the management of critically ill and hemodynamically unstable patients.<sup>716</sup> Peritoneal dialysis, although not very effective in an acute setting, has been utilized effectively in cardiac arrest complicating acute hyperkalemia.<sup>717</sup> Peritoneal dialysis is capable of removing significant amounts of  $\text{K}^+$  (5 mmol/hour or 240 mmol in 48 hours) using 2-L exchanges, with each exchange taking almost an hour. However, hemodialysis is the preferred mode when rapid correction of a hyperkalemic episode is desired.<sup>559</sup>

An average 3- to 5-hour hemodialysis session removes approximately 40 to 120 mmol of  $\text{K}^+$ .<sup>27,559,718-724</sup> Approximately 15% of the total  $\text{K}^+$  removal results from ultrafiltration, with the remaining clearance from dialysis.<sup>688,721</sup> Of the total  $\text{K}^+$  removed, about 40% is from extracellular space, and the remainder is from intracellular compartments.<sup>719,721,722</sup> In most patients, the greatest decline in serum  $\text{K}^+$  (1.2–1.5 mmol/L) and the largest amount of  $\text{K}^+$  removed occur during the first hour; the serum  $\text{K}^+$  usually reaches its nadir at about 3 hours. Despite a relatively constant serum  $\text{K}^+$ ,  $\text{K}^+$  removal continues until the end of the hemodialysis session, although at significantly lower rate.<sup>27,720,721</sup>

The amount of  $\text{K}^+$  removed depends primarily on the type and surface area of the dialyzer used, blood flow rate, dialysate flow rate, dialysis duration, and serum to dialysate  $\text{K}^+$  gradient. However, about 40% of the difference in removal cannot be explained by the previously mentioned factors and may instead be related to the relative distribution of  $\text{K}^+$  between intracellular and extracellular spaces.<sup>719</sup> Glucose-free dialysates are more efficient in removing  $\text{K}^+$ .<sup>719,722</sup> This effect may be caused by alterations in endogenous insulin levels, with concomitant intracellular shift of  $\text{K}^+$ ; the insulin level is 50% lower when glucose-free dialysates are utilized.<sup>719</sup> Furthermore, these findings imply that  $\text{K}^+$  removal may be greater if hemodialysis is performed in a fasting state.<sup>688</sup> Treatment with  $\beta_2$ -agonists also reduces the total  $\text{K}^+$  removal, by approximately 40%.<sup>682</sup> The change in pH during dialysis has been thought to have no significant effect on  $\text{K}^+$  removal.<sup>559,688</sup> A study evaluated this issue in detail, examining the effect of dialysate bicarbonate concentration on both serum  $\text{K}^+$  and  $\text{K}^+$  removal. Dialysates with bicarbonate concentrations of 39 mmol/L (high), 35 mmol/L (standard), and 27 mmol/L (low) were utilized. The use of high concentration of bicarbonate was associated with a more rapid decline in serum  $\text{K}^+$ ; this was statistically significant for high versus both standard and low bicarbonate dialysates, at 60 and

240 minutes. However, the total amount of  $\text{K}^+$  removed was higher with the low bicarbonate dialysate ( $116.4 \pm 21.6$  mmol/dialysis) in comparison with standard ( $73.2 \pm 12.8$  mmol/dialysis) and high ( $80.9 \pm 15.4$  mmol/dialysis) bicarbonate dialysates; all statistically not significant.<sup>725</sup> Therefore whereas high-bicarbonate dialysis may *acutely* have a more rapid effect on serum  $\text{K}^+$ , this advantage is potentially mitigated by a lesser total removal of the ion over the course of a typical treatment session.

One of the major determinants of total  $\text{K}^+$  removal is the  $\text{K}^+$  gradient between the plasma and dialysate. Dialysates with a lower  $\text{K}^+$  concentration are more effective at reducing plasma  $\text{K}^+$ .<sup>720,724</sup> Many nephrologists utilize the “rule of 7’s” to set the dialysate  $\text{K}^+$  concentration; the plasma  $\text{K}^+$  plus the dialysate  $\text{K}^+$  should equal approximately 7. However, this then entails dialysis with 0 or 1.0 meq/L  $\text{K}^+$  dialysate (“0K” and “1K” bath) in patients with serum potassium concentrations that exceed 6 to 7.0 mmol/L. However, a rapid decline in plasma  $\text{K}^+$  due to 0K or 1K dialysates can be deleterious, via several mechanisms. First, an acute decrease in plasma  $\text{K}^+$  can be associated with rebound hypertension (i.e., a significant increase in blood pressure 1 hour after dialysis),<sup>718</sup> which is attributed in part to the peripheral vasoconstriction that is a direct result of the change in plasma  $\text{K}^+$ .<sup>718</sup> Second, a low plasma  $\text{K}^+$  can alter the rate of tissue metabolism—the so-called Solandt effect<sup>726</sup>—and decrease tissue oxygen consumption, promoting arteriolar constriction.<sup>718</sup> This vasoconstriction, in turn, may reduce the efficiency of dialysis<sup>27</sup>; a randomized, prospective study did not, however, confirm this finding.<sup>720</sup> The difference may have been due to the glucose content of the dialysate (i.e., 200 mg/dL in the former and zero in the latter study); differences in circulating insulin may have had additional, unrelated effects on muscle blood flow.<sup>727</sup> Finally, dialysates with a low  $\text{K}^+$  concentration may increase the risk of significant arrhythmia and sudden cardiac death.<sup>724,728-730</sup>

Several studies have found an increased incidence of significant arrhythmia with hemodialysis, occurring during and immediately after treatment<sup>729,731,732</sup>; an incidence of up to 76% has been reported.<sup>733</sup> In addition to ventricular arrhythmias, hemodialysis can provoke atrial fibrillation, particularly in patients with lower dialysate  $\text{K}^+$  concentration.<sup>734</sup> Historically, some investigators did not consider the hemodialysis procedure to be significantly arrhythmogenic,<sup>735-737</sup> whereas others had suggested a relationship exists between decreases in  $\text{K}^+$ , dialysate  $\text{K}^+$ , and the incidence of significant arrhythmias.<sup>729</sup> Given the more recent data,<sup>728,730</sup> it seems prudent to recommend that dialysates with a very low  $\text{K}^+$  (0 or 1 mmol/L) be used cautiously, if at all. Higher-risk patients include those receiving digoxin; those with a history of arrhythmia, coronary artery disease, left ventricular hypertrophy, or high systolic blood pressure; and those of an advanced age. Continuous cardiac monitoring for all patients dialyzed against a 0 or 1 mmol/L  $\text{K}^+$  bath is strongly recommended.<sup>516</sup>

Given the risk of inducing arrhythmias with low-potassium dialysates, an alternative approach has been proposed for the treatment of significant hyperkalemia.<sup>516,738</sup> In this regimen, dialysis is initiated with a 3 to 4 meq/L potassium bath, which will immediately lower the plasma potassium

concentration in a slower and perhaps safer manner.<sup>729</sup> The potassium concentration in the dialysate may then be lowered in a stepped fashion with each subsequent hour. A more sophisticated approach utilizes “potassium profiling” to maintain a constant potassium gradient during dialysis.<sup>738-740</sup> Potassium profiling results in more sustained, even removal of potassium<sup>739</sup> than dialysis against a fixed-potassium bath (2.5 mEq/L), with less effect on ventricular ectopy.<sup>738-740</sup>

For the management of severe hyperkalemia (serum K<sup>+</sup> ≥7.0 mEq/L), we favor the use of stepped reduction or potassium profiling of dialysate potassium concentrations. We rarely encounter the need to utilize 1K or 0K dialysate baths, which we also avoid at the beginning of dialysis sessions for acute hyperkalemia. We recommend restricting the upfront use of these low-potassium baths to patients with life-threatening hyperkalemic arrhythmias and/or life-threatening conduction abnormalities.

A rebound increase in plasma K<sup>+</sup> can occur after hemodialysis. This phenomenon can be especially marked in cases of massive release from devitalized tissues (e.g., tumor lysis and rhabdomyolysis), requiring frequent monitoring of serum K<sup>+</sup> and further hemodialysis. However, a rebound increase may also occur in ESRD patients during regular maintenance hemodialysis, despite technically adequate treatment,<sup>721</sup> particularly in those patients with a high predialysis K<sup>+</sup>. Factors attenuating K<sup>+</sup> removal and thus increasing the risk and magnitude of postdialysis rebound include pretreatment with β<sub>2</sub>-agonists<sup>682</sup>; pretreatment with insulin and glucose, eating early during the dialysis treatment<sup>688</sup>; a high predialysis plasma K<sup>+</sup><sup>721</sup>; and higher dialysate Na<sup>+</sup> concentrations.<sup>723</sup> Notably, a study of emergency hemodialysis treatments for hyperkalemia failed to detect an association between the predialysis administration of pretreatment expected to cause a shift in serum K<sup>+</sup> and either recurrent hyperkalemia or the need for repeat dialysis within 24 hours.<sup>741</sup>

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